

Astrophysics III: Galactic Astrophysics

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Lectures:

Wednesday: 13:45-15:30, HPV G5, H nggerberg

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Exercises:

Thursdays: 15.45-17.30 (HIT J52/J51)

Friday: 07:45-09:30 (HIT J 51)

Lecture material will be distributed via Moodle

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Chapter 1

Introduction

1.1 The Milky Way and the Universe

This lecture concentrates on the physical properties of the Milky Way galaxy and the processes which are important to understand its current structure and properties. Another strong focus is set on observational data which provide the basic empirical information for our models and theories of the Milky Way.

The place of our Galaxy in the Universe is roughly illustrated in the block diagram in Fig. 1.1.

- The Milky Way is a quite normal spiral galaxy among billions of galaxies in the observable Universe.
- The galaxies were born by the assembly of baryonic matter in the growing potential wells of dark matter concentrations in an expanding Universe. This process started about 14 billion years ago with the big bang. The galaxies evolved with time by assembling initially gas rich matter fragments, going through phases of strong star formation, having phases of high activity of the central black hole, and many episodes of minor and perhaps also major interactions with other galaxies. Although the Milky Way belongs to one of the frequent galaxy types, it represents just one possible outcome of the very diverse galaxy evolution processes.
- Initially, the big bang produced matter only in the form of hydrogen, helium and dark matter. The heavy elements which we see today were mainly produced in galaxies from H and He by nuclear processes in previous generations of intermediate and high mass stars (see Fig. 1.1). Stars form through the collapse of dense, cool interstellar clouds. Then they evolve because of nuclear reactions until they lose a lot of their mass at the end of their evolution in stellar winds or supernova (SN) explosions. This matter, enriched in heavy elements, is mixed with the interstellar gas in the Milky Way which may form again a new generation of stars. The remnants of the stellar evolution, mostly white dwarfs (WD) and neutron stars (NS), contain also a lot of heavy elements which are no more available for the galactic nucleosynthesis cycle.
- Many galaxies, including the Milky Way, have a super-massive black hole (SM-BH) in their center. The black hole grows by episodic gas accretion which may be triggered by galaxy interaction. Supernovae explosions, active phases of the central black hole,

or galaxy interactions are responsible for the loss of interstellar matter of a galaxy to the intergalactic medium. On the other side cold intergalactic matter (IGM), which could be already enriched more or less with heavy elements from another galaxy or previous phases of the Milky Way, can fall onto the Milky Way and enhance further the gas content.

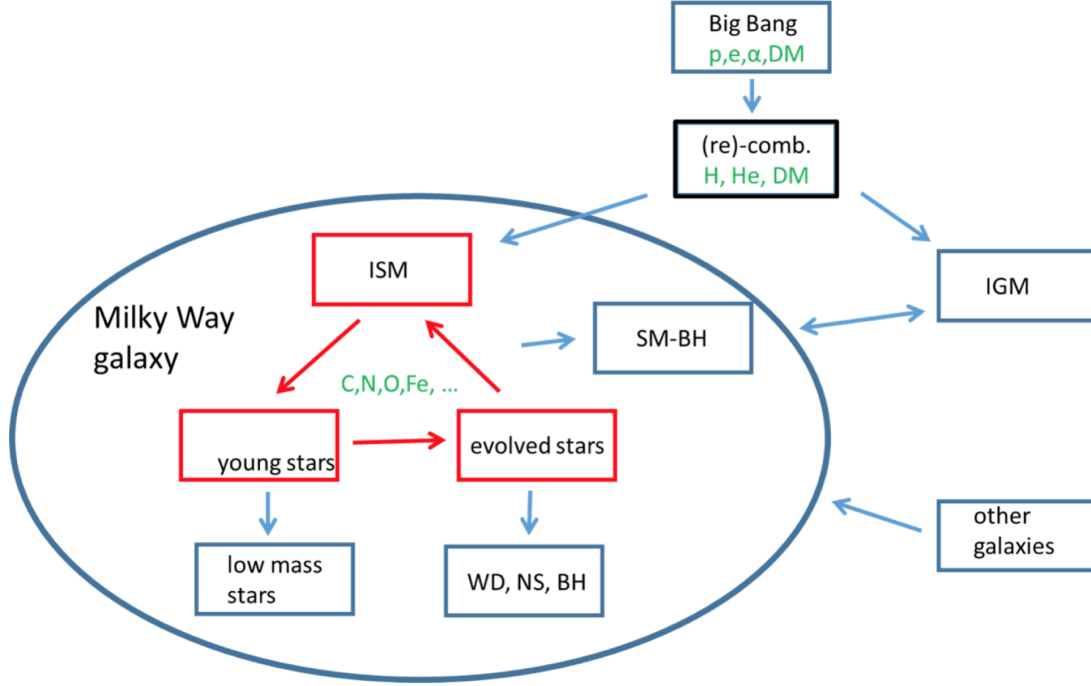


Figure 1.1: The Milky Way in relation to the big bang, the intergalactic matter (IGM), the internal interstellar matter (ISM), different types of stars (WD: white dwarfs; NS: neutron stars), the central, super-massive black hole (SM-BH), and other interacting galaxies.

1.2 Short history of the research in galactic astronomy

Our knowledge on the Milky Way is constantly improving. The Milky Way research profits also a lot from new results gained in other fields in astronomy, like stellar evolution theory, interstellar matter studies, extra-galactic astronomy, or dark matter research. Most important for the progress is the steady advance in observational techniques. The following Table 1.1 lists a few milestones in the evolution of our knowledge in Galactic astronomy.

Table 1.1: Chronology of important studies in Galactic astronomy.

year	important concept, theory, event, or observation
1610	Galileo resolves with his telescope the diffuse light of the Milky Way into countless faint stars.
around 1750	Thomas Wright and Emmanuel Kant describe the Milky Way as a disk of stars with the sun in its center. Kant also speculates that there might exist other Milky Ways similar to our own and that some of the known nebulae could be such galaxies, or “island universes”.
1785	Herschel counts stars in many hundred directions and concludes that the sun is close to the center of a flattened elliptical system which is 5 times larger in the Milky Way plane when compared to polar directions.
1838	Bessel measured for the first time the distance to a star, 61 Cyg at 3.5 pc, based on measurements of the yearly parallax.
1845	Lord Rosse describes for the first time a spiral structure in a nebula (M51) which could be an external galaxy.
around 1890	Photography is introduced in astronomy and this allowed to record thousands or millions of stars on a single plate. Herschel’s Milky Way concept was quantified more accurately by the photographic studies of J. Kapteyn. In the Kapteyn model (1920) the sun is about 650 pc away from the galactic center. The star density drops steadily from the center to about 10 % of the central density at 2.8 kpc in the galactic plane and at 550 pc in polar direction (5:1 ratio).
1919	Shapley studies the distribution of the globular clusters and finds that they are equally frequent above and below the galactic plane but strongly concentrated towards the constellation Sagittarius. Shapley concludes that the sun is far away from the galactic center (he estimated 15 kpc, instead of 8 kpc, because the interstellar extinction was not known yet).
1923	Hubble detects Cepheid variables in M31 (Andromeda galaxy) and this provides very strong evidence that nebulae with spiral structure, but also other nebulae, are galaxies like our Milky Way.
around 1928	Lindblad and Oort develop and prove the basic dynamical model for the Milky Way, in which most stars and the gas in the galactic disk rotate around the galactic center with a speed of about 200 km/s.

1930	Robert Trumpler describes the interstellar absorption due to interstellar dust. The extinction is in the disk plane about 1.8 mag / kpc in the V-band (reduces radiation flux by about a factor of 5/kpc). This effect explains many discrepancies of earlier studies.
1944	W. Baade notices that there exist different populations of stars in galaxies and in the Milky Way. Population I stars are young stars located in the spiral arms and population II stars are old stars predominant in elliptical galaxies, in the bulges of disk galaxies, and in globular clusters.
1951	Ewen and Purcell detect with Radio observations the H I 21 cm line emission which was predicted by van de Hulst in 1944. This line allows the observation of the diffuse interstellar gas in the Milky Way.
around 1970	Vera Rubin and others describe the galaxy rotation problem based on spectroscopic observations of disk galaxies. Since then more and more evidence was collected that this initially unexpected effect is due to the presence of dark matter as postulated first by Fritz Zwicky in 1933 for galaxy clusters.
around 1995	Sensitive near-IR observations provide firm proof for the existence the central super-massive black hole in our Galaxy with measurements of the Keplerian motion of surrounding stars.
2014	The GAIA satellite starts with the measurements of accurate distances, positions and proper motions of about a billion of stars in the Milky Way and much more. This mission is now strongly improving our knowledge about many aspect of the Milky Way.
June 2022	Gaia Data Release 3 (Gaia DR3) including object classifications for 1.59 billion sources and astrophysical parameters (T_{eff} , $\log g$, $[M/H]$, A_G , distance, etc.) from BP/RP spectra for 470 million objects.
January 2025	Official end of Gaia mission.
December 2026 (tbc)	GAIA Data Release 4 (based on 66 months of data).

1.3 Lecture contents and literature

Plan for this lecture: Important topics to be covered by this lecture are:

- components of the Milky Way,
- galactic dynamics,
- physics of the interstellar medium,
- star formation,
- origin and evolution of the Milky Way.

Textbooks:

- *Galactic Astronomy*. J. Binney & M. Merrifield, M. 1998, Princeton Series in Astrophysics

An introduction in galactic astronomy.

- *Galactic Dynamics*. J. Binney & S. Tremaine 2008 (2nd edition), Princeton Series in Astrophysics

The standard textbook for galactic dynamics.

- *Physical Processes in the Interstellar Medium*. L. Spitzer, Wiley & Sons, 1978

The classic collection of basic concepts, but the relation to observations are all outdated.

- *Physics of the Interstellar and Intergalactic Medium*. B.T. Draine, Princeton Univ. Press, 2011
Very comprehensive, at a graduate student level.
- *Astrophysics of Gaseous Nebulae and Active Galactic Nuclei*. D. Osterbrock, University Science / Oxford Univ. Press, 1989 (2nd ed.)
Easily understandable textbook with a strong focus on atomic physics and emission line spectroscopy.

Review articles or collection of review articles on galactic astronomy: The review articles provide usually more detailed and more actual information on specific topics with the drawback that they are often more rapidly outdated than textbooks.

- *The Milky Way as a Galaxy*. G. Gilmore, I. King, P. van der Kruit, Saas-Fee Advanced Course 19, 1989, Geneva Observatory.
- *The Galactic Interstellar Medium*. W.B. Burton, B.G. Elmegreen & R. Genzel, Saas-Fee Advanced Course 21, Springer, 1992

On-line sources:

- **http://adsabs.harvard.edu/abstract_service.html**
NASA astrophysics database system contains essentially all scientific articles in astronomy and astrophysics. Many articles can be downloaded from this site and essentially all articles are available from an ETH account.

Part I

Our galaxy at large (Prof. S. P. Quanz)

Chapter 2

Components of the Milky Way Galaxy

This chapter gives an overview of the two major baryonic constituents in our Galaxy; the stars, and the interstellar matter. This discussion describes mainly observational data which characterize well the Galaxy, its appearance, structure and dynamics. The first section gives an overview of basic data of the Milky Way from which we can derive the distribution and motion of the stars and the interstellar matter. This includes a description of the GAIA mission, which takes currently high precision measurements of hundreds of million of galactic stars. This will provide in the coming years a major advance or even a revolution for the Milky Way research.

Section 2.2 describes basic properties of stars and star clusters and Section 2.3 discusses how stars can be used as test particles for tracing the galactic structure and the local dynamics.

In Section 2.4 the main components of the interstellar matter are briefly described. Emission lines observations of the interstellar gas are very important in providing the large scale structure and the overall rotation of the galactic disk. Later, in Chapter 4, follows a much more detailed treatment of the physics of the interstellar matter.

2.1 Basic data: galactic maps and the GAIA mission

2.1.1 Geometric components

The Milky Way is visible as a straight band extending along a great circle on the celestial sphere from a declination of $+63^\circ$ in the northern constellation Cas (Cassiopeia) to -63° in the southern constellation Crux (Cru). The Galactic center is in the direction of Sgr (Sagittarius) at the position $\alpha = 17^h 46^m$, $\delta = -28^\circ 56'$ in equatorial coordinates.

The galactic center is the zero point for the galactic coordinate system with longitude angle ℓ ($0^\circ \leq \ell \leq 360^\circ$) and latitude angle b ($-90^\circ \leq b \leq 90^\circ$). The galactic system is shown in Slide 2-1 within the equatorial coordinate system. Longitude increases from the center towards NE and the galactic anti-center is in Auriga (Aur). The galactic North pole is in Com (Coma Berenices) and the South pole in Scl (Sculptor).

The galactic structure is best illustrated in maps in galactic coordinates. Slide 2-2 to 2-5 show modern all-sky maps (Mollweide or equal-area projections) of the Galaxy

in different wavelength bands. They provide views of the geometric structures and the distribution of different matter components.

The distribution of stars is best visible in the near-IR map in Slide 2-2 because the absorption by the interstellar dust in this wavelength band is small. The stars trace nicely the galactic disk and the elongated central bulge region.

The distribution of cold gas can be seen in the radio map for the H I line emission in Slide 2-3. H I is a very good tracer of the diffuse, neutral interstellar gas.

The dust absorbs the UV and visual light. Therefore, there are “dark” lanes and holes in the visual map along the Milky Way disk (Slide 2-4), where the dust in the solar neighborhood hides the background stars. The large scale distribution of the dust is better visible in the far-IR wavelength range (Slide 2-5), where the dust re-emits the absorbed radiation energy.

Schematically, the Milky Way can be divided into the components disk, bulge, and halo (see Fig. 2.1).

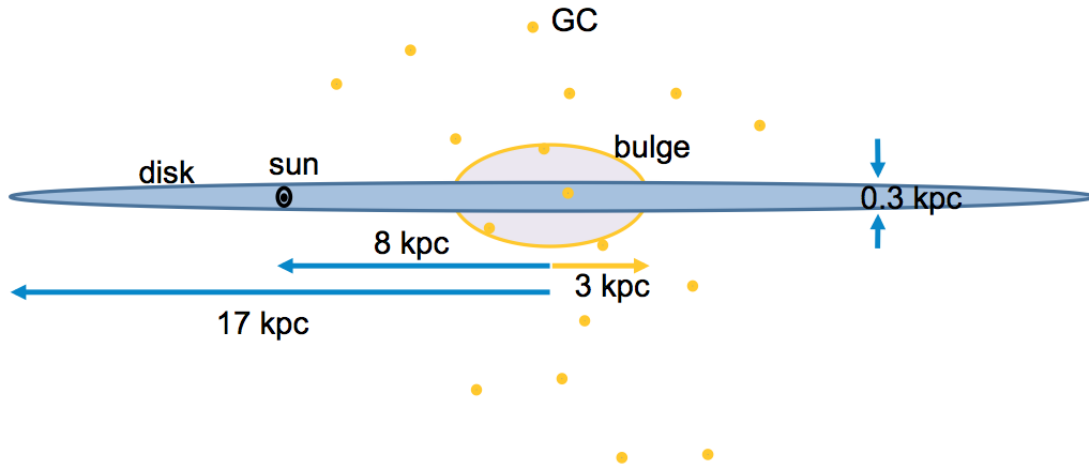


Figure 2.1: Schematic side view of the Milky Way.

Disk. The disk consists of stars, open star clusters and associations, H II regions, molecular clouds, and diffuse gas and dust. There is an overall galactic rotation with a velocity of about $v \approx 220$ km/s. The disk extends from about 3 – 17 kpc from the galactic center and the sun is located at about $r = 8$ kpc. The width of the disk, as measured from the star density, is of the order 100 pc at the location of the sun.

Bulge. The galactic bulge is a bar extending to about 3 kpc from the center. It consists mainly of old, metal-rich stars with randomly oriented orbits around the galactic center. There is essentially no cold gas in the bulge except for the center of the galaxy where we see a small gas disk with a radius of about 100 pc. In the very center of the Galaxy is a super-massive black hole.

Halo. The extended galactic halo has a much lower density of baryonic matter than the disk and the bulge. An important baryonic component of the halo are the roughly 150 globular clusters. They reside in a spherical distribution around the galactic center. About half of the globular clusters lie within 2 kpc from the galactic center but some are also further away than 10 kpc. There exists also a (low density) population of halo stars with a distribution similar to the globular clusters.

The nearest dwarf galaxies are also located in the galactic halo. The Canis Major and Sagittarius dwarf galaxies are currently colliding with the Milky Way at a distance of about ≈ 10 kpc from the galactic center. The galactic halo contains further clouds of neutral H I gas within a hot, low density gas.

The main mass component of the halo and the Milky Way is dark matter. It extends to a radius of about 100 kpc from the galactic center and dominates the galactic gravitational potential on large scales.

2.1.2 The GAIA revolution

Galactic astronomy is currently revolutionized by the results from the GAIA space mission. The GAIA instrument is measuring now for about a billion of stars very accurate positions (Slide 2-6), parallax distances, proper motions, radial velocities, photometric brightnesses, colors, and photometric variabilities, as well as spectral types. More accurately, GAIA will measure the following:

- photometry, colors, stellar positions, proper motion, parallax distances for all (≈ 1 billion) stars down to magnitude $m_V \approx 20$,
- radial velocities with a precision of a few km/s for all stars (about 100 million) down to $m_V \approx 17^m$,
- spectroscopy for millions of stars will be obtained for metallicity determinations.

The data quality of the GAIA mission will be extremely good. We pick as an illustration only one example, the measurements of distances by the annual parallax:

- about 340'000 stars down to $m_V = 10^m$ will have a parallax uncertainties of about 5–10 μas (micro-arcsec). The parallax of a star at 1 kpc is 1000 μas allowing thus distance measurements with a precision better than 1 %.
- a precision of 25 μas will be achieved for 30 million stars with $m_V < 15^m$. This limit includes very bright stars in the galactic bulge, many halo stars, and countless stars in the Magellanic Clouds.
- for 1 billion stars with $m_V < 20^m$ the achievable precision is 300 μas . This will provide accurate distances for all faint stars to distances up to 1 kpc.

This should be compared with the (best) previous data from the Hipparcos satellite, which obtained distance parallaxes with a precision of a few mas (milli-arcsec) for about 100'000 stars with $m_V < 7.5^m$. This provided a mapping of all bright stars to about 100 pc. GAIA will go about 300 times further in distance.

Beside this, GAIA will also detect about $\approx 100\,000$ quasars, $\approx 100\,000$ asteroids, detect the reflex motion of ≈ 1000 stars because of the presence of an extrasolar planet, measure the astrometric light bending due to General Relativity by the sun and planets. GAIA has also the potential to uncover new phenomena we are not aware of yet.

Expected results for galactic astronomy. GAIA will use many of the described methods discussed in this chapter for the study of the Milky Way. The much improved quality of the data will clarify or at least provide important progress for the following topics:

- we will get a synoptic picture of the evolution of our Galaxy from its detailed geometric and dynamic structure and the distribution of stellar metallicities as function of age,
- trace accurately the distribution of the invisible dark matter from the motion of stars out to distances $\gtrsim 10$ kpc from the sun,
- map the spiral structure and define its dynamics in much detail from the distribution and velocities of young stars,
- measure the bar and inner bulge dynamics,
- provide a comprehensive inventory of galactic star clusters and associations, measure the velocity dispersion of their star and identify the processes which leads to the cluster disruption,
- measure the distribution of dust with photometric measurements of the interstellar absorption (reddening) for millions of stars,
- describe the interaction of halo stars and globular clusters with the galactic disk based on the positions and motions of halo stars,
- identify the star streams which are the debris of dwarf galaxies which were tidally disrupted by an interaction with the Milky Way.

Beside this, there are countless other questions and problems in galactic astronomy which will be solved by the data of this mission.

Current status of the mission. GAIA was launched in December 2013 and has started its science operation in mid 2014. Measurements of 5 years were planed to reach the science results. Because the instrument is still working very well the mission duration was extended and operation might be possible until 2025. This extension will further improve the quality of the final scientific data. The GAIA team provided already 3 data releases with the most basic results (source list, positions, distances, proper motion, brightness colors, etc.) which are available for the whole astronomical community.

Measuring principles. GAIA has two telescopes which observe simultaneously the sky in two observing directions with a fixed angle of 106.5° between. The spacecraft rotates continuously around an axis perpendicular to the line of sight of the two telescopes and together with a slight spacecraft precession the whole sky is scanned many times (≈ 50) during the satellite life-time.

The objects from the two observing directions are registered by the same detector array so that their relative positions can be determined accurately (Slide 2.7).

The two telescopes have primary mirrors of 1.45×0.5 m each. The detector system consists of 106 CCD detectors with 4500×1966 pixel each, what gives in total a system with 10^9 pixels.

Because of the spacecraft rotation all stars move in the same direction over the focal plane. First the stars “hit” the wave front sensor and telescope angle monitor. The sky mapper detects the targets and this defines then the data to be stored by the following

systems in order to reduce the data downlink rate. The large array is used for accurate position measurements, then low resolution prism spectro-photometry is taken, before the high resolution spectrograph takes $\lambda/\Delta\lambda \approx 10000$ spectra for radial velocity measurements and the spectral characterization of the brighter targets (Slides 2.8 to 2.10).

2.2 Stars

The stars are a major component of the Milky Way. Stars are ideal test particles which provide accurate positions, density distributions and motion information for the characterization of the Galactic potential and dynamical processes. In addition one can estimate for certain stars their age and/or their metallicity which provide further dynamical but also evolutionary information about the Milky Way system. On the other side the large scale Milky Way structure has a strong impact on the star formation which takes place in dense molecular clouds.

In this section the properties of stars are described with the particular focus on parameters which provide diagnostic information about the Milky Way system. Stellar astrophysics is a main topic of the ETH lecture Astrophysics I. Slide 2-11 provides as a reminder a short description of the evolution of a low and a high mass star together with the corresponding (schematic) evolutionary tracks in the theoretical Hertzsprung-Russell diagram. In the following we summarize basic formulae and a few important points on stellar parameters and evolution.

Stars can be characterized quite well by a few key parameters. The most basic quantities are L luminosity, R radius, T_{eff} effective surface temperature, M mass, and τ age. Another important parameter for galactic studies is the metallicity (e.g. Z). Further parameters are binarity and the corresponding binary parameters, stellar rotation, and magnetic fields. There exist several important relations between stellar parameters.

Black-body laws: For a sphere radiating like a black body there is according to the *Stefan-Boltzmann law*:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4. \quad (2.1)$$

The Planck curve describes the spectral energy distribution of a black body

$$B_{T_{\text{eff}}}(\lambda) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k T_{\text{eff}}} - 1}. \quad (2.2)$$

The wavelength spectrum has its maximum flux $B_{\text{max}} = B_{T_{\text{eff}}}(\lambda_{\text{max}})$ according to *Wien's law* at

$$\lambda_{\text{max}} = \frac{2.9\text{mm}}{T_{\text{eff}}[\text{K}]}. \quad (2.3)$$

For $\lambda \gg \lambda_{\text{max}}$ the spectral energy distribution can be described by the *Rayleigh-Jeans approximation*:

$$B_{T_{\text{eff}}}(\lambda) \approx \frac{2c}{\lambda^4} k T_{\text{eff}}, \quad (2.4)$$

2.2.1 Properties of main-sequence stars

Main sequence stars burn hydrogen to helium. This phase lasts about 90 % of the nuclear burning life time of a star. Therefore about 90 % of all stars are main sequence stars and their properties are therefore particularly relevant.

Mass-luminosity relations on the main sequence. The luminosity of main sequence stars is a strong function of mass which is described by a power law function with different exponent α for different mass ranges:

$$\frac{L}{L_{\odot}} \approx a \left(\frac{M}{M_{\odot}} \right)^{\alpha}, \quad (2.5)$$

where

$$\begin{aligned} \alpha = 2.3, \quad a = 0.23 & \quad \text{for } M < 0.43 M_{\odot}, \\ \alpha = 4.0, \quad a = 1.0 & \quad \text{for } 0.43 M_{\odot} < M < 2 M_{\odot}, \\ \alpha = 3.5, \quad a = 1.5 & \quad \text{for } 2 M_{\odot} < M < 20 M_{\odot}, \\ \alpha = 1.0, \quad a = 3200 & \quad \text{for } M > 20 M_{\odot}. \end{aligned}$$

Main sequence lifetime. The main sequence lifetime of a star ends when about 10 % of all H is burnt to He. High mass stars have a much larger luminosity and therefore they burn their fuel much faster than low mass stars. To first order one can write for example for higher mass stars

$$\tau_{\text{ms}} \propto \frac{M}{L} \propto \frac{1}{M^{2.5}} \quad \text{for } 2 M_{\odot} < M < 20 M_{\odot} \quad (2.6)$$

or for low mass stars

$$\tau_{\text{ms}} \propto \frac{1}{M^{1.3}} \quad \text{for } M < 0.43 M_{\odot}. \quad (2.7)$$

Stellar parameters for main-sequence stars. The following table lists main sequence parameters for different spectral types of stars.

Table 2.1: Parameters for main sequence stars: mass, luminosity, radius, effective surface temperature and main sequence life time.

sp.type	O5 V	B0 V	A0 V	G0 V	M0 V	M8 V
$M/M_{\odot}$	60	18	3.0	1.1	0.50	0.06
$L/L_{\odot}$	$8 \cdot 10^6$	$7 \cdot 10^4$	54	1.5	0.080	$1.2 \cdot 10^{-3}$
$R/R_{\odot}$	12	7.5	2.5	1.1	0.50	0.10
T_{eff} [K]	41'000	30'000	9500	6000	3800	2600
τ_{ms} [yr]	$8 \cdot 10^5$	$4 \cdot 10^6$	$6 \cdot 10^8$	$7 \cdot 10^9$	$6 \cdot 10^{10}$	$5 \cdot 10^{11}$

The parameters given in Table 2.1 are only approximative. The given values allow to construct a $\log T_{\text{eff}} - \log L/L_{\odot}$ plot or a “theoretical Hertzsprung-Russel diagram”. Detailed studies show that there are many subtle dependencies of the basic stellar parameters on e.g. age, metallicity, or rotation rate, but this is beyond the scope of this lecture.

Initial mass function (IMF). The initial mass function describes the mass distribution $N_S(M)$ of newly formed stars per mass bin ΔM . This distribution is quite universal. The standard IMF derived from stellar luminosity statistics and assuming that the main-sequence lifetime ends after burning of about 10 % of the hydrogen (Salpeter 1955) can

be described by a power law distribution

$$\frac{dN_S}{dM} \propto M^{-2.35} \quad \text{for } M > 0.5 M_\odot. \quad (2.8)$$

This relation is often given as a logarithmic power law of the form

$$\frac{dN_S}{d \log M} \propto M^{-1.35} \quad \text{because} \quad \frac{dN_S}{dM} = \frac{dN_S}{d \log M} \frac{d \log M}{dM} = \frac{1}{M} \frac{dN_S}{d \log M}.$$

This is equivalent to a linear fit with slope -1.35 in $\log M$ - $\log N_S$ diagram (Figure 2.2). This law indicates, that the number of newly formed stars with a mass between 1 and 2 $M_\odot$ is about 20 times larger than the stars with masses between 10 and 20 $M_\odot$. One may also say that twice as much gas from a star-forming cloud ends up in stars between 1 and 2 $M_\odot$ when compared to stars with masses between 10 and 20 $M_\odot$. For low mass star the IMF power law has a steep cut-off for $M < 0.5 M_\odot$ where the general law does not apply.

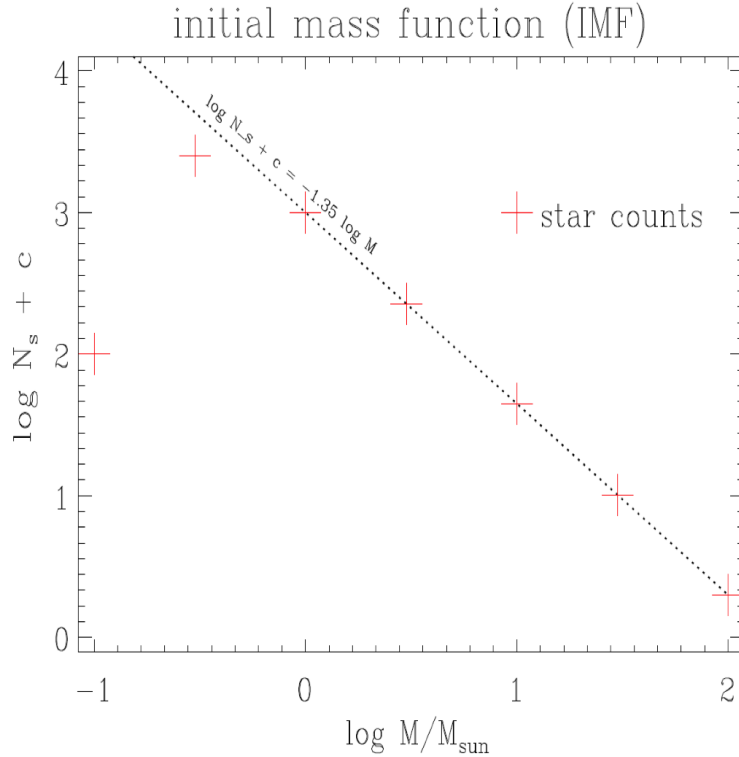


Figure 2.2: Schematic illustration of the initial mass function (IMF) for stars.

Discussion on main sequence stars. Luminosity, effective surface temperature, and the life time of main-sequence stars are very important for the interpretation of stellar populations. The following points can be made:

- high mass stars are born much less frequently than low mass stars,
- high mass stars, although rare, dominate the luminosity of a new-born population of stars (a young association or star cluster),

- high mass stars are blue stars and therefore a young population has a blue color,
- after some time (e.g. > 1 Gyr) the yellow-red low mass stars dominate the main-sequence population because all short-lived high-mass stars are gone,
- the total luminosity of a stellar population decreases steadily with age.

2.2.2 Observational Hertzsprung-Russell diagrams

A stellar population can be characterized well if the stars can be placed into the Hertzsprung-Russell diagram (HR-diagram) or color-magnitude diagram. This requires the measurement of the absolute brightness which can be related to the absolute luminosity and the color index which can be related to the surface temperature. This information follows from accurate photometry and distance determinations. In astronomy many different photometric systems are used and each requires accurate calibration procedures. This subsection provides only a simplified description of the basic principles.

Measurements of magnitudes and colors. Photometric measurements are carried out typically in wavelength bands which are specific for each instrument used. As general photometric reference the Vega magnitude system is used.

All photometric measurements are related to the star Vega (α Lyr) by the definition that Vega has an apparent magnitude of

$$m_{\lambda}(\text{Vega}) = 0.0^m \quad (2.9)$$

in all photometric bands in the wavelength region from about 150 nm to 15 μm (UV - visual - IR range).

Photometric magnitude is a logarithmic quantity which relates the relative flux ratio of two measurements f_1 and f_2 by the relation

$$m_1 - m_2 = -2.5 \log \frac{f_1}{f_2} . \quad (2.10)$$

This means that a star with $m_2 = 2.5^m$ is 10 times fainter than a star with $m_1 = 0^m$.

Apparent colors or color indices CI between two wavelength filters λ_1 and λ_2 are also quantified as magnitude difference

$$\text{CI} = m_{\lambda_1} - m_{\lambda_2} , \quad (2.11)$$

e.g. the color B–V is the difference between the standard Johnson blue filter and visual filter $m_B - m_V$. B–V is positive for a star which is more “red” than Vega and negative for a star which is more “blue”. Colors for other filter pairs are defined according to the same principle.

Distances and interstellar extinction. The apparent magnitude m measured for stars must be converted in the next step into absolute stellar magnitudes M and intrinsic stellar colors. For this one needs to take into account the distance of the star and the possible interstellar extinction.

The relation between the apparent flux f_{λ} and absolute flux F_{λ} of a star depends on the distance d and the interstellar extinction τ_{λ}

$$f_{\lambda}(d) = \frac{F_{\lambda}}{4\pi d^2} e^{-\tau_{\lambda}} . \quad (2.12)$$

This relation can be expressed in magnitudes. For this the absolute magnitude M_λ is introduced, which is the apparent magnitude of an object at a distance of 10 pc without interstellar extinction:

$$M_\lambda = m_\lambda(f_\lambda(10 \text{ pc})) . \quad (2.13)$$

For example, our sun has an absolute magnitude of $M_V = +4.5^m$ in the visual band. Vega is at a distance of about 10 pc and therefore also the absolute magnitude of Vega is approximately

$$M(\text{Vega}) \approx 0^m .$$

The general formula for the conversion of the apparent magnitude m of a star into absolute magnitudes M is given by the following formula:

$$m_\lambda = M_\lambda + 5 \log d [\text{pc}] - 5 + A_\lambda . \quad (2.14)$$

In this equation there are two terms:

- the *distance modulus*: $5 \log d [\text{pc}] - 5$ which follows from

$$m_\lambda - M_\lambda = -2.5 \log \frac{f_\lambda(d)}{f_\lambda(10 \text{ pc})} = -2.5 \log \frac{(10 \text{ pc})^2}{(d [\text{pc}])^2} = -(5 - 5 \log d [\text{pc}]) , \quad (2.15)$$

- and the interstellar extinction: $A_\lambda \geq 0^m$.

The interstellar extinction is due to small $< 1 \mu\text{m}$ interstellar dust particles. Their absorption is stronger in the blue than in the visual $A_B > A_V$ and therefore the light is reddened. On average the following relation approximates quite well the extinction effect:

$$E_{B-V} = A_B - A_V \approx \frac{A_V}{3.1} . \quad (2.16)$$

The color effect $E_{B-V} = A_B - A_V$ is according to this relation roughly proportional to the absolute extinction A_V and therefore one can use the reddening of a star as a measure for the extinction. The reddening follows from the measurements of the apparent color $m_B - m_V$ for a star for which the intrinsic color $M_B - M_V$ is known, for example from its spectral type. This method can also be applied to photometric measurements in other filters. Typically, the extinction is about $A_V \approx 1.8^m/\text{kpc}$ in the galactic disk and $A_V < 0.2^m$ for extragalactic observations in the direction of the galactic poles.

HR-diagram for the stars in the solar neighborhood. HR-diagrams for nearby stars have two advantages:

- the distances d are well known from parallax measurements, and
- the interstellar extinction is small $A_V < 0.2^m$ and can be neglected.

Slide 2-12 shows the Hertzsprung-Russell diagram as determined from data of the Hipparcos satellite. Hipparcos obtained between 1990 and 1993 accurate distances and photometry for about 100'000 stars up to a distance of about 120 pc and covered all stars brighter than $m_V = 7.2^m$ and selected additional stars of interest. Slide 2-12 shows the location of about 17'000 single stars in the HR-diagram which could be measured with the highest precision.

The Hipparcos HR-diagram has the following characteristics:

- the nearby stars are a good average sample for the stars in the Milky Way,
- with Hipparcos it was possible to measure accurately the location of the main-sequence for low mass stars down to an absolute magnitude of $M_V = 12^m$ (for selected nearby stars),
- Hipparcos obtained for each star about 50 – 150 photometric measurements and could therefore measure the photometric variability and use well defined averaged values,
- the Hipparcos sample is not contaminated by foreground or background stars because the distances are well known for all objects,
- the nearby stars are well known and the sample can be cleaned from binaries which can spoil the stellar photometry of supposed single stars,
- a significant disadvantage of the Hipparcos sample is the lack of rare high mass objects.

The *Hipparcos HR-Diagram* (Slide 2-12) shows that the local population of bright stars is mainly composed of low to intermediate mass main sequence stars ($M \approx 0.5 - 3 M_\odot$) and a significant population of evolved stars on the red giant branch. There are also some very low mass main sequence stars $M_V > 12^m$ and white dwarfs in the Hipparcos sample. There are many more such faint stars in the solar neighborhood, but they were too faint for the Hipparcos satellite.

The *GAIA HR-Diagram* shown in Slide 2-13 is far more complete, because it covers also “all” faint stars down to the absolute brightness limit of 15 mag. The shown stellar distribution illustrates that low luminosity objects, white dwarfs and low mass main sequence objects, are very frequent in the solar environment.

HR-diagram for stellar clusters. The stars in a stellar cluster have all essentially the same distance (same distance modulus $m - M$) and a similar amount of interstellar extinction A_λ . For this reason it is possible to determine observationally all features of the HR-diagram or color-magnitude diagram for the stellar population in the studied cluster without knowledge of the exact distance and interstellar extinction.

Slide 2-14 shows color – apparent magnitude diagrams of the nearby Hyades and Pleiades clusters. We see for both clusters the stars on the main sequence, but simply shifted relative to each other because of the different distance modulus. The distance moduli are $m - M = 3.3^m$ (46 pc) for the Hyades and 5.65^m (135 pc) for the Pleiades.

This distance difference is taken into account in the color – absolute magnitude diagram from GAIA (Slide-14). The high precision of GAIA illustrates the narrowness of the main sequence and the population of unresolved binaries above the main sequence. A binary with two stars of equal mass, brightness and color is just 0.75 mag (twice) brighter than a single star. A low mass companion is cooler and less luminous and the unresolved object appears in this diagram slightly shifted upwards and to the right with respect to the expected location of the primary.

The main sequence turn-off point or the upper end of the main sequence is a key parameter of the color-magnitude diagram which provides the age of the cluster. One can assume that all stars in a cluster have essentially the same age (t_{cl} : cluster age) with a main sequence lifetime as described in the previous section (first order $\tau_{ms}(M) \propto 1/M^2$). In young clusters the main-sequence extends to very bright stars while in older clusters

all high mass stars have already evolved away from the main-sequence ($t_{\text{cl}} > \tau_{\text{ms}}(M)$). In the case of the Hyades the turn-off point is around $M_V = +0.5^m$, while it is around $M_V = -2.5$ for the Pleiades.

The distribution of cluster stars in the color-magnitude diagrams provide very important information about stellar evolution because all stars have the same age. This allows to trace and establish the exact evolution of stars within the HR-diagram.

One difficulty to be considered for the analysis of observational color-magnitude diagrams is the contamination of the cluster sample by foreground or background stars. For this reason the data of rich clusters in low density fields (location at high galactic latitude or fields with high background absorption) provide better results with less contamination.

2.2.3 Stellar clusters and associations

Galactic clusters. There are more than 1000 galactic clusters (or open clusters) known and the total number is estimated to be about 100'000. Galactic clusters have a radius of the order of 10 pc and a wide range of star densities ranging from 0.3 stars/pc³ for the Hyades to about 1000 stars/pc³ at the center of the richest clusters. For comparison, the star density in the solar neighborhood is about 0.1 stars/pc³. Dense clusters are dynamically bound by the mutual gravitational attraction of the cluster stars, while lower density systems are in the process of dissolving themselves. The total masses of galactic clusters lie in the range of about 100 to 3000 $M_{\odot}$. The integrated brightness is typically $M_V \approx -5$, but can also be as high as $M_V \approx -10$ for the most extreme cases. Table 2.2 lists parameters and Slide 2-15 shows pictures of some well-known galactic clusters.

Table 2.2: Parameters for galactic clusters

name	dist. [pc]	age [Myr]	N_{stars}	turn-off stars
M67	900	4000	≈ 1000	F5
Hyades	46	625	≈ 200	A7
Pleiades	135	125	≈ 1000	B6
Orion (NGC 1976)	410	< 0.5	≈ 2500	O6

A few comments on the open clusters shown in Slide 2-15 (see also Table 2.2):

- M67 is one of the oldest open clusters known. It is the nearest of the old open clusters and therefore well studied. The main sequence turn-off is around spectral type F. Because of its age it contains more than 100 white dwarf stars.
- The Hyades is the nearest open cluster. The bright red star, α Tau, is a foreground object and does not belong to the cluster. The Hyades cluster shows a strong mass segregation. The central 2 pc of the cluster contains only systems with masses $> 1 M_{\odot}$ or white dwarfs. The cluster contains about 20 A, 60 F, 50 G, 50 K dwarfs, and about 10 white dwarfs but only about 15 M stars. It seems that lower mass stars have been lost.
- The Pleiades is the nearest cluster which is dominated by blue stars. It is a rich cluster with more than 1000 members. Because it is so close and young the full main sequence from B-stars down to brown dwarfs could be mapped.
- The Orion-(Trapezium) cluster, or NGC 1976, is part of the nearest high mass star forming region including the famous Trapezium stars and the Orion nebula. The brightest star, θ^1 Ori C is an O6 V star, which is responsible for the ionization of the Orion Nebula. The stars are younger than < 1 Myr and many stars are still forming or they are in their pre-main sequence phase. For such young clusters one cannot indicate a well defined age, because the duration of the star-formation process is of the same order as the cluster age. The presence of thick interstellar clouds make the derivation of the cluster parameters quite difficult because many stars are because of the dust not visible in the V-band. In any case the stellar density of the Orion-Nebula cluster is with ≈ 10000 stars/pc³ very high.

Clusters ages. Slide 2-16 shows schematically the distribution of star from different open clusters in the HR-diagram and measurements from GAIA for stars from 14 open clusters color coded with their age t_{cl} . Clearly visible is the difference in the main-sequence turn-off which is a good indicator of stellar age. The HR-diagram of young clusters has a main-sequence extending to O or early B stars, some A-F supergiants in the Hertzsprung gap (the low star density region in the HR-diagram between main sequence and red giant branch), and a concentration of M supergiants. Intermediate age clusters show still some late B or A stars on the main sequence and then a well developed red giant branch separated by a smaller Hertzsprung gap. Older galactic clusters (≈ 1 Gyr) show a main-sequence turn-off in the F-star region and a continuous sub-giant branch (without Hertzsprung gap) extending into a lower brightness red giant branch.

There exist almost no galactic open clusters with ages larger than 1 Gyr (thus, M67 is an exception). If the cumulative age distribution of galactic clusters is plotted then the 50 % mark is around 300 Myr (see Fig. 2.3). The large number of galactic cluster and their age distribution indicates the following *evolutionary scenario for galactic clusters*:

- new clusters are continuously formed in the galactic disk,
- after formation they loose stars and dissolve with time mainly due to dynamical interaction with interstellar clouds (see next Chapter),
- older clusters ($\tau > 1$ Gyr) are very rare because they were all disrupted,
- it is estimated that a significant fraction (≈ 10 %) of the stars in the Milky Way disk were initially formed in clusters.

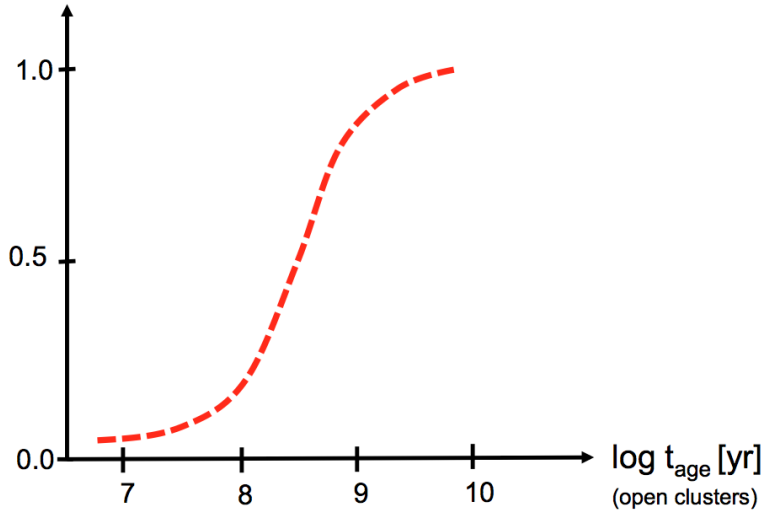


Figure 2.3: Cumulative distribution of cluster ages (according to Binney and Tremaine based on data from Piskunov et al. (2006)).

Stellar associations and groups. A stellar association or group is a very loose assembly of about 100 or less stars which are not dynamically bound. The space density is lower than the typical density in the galactic disk, with perhaps 100 stars within a volume of 10^6 pc^3 . Associations and groups can often be identified because of a small concentration of young, rare stars. Two types of associations are well known:

- O- or OB-associations with an enhanced density of massive main sequence stars,
- T-associations, which contain an over-density of variable T Tauri type pre-main sequence stars.

The nearest examples are the Sco-Cen OB association and the Taurus-Auriga T association.

Associations are just transients groups of newly formed stars in the galactic disk (spiral arm) population. They are in the process of dispersing from a star forming region into the “galactic field”. Nearby associations may cover a very large sky region and therefore it is difficult to identify associations and quantify their frequency and lifetime in the galactic disk. However, it is estimated, that the majority ($\gtrsim 50\%$) of stars in the Galaxy were formed in loose associations.

2.2.4 Globular clusters.

Globular clusters are spherical systems which contain typically 10^5 to 10^6 stars and a mass of $10^5 - 10^6 M_\odot$ in a volume with a radius of $r \approx 20 - 50$ pc. They have a high central star density of 100 to $> 10^4$ stars/pc³ and are dynamically very stable and long lived. The absolute brightness of globular clusters is on average $M_V \approx -8.5^m$. There are about 150 globular clusters known in our galaxy, and they are distributed in the galactic halo. Two examples for the globular clusters are shown in Slide 2-17. ω Cen is one of the brightest and best studied globular clusters. NGC 6522 is an example of an object very close to the galactic center, located in the low extinction region called “Baade’s window”, where the contamination by foreground and background stars is a severe complication for the investigation of this globular cluster.

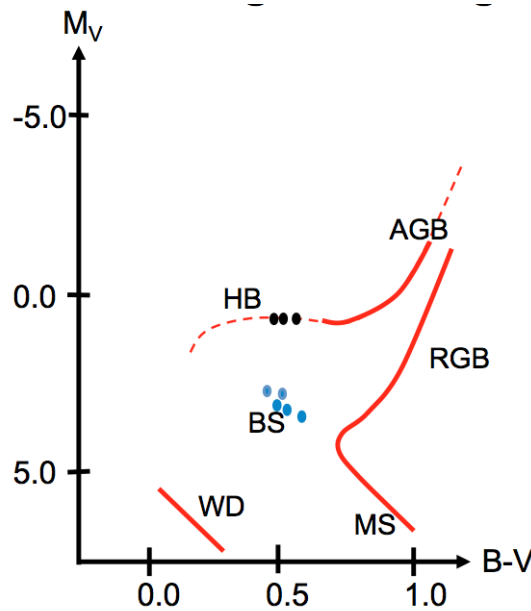


Figure 2.4: Schematic HR-Diagram for globular clusters.

The Hertzsprung-Russell diagrams of globular clusters are special because they contain

only old low mass stars. Figure 2.4 shows a schematic HR-diagram for globular clusters which has the following characteristics:

- the main sequence (MS) turn-off point is in the region of F and G stars, or at stellar masses $0.9 - 1.3 M_{\odot}$ indicating an age of the order 10 Gyr,
- there is a subgiant branch which joins the main sequence with the giant branch (RGB),
- near $M_V \approx +0.5$ there is a horizontal branch (HB), which contains pulsating RR Lyr stars and some blue hot stars. The HB extends toward the red until it rises in the so-called asymptotic giant branch (AGB) lying just above the giant branch.
- above the main-sequence turn-off point there are a few so-called blue stragglers (BS) stars, which are too bright for normal main-sequence stars with the age of the cluster,
- the horizontal branch may extend into the white dwarf cooling track for clusters which were observed with very high sensitivity.

Stellar evolution of globular cluster stars. According to the stellar evolution theory the stars with an initial mass just above the main-sequence turnoff stars have evolved to the red giant branch. Stars with even higher initial mass are now in the core helium burning phase on the horizontal branch. Even higher initial mass stars are either evolving up along the asymptotic red giant branch or they have already lost their envelope due to stellar winds so that their hot core becomes visible. They then evolve to the blue part of the horizontal branch where they stop nuclear burning and enter the white dwarf cooling track.

The “blue stragglers” stars are special cases. They were probably low mass binaries which merged after some time ($\approx$ Gyr) to a higher mass, rapidly rotating star. These stars are therefore still on the main sequence because of the late merging event. All single stars with the same mass as blue stragglers have already evolved to an advanced evolutionary stage.

Metallicity of globular clusters. Globular clusters are very old, have a special space distribution and often a very low metallicity. A low metallicity means that the abundance of heavy elements is 10 to 100 times lower than in the sun. This indicates that all the star in a globular cluster were born in a well mixed gas clouds and that no additional stars were formed in a later generation from gas with enhanced metallicity.

An indicator for low metallicity is the color of the main sequence. High metallicity stars have atmospheres with more heavy elements (e.g. Fe) producing many absorption lines in the UV and blue spectral region (see Slide 2–18). For this reason they emit for a given luminosity less blue light because the UV and blue radiation cannot escape from the deep, hot layers of the stellar atmosphere. The radiation escapes only from higher cooler layers and the resulting spectral energy distribution is redder than for low metallicity stars.

For this reason, the main sequence of globular clusters is shifted in the HR-diagram towards the blue. The stars appear for a given color less luminous (in fact they are for a given luminosity just more blue) and are therefore called subdwarfs (main-sequence stars are dwarfs). A subdwarf branch indicates therefore a low metallicity.

A similar line opacity effect occurs for the red giant branch. For metal poor clusters the red giant branch is shifted significantly to the blue. Slides 2–13 and 2–20 illustrate

the HR-location of the main-sequence and the giant branch for clusters with different metallicities.

With modern large telescopes like the VLT it is possible to take accurate spectroscopic measurements of individual stars in globular clusters so that the elemental abundances can be derived from a detailed spectral analysis (see Slide 2–21).

Origin of globular clusters. The metal-poor globular clusters are probably relics of the Milky Way formation process, because they are old and have preserved the gas abundance pattern which dominated in the early Universe.

The globular clusters with higher elemental abundances ($\approx$ solar) may have formed during phases of extreme star formation, e.g. induced by a galaxy merging event. The bright globular cluster ω Cen may be the dense center of a tidally disrupted galaxy.

Similar evolutionary histories are put forward for globular clusters seen in other galaxies. It should be noted that these are only tentative evolutionary scenarios because our understanding of globular clusters is still incomplete.

2.2.5 Age and metallicity of stars

Stars serve as test bodies for deriving the galactic dynamics and the galactic gravitational potential. In addition we can also derive or at least constrain the age and metallicity of the stars. This provides information about the evolution of the distribution and dynamics of stars from their formation in an interstellar cloud to the present day. Similarly we can use the metallicity of stars as a second parameter for constraining the time and region where they were formed. Thus, selecting stars with a certain age or metallicity can provide important information about earlier epochs and long term evolutionary processes of our Galaxy.

Stellar ages. The age of a star or a stellar group can be estimated from the following age indicators:

- the determination of the main-sequence turn-off age for stellar clusters or groups is a very reliable age indicator for ages from 10 Myr to 10 Gyr,
- high mass stars, such as O stars and early B stars, as well as classical Cepheids, bright giants, or Wolf-Rayet stars are always young $\tau < 100$ Myr objects,
- the stellar rotation speed and coronal activity are useful age indicator for low mass stars of spectral type G, K, and M; fast rotating, active stars are relatively young $\tau \lesssim 1$ Gyr, while slowly rotating, quiet stars are old $\tau \gtrsim 1$ Gyr,
- low luminosity red giants, planetary nebulae, and white dwarfs are typically evolved intermediate or low mass stars which are older than $\tau > 500$ Myr,
- RR Lyr variables are very old $\tau \approx 10$ Gyr objects and they are very reliable indicators for an old population.

Stellar metallicity. The metallicity of a star is often indicated with one of the following two parameters:

- Z is the mass fraction of all elements heavier than H ($= X$) and He ($= Y$). The sun has $X = 0.70$, $Y = 0.28$ and $Z = 0.02$, a metal rich galactic disk star has $Z = 0.05$, while stars in metal-poor globular cluster have $Z \approx 0.002 - 0.0002$.

- $[\text{Fe}/\text{H}]$ is the logarithmic iron abundance relative to hydrogen and in relation to the solar value

$$[\text{Fe}/\text{H}] = (\log \text{Fe}/\text{H})_{\text{star}} - (\log \text{Fe}/\text{H})_{\odot}.$$

A globular cluster star (as example) with an iron underabundance of 100 with respect to the sun has the value $[\text{Fe}/\text{H}] = -2.0$. Often the $[\text{Fe}/\text{H}]$ value is a good indicator of the overall metallicity of a star. This definition can also be used to quantify specific elemental abundance ratios for stars such as e.g. $[\text{Ca}/\text{Fe}]$ or $[\text{O}/\text{Si}]$ and others.

The best method for a metallicity determination are spectroscopic abundance determinations from high resolution spectra of well understood stars. These are stars where the elemental abundances of the surface layers are representative for the initial composition of the star. Many main-sequence stars, but also certain giant stars fulfill this criterion. Slide 2–21 illustrates the dependence of the line strengths with metallicity or the presence of specific abundance patterns (e.g. for HE 0107-5240). High resolution spectroscopy requires time consuming observations and stars $m_v > 17^m$ might be too faint even for a large telescope.

For stars with well known distances, clusters, or for stellar groups the metallicity can also be derived from photometry as described in the subsection for globular clusters (or Slide 2-19). The method is based on the strength of line opacities in the blue-UV spectral region, which is high for high metallicity objects and weak for low metallicity objects. The corresponding color effect provides then a measure for the metallicity. This technique is very powerful if a cluster is investigated for the presence of two populations of stars with different metallicities. With accurate GAIA distances the metal poor stars in the field (not part of a cluster) can be directly identified from their colors. This is very useful for the spatial distribution of low metallicity stars in the Milky Way.

Metallicity gradients in the Milky Way. Metallicity determinations from cluster photometry and spectroscopic studies provide a quite detailed picture of the different metallicity gradients in the Galaxy:

- for young disk stars in the Milky Way disk there is a radial metallicity gradient from higher $[\text{Fe}/\text{H}] \gtrsim 0.0$ for regions closer to the galactic center than the sun and lower $[\text{Fe}/\text{H}] \lesssim 0.0$ further out; the metallicity gradient is of the order

$$\frac{\Delta[\text{Fe}/\text{H}]}{\Delta R} \approx \frac{-0.05}{\text{kpc}}.$$

- old galactic open clusters have a lower metallicity than young clusters and the temporal gradient is of the order

$$\frac{\Delta[\text{Fe}/\text{H}]}{\Delta \tau_{\text{cl}}} \approx \frac{-0.05}{\text{Gyr}}.$$

- globular cluster have typically a much lower metallicity, if they are located at large galacto-centric distances; a rough statement for the metallicity is:

$$[\text{Fe}/\text{H}] > -1.0 \text{ for clusters at } d \lesssim 3 \text{ kpc},$$

$$[\text{Fe}/\text{H}] < -1.0 \text{ for clusters at } d \gtrsim 3 \text{ kpc}.$$

- the metallicity of the galactic bulge is not well known, but it is approximately solar ($[\text{Fe}/\text{H}] \approx 0.0$).

2.2.6 Cepheids and RR Lyr variables as distance indicators

Distance determinations are required for the 3-dimensional mapping of the distribution of objects. The GAIA mission extended accurate trigonometric distance determinations from about 150 pc to roughly 5 kpc. For more distant objects other methods have to be used.

A very basic method for the determination of the distance modulus $m - M$ is the main-sequence fitting for stellar clusters. This method works well for good observations of clusters, where the main sequence stars can be observed over a significant color range. This requires photometry of F-G stars in open clusters because all O, B, and A stars have similar colors. For globular cluster one needs to reach even K dwarfs for the main sequence fitting.

Pulsating Cepheid variables provide a very powerful alternative for the distance determination because their pulsation period is an indicator of the stellar type and its absolute luminosity. The calibration of the period-luminosity or P-L relation has a very interesting history since the first detection of such a relation for Cepheids in the Small Magellanic Cloud by Henriette Leavitt in 1912. Initially the size of our Galaxy, or the distance to M31 based on Cepheid distances were estimated wrongly by about a factor of two by Shapley, Hubble and others until Baade recognized in 1952 that there are two different types of Cepheids:

- the population II metal poor, old, low mass RR Lyr variables with periods $P \lesssim 1^d$ and a pulsation brightness amplitude of $\Delta m \approx 1^m$. They are low mass $\approx 0.7 M_\odot$ horizontal branch stars which are in their helium burning phase. RR Lyr variables are further divided into subgroups which are defined according to subtle differences in evolutionary phase and metallicity.
- the population I, metal rich, young, high mass classical Cepheids have periods in the range $3^d \leq P \leq 40^d$. They are evolved high mass stars crossing the HR-diagram.
- in addition there are several other groups of Cepheid-type pulsating variables like W Vir stars, δ Scuti stars (main sequence pulsators), or RV Tau variables, which are not discussed here.

Cepheids are A to K giants or supergiants located in the (vertical) pulsation instability strip in the middle of the Hertzsprung-Russell diagram. These stars pulsate because of an opacity effect or *κ -mechanism* due to He-ionization. The process works as follows:

- the slightly enhanced temperature in the stellar envelope leads to the additional ionization of He^+ to He^{+2} ,
- the ionization to He^{+2} enhances the electron density and therefore the e^- -scattering opacity and the outward radiation transport (= energy transport) is reduced, the star heats further up and starts to expand,
- with the expansion the gas density, the opacity per volume element, and the temperature drops, He^{+2} recombines to He^+ , the opacity drops further and the radiation is transported more efficiently,
- the stellar envelope cools rapidly, contracts, heats up and the He^+ -ionization increases again,
- the opacity and temperature rises again, and a new cycle begins.

For pulsating variables there exists a simple relationship between the mean density of a star and the pulsation period:

$$P \left(\frac{\bar{\rho}}{\bar{\rho}_{\odot}} \right)^{1/2} \approx Q, \quad (2.17)$$

where P is the pulsation period, $\bar{\rho} \propto M/R^3$ the mean density, and Q the pulsation constant. Q is about 0.03 – 0.05 days or about 1 hour, which corresponds roughly to the pulsation period of a sun like star.

The relation given above indicates basically that the pulsation period P is roughly the time required for a sound or pressure wave to move with (isothermal) sound speed c_s through the star,

$$P \approx \frac{2R}{c_s} \approx \frac{2R}{\sqrt{\frac{p}{\rho}}} \overset{\bar{p} \approx \bar{\rho} \frac{GM}{R}}{\approx} \frac{2R}{\sqrt{\frac{GM}{R}}} \overset{M = \bar{\rho} \frac{4\pi R^3}{3}}{\approx} \frac{2R}{\sqrt{G\bar{\rho} \frac{4\pi R^2}{3}}} \approx \frac{1}{\sqrt{G\bar{\rho}}}$$

using approximations for the mean pressure and density of stars. This pulsation time scale is comparable to the dynamical time scale (free fall time of particles) for a star. The pulsations because of the He^+ ionization instability occurs for stars in the middle of the HR-diagram (the so-called instability strip), because they have similar surface temperatures and therefore also similar internal temperatures where these pulsations can occur.

Cepheid variables are ideal objects for distance determinations. They are bright objects and it is easy to identify pulsating variables in a crowded field of stars with repeated observations. The properties of Cepheids variables have been studied in much detail. For this lecture we consider only a rough relation for their absolute magnitude:

- The classical Cepheids (pop. I) are F to K supergiants and the most luminous Cepheids have an absolute magnitude $M_V \approx -6^m$. They belong to the brightest stars (in m_V) in a galaxy. A simple empirical period-luminosity relation valid for $3^d \leq P \leq 40^d$ is

$$M_V(\text{max}) = -2.0^m - 2.8 \log P [\text{d}].$$

- The RR Lyr variables are old (pop. II) A to G horizontal branch stars which are in the He-burning phase. Their absolute magnitude is

$$M_V \approx +0.5^m.$$

In globular cluster or in a similarly old stellar population they are about 4 mag brighter than the main-sequence turn-off (Slide 2-20).

Classical Cepheids are and will remain in the near future important distance indicator for young clusters in the Milky Way and the distances to other galaxies. They are an important part for the distance ladder in extra-galactic astronomy and cosmology.

The RR Lyr variables are important tracers of the old galactic population, and therefore ideal for globular cluster studies and for determinations of distances to objects in the galactic halo.

2.2.7 Star count statistics

Star counts provide information about the distribution and frequency of stars in our galaxy. This is a very basic technique in Astronomy, which was introduced initially for studies of the Milky Way and is now also applied to other objects like galaxies in cosmology and extra-galactic astronomy, or asteroids for solar system studies.

Different types of star counts are used for studies in Galactic astronomy.

- Determination of the number of all stars brighter than a given limit in a brightness limited sample; the counts for different sky regions provide the overall geometric distribution of objects.
- Determination of the number of stars in a volume limited sample (e.g. out to a distance d_{lim} , or a particular cluster) for determining the volume density of stars which can then be compared with the volume density of other regions in the Milky Way. This requires knowledge of the distances to the objects.

These types of studies can be refined by the determination of the space distributions for different stellar types. The distribution of stars can be described by:

- $A(m, S)$: the differential star counts, which is the number of stars of type S , at apparent magnitude m , per unit magnitude interval (e.g. from $[m - 0.5, m + 0.5]$ and per solid angle $d\omega$, e.g. square degree.
- $N(m_{\text{lim}}, S)$: the integrated star counts for stars of type S down to the magnitude limit m_{lim} , e.g. $m_{\text{lim}} + 0.5$, and $d\omega$:

$$N(m_{\text{lim}}, S) = \int_{-\infty}^{m_{\text{lim}}} A(m, S) dm. \quad (2.18)$$

Homogeneous distribution. We calculate first the volume limited number of stars for a homogeneous distribution for a given star density D [stars/pc³] as function of distance r :

$$N(r_{\text{lim}}) = \omega D \int_0^{r_{\text{lim}}} r^2 dr = \frac{\omega D}{3} r_{\text{lim}}^3$$

The corresponding magnitude limited number follows then from the relation between radius limit in [pc] and magnitude limit:

$$m_{\text{lim}} = M + 5 \log r_{\text{lim}} - 5 \quad \text{or} \quad r_{\text{lim}} = 10^{0.2(m_{\text{lim}} - M + 5)}$$

Combining these two equation yields $N(m_{\text{lim}}) = 10^{0.6 m_{\text{lim}} + C}$ or

$$\log N(m_{\text{lim}}) = 0.6 m_{\text{lim}} + C, \quad (2.19)$$

where C is a constant that depends on D , ω , and M .

This equation states that:

- a homogeneous distribution of stars produces a line in a m_{lim} vs. $\log N$ star count diagram,
- for a homogeneous distribution the number of stars increase by $\Delta \log N = 0.6$ or a factor 4.0 if the brightness limit m_{lim} is one magnitude deeper.

Realistic distributions. In reality, a detailed count statistics needs to consider that the stellar luminosity function and the star density is a function of distance and that there is interstellar absorption.

- $\Phi(M, r, A_\lambda(r), S)$: the luminosity function Φ of a selected stellar type S is in general also a function of the distance and should consider the interstellar absorption A_λ along the line of sight,
- $D(r, S)$: a star density which depends on the distance within the selected field.

The general formula for the differential star density is:

$$A(m, S) = \omega \int_0^\infty D(r, S) \Phi(M, r, A_\lambda(r), S) r^2 dr. \quad (2.20)$$

This is essentially a convolution integral of the density function and the luminosity function. The effective width of the luminosity functions determines the range of distances that can contribute to the observed number of stars with magnitude m . Obviously, it is not easy to “deconvolve” the problem and derive the line of sight distribution D from the absorption affected luminosity function Φ .

Selecting carefully the stellar type and understanding the selection bias is the real challenge for the interpretation of star count data. The analysis can be strongly simplified if the selected star type S fulfills certain conditions:

- $D(r, S)$ can be well determined, if the luminosity function does not depend on the distance and if also the interstellar extinction can be neglected $\Phi(M, r, A_\lambda(r), S) = \Phi(M, 0, 0, S)$. This is essentially the case for:
 - stellar types with narrow luminosity functions like e.g. F, G and K-type main sequence stars or RR Lyr variables, and
 - sight-lines perpendicular to the galactic plane which are barely affected by interstellar extinction.
- The properties of the luminosity function $\Phi(M, r, A_\lambda(r), S)$ can be quite well determined if the stellar density does not depend on distance $D(r, S) = D(0, S)$. This is essentially the case for
 - the determination of $\Phi(M, 0, 0, S)$ in the solar neighborhood where changes in the luminosity function and effects due to interstellar extinction can be neglected,
 - the determination of the luminosity function $\Phi(M, r_c, A_{\lambda,c}, S)$ of a cluster where all stars have essentially the same distance and extinction.

The following paragraphs summarize a few basic results of stellar count statistics.

Integrated star counts. The star counts show that the numbers of stars are higher in the galactic plane when compared to the galactic poles (see Table 2.3 and Fig. 2.5). The difference is about a factor of 5 for stars brighter than $\lesssim 10^m$ in agreement with the historical results from Herschel and Kapteyn. For fainter magnitudes the stellar density is much higher in the galactic plane when compared to the poles.

Table 2.3: Integrated star counts in the solar neighborhood per deg^2 and mag in the Galactic plane $N(m, 0^\circ)$ and towards the North Galactic pole $N(m, 90^\circ)$, the ratio of these two values, and the total number of stars $N_{\text{tot}}(m)$ over the entire sky.

m_V	$\log N(m, 0^\circ)$	$\log N(m, 90^\circ)$	$N(m, 0^\circ)/N(m, 90^\circ)$	$\log N_{\text{tot}}(m)$
5	-1.08	-1.69	4.1	3.20
10	1.25	0.55	5.6	5.52
15	3.42	2.27	14	7.56
20	5.0	3.4	40	9.0

Another important result is that the number counts increase less than expected for a homogeneous star distribution (factor 4 per mag or $4^5 = 1024$ per 5 mag). In the case of the polar direction this is due to strong decrease in stellar density with distance. In the galactic plane it is mainly due to the interstellar absorption.

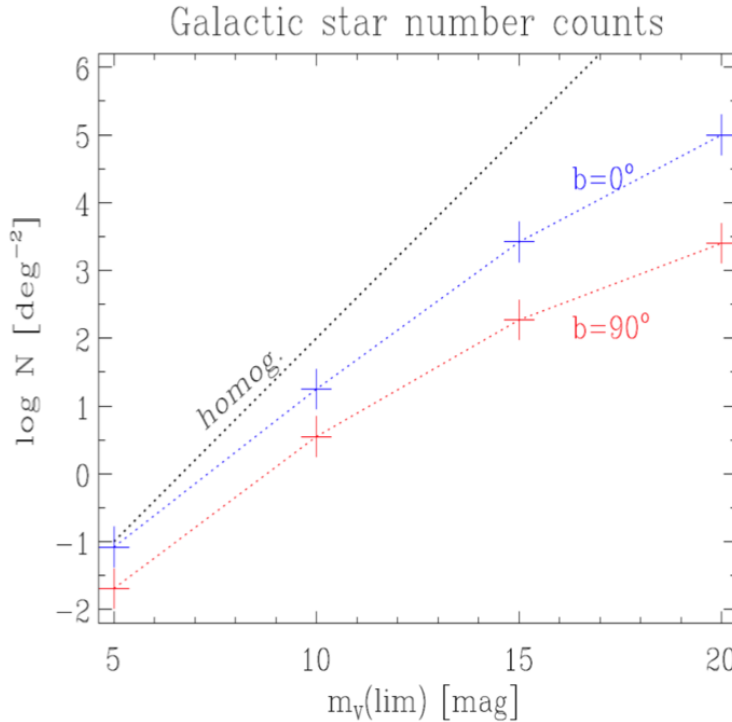


Figure 2.5: Total star number counts for stars in the galactic plane and towards the galactic poles and comparison with the slope of a homogeneous star distribution.

Table 2.3 list also the total number of stars over the entire sky. The celestial sphere has $41\,253 \text{ deg}^2$ or $\log 41253 = 4.6$, and therefore the total star counts lie in the range

$$\log N(m, 90^\circ) + 4.6 < \log N_{\text{tot}}(m) < \log N(m, 0^\circ) + 4.6.$$

The luminosity function and the integrated luminosity and mass for the stars in the solar neighborhood are presented in Fig. 2.6 or Table 2.4. This statistics show that the most frequent stars are (low mass) M-type main sequence stars and white dwarfs with an absolute magnitude of about $M_V \approx 15$ which is about 10 mag fainter than the sun (Slide 2-13).

The integrated luminosity or the integrated mass of the stars with the same absolute magnitude M_V are not well represented by the luminosity function. The luminosity of the stars in the solar neighborhood is mainly produced by stars with $M_V \approx 0^m$ which are A and F main sequence stars and G and K giants. Contrary to this, the mass is in the stars with $M_V \approx +5$ to $+15$ which are the low mass main sequence stars (K and early M) and the white dwarfs.

Table 2.4: General luminosity function $\Phi(M_V)$, integrated luminosity $L/L_\odot(M_V)$, and integrated mass $M/M_\odot(M_V)$ per 10^3 pc^3 and magnitude for the stars in the galactic disk near the sun.

M_V	$\Phi(M_V)$	$L/L_\odot(M_V)$	$M/M_\odot(M_V)$
-5	$6 \cdot 10^{-5}$	0.6	0.002
0	0.1	11.2	0.4
+5	3.4	3.4	3.7
+10	7.8	0.1	3.4
+15	12.5		2.0
+20	3.0		0.2
total	131	54	44

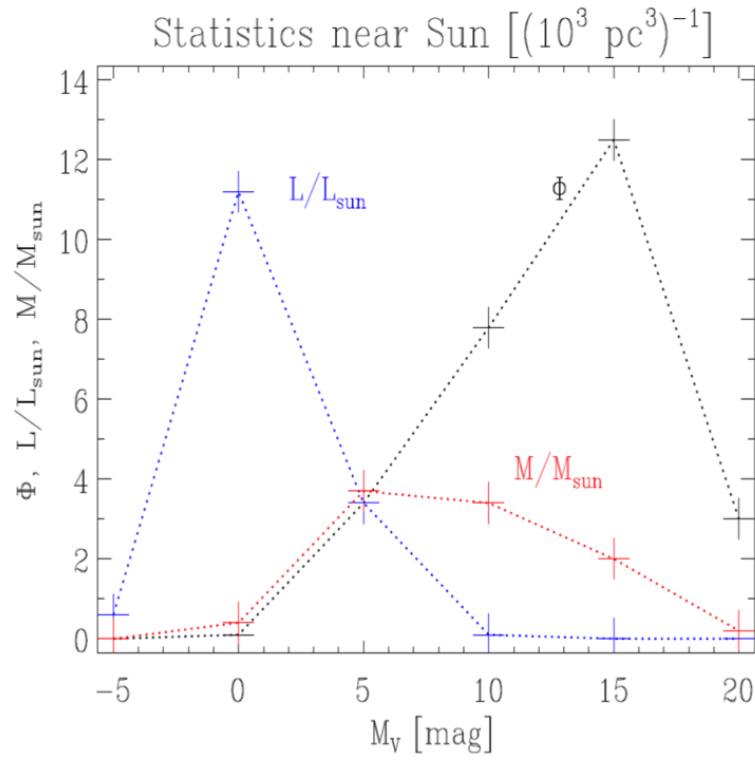


Figure 2.6: Star counts luminosity function, integrated luminosity and mass for the stars in the solar neighborhood.

Mass to light ratio. The total values for the luminosity and mass of stars can be used to determine a mass to light ratio for the stellar population in the solar neighborhood:

$$R_{M/L} = \frac{(M/M_{\odot})}{(L/L_{\odot})} = 0.8.$$

For a young cluster this value is much smaller and for a globular cluster much larger.

The volume density of different stellar types in the solar neighborhood are listed in Table 2.5. The table gives number counts for main sequence stars, red giants and white dwarfs. The used volume of 10^6 pc^3 corresponds to a sphere with a radius of 62 pc and it contains more than 10^5 stars. However, essentially all M type main sequence stars and white dwarfs are faint stars $M_V > 10^m$. Observations which pick only objects with $m_V < 10^m$, e.g. the HD-star catalog or the Hipparcos catalog miss all these faint stars, or more than 80 % of all stars. Therefore one needs to go very deep to produce a complete star catalog.

Table 2.5: Mean number densities $N(S)$ in stars/ 10^6 pc^3 for different spectral types.

Spec.Type	main seq.	giants	white dwarfs
O stars	0.02		
B stars	100		6300
A stars	500		10000
F stars	2500	50	5000
G stars	6300	160	5000
K stars	10000	400	2500
M stars	63000	30	
total	82500	640	28800

The star numbers in Table 2.5 for the solar neighborhood indicate:

- the distribution of main-sequence stars has a very larger fraction of low mass stars which can be expected from the IMF and main-sequence lifetime,
- evolved giants are of the order 10 times less frequent than main-sequence stars of spectral type B, A, F and early G, and this represents roughly the 10 times shorter red giant phase (He-burning stars) when compared to the main-sequence life-time (H-burning stars).
- there is a large number of white dwarfs, the remnants of previous B to early G main-sequence stars. The high number of white dwarfs proves that there were already several previous generations of stars in the galactic disk.

These points illustrate that the number counts in the solar neighborhood are very important for quantifying the density of the faint low mass stars and white dwarfs in the Milky Way disk.

The star distributions vertical to the disk shows a very strong dependence on stellar type. This property is not surprising because the cold gas, the star forming regions, and the young stars are strongly concentrated towards the disk mid-plane, much more than the overall star distribution. Therefore, older stars must have a wider distribution than the young stars.

With number counts for different stellar types perpendicular to the disk one can derive in detail their vertical or “z”-distribution. The distribution can be approximated with an exponential law

$$D(z, S) = D(0, S) e^{-z/\beta}, \quad (2.21)$$

where β is the disk scale height. Table 2.6 gives disk scale heights β_S and disk surface densities Σ_S for various stellar types.

Table 2.6: Vertical scale heights β_S perpendicular to the Galactic disk for various stellar types and other tracers. For frequent stars also the disk surface density Σ_S is given

stellar type	β_S [pc]	Σ_S [stars/pc ²]	stellar type	β_S [pc]
O main seq.	50	$1.5 \cdot 10^{-6}$	clas. Cepheids	50
B main seq.	60	$6 \cdot 10^{-3}$	open cluster	80
A main seq.	120	$6 \cdot 10^{-2}$	interstellar gas	120
F main seq.	190	0.6	planetary nebulae	260
G main seq.	340	2.0	RR Lyr variables	2000
K main seq.	350	3.5	subdwarfs	2000
M main seq.	350	20	globular clusters	3000
G giants	400	0.06		
K giants	270	0.0012		
white dwarfs	500	12.5		

Table 2.6 shows for normal stars a clear correlation between average age and disc scale height. This indicates that older objects have a larger vertical dispersion. Exceptions are the RR Lyr variables, the subdwarfs, and the globular clusters which belong to the halo, and they have therefore a much larger disk scale height.

Another interesting fact is that the surface density of white dwarfs is more than 50 % of the M dwarfs. The average white dwarf has a mass of $\approx 0.5 M_\odot$, while the mean M-dwarf mass is more like $\approx 0.3 M_\odot$ and therefore both groups of stars contribute a similar amount to the stellar mass of the galactic disk. Roughly the mass share of the stars in the Galactic disk is:

- ≈ 30 % M dwarfs,
- ≈ 30 % white dwarfs,
- ≈ 30 % G, and K main sequence stars.

2.3 Stellar Dynamics

All stars in the Milky Way move in the galactic potential. Most disk stars move with the same circular direction in more or less co-planar orbits around the galactic center. Halo stars and stars in the galactic bulge have orbits with a wide distribution of orbital plane orientations and eccentricities. The orbits of stars can also be stochastically deflected by small scale mass concentrations due to stellar clusters or massive interstellar clouds, or by the dynamical interactions between individual stars.

2.3.1 Velocity parameters relative to the sun

The space motion v_s of a star relative to the sun consists of the radial velocity component v_r and the tangential or transverse velocity component v_t according to

$$v_s = \sqrt{v_r^2 + v_t^2}. \quad (2.22)$$

Typical relative space velocities for stars are:

- about 220 km/s for the orbital motion around the galactic center,
- about 5 – 50 km/s for the velocity dispersion of corotating stars in the disk,
- about 0.2 – 20 km/s for the velocity dispersion in groups and clusters,
- up to 500 km/s for stars on counter-rotating orbits,
- up to and beyond 1000 km/s for stars in close orbits around the super-massive black hole in the galactic center.

Thus one needs to reach a measuring precision of about ± 1 km/s for the investigation of the velocity dispersion in the galactic disk and in clusters. A lower precision (± 10 km/s) is sufficient for the investigations of the galactic rotation. The observations provide the radial velocity and the angular proper motion:

The radial velocity v_r is measured in km/s via the Doppler shift of spectral lines of the stars.

$$\frac{v_r}{c} = \frac{\Delta\lambda}{\lambda_0} = \frac{\lambda - \lambda_0}{\lambda_0}.$$

The measured values must be corrected by up to ± 30 km/s for the Earth motion around the sun in order to get the stellar motion relative to the sun. A positive v_r means that the object is moving away from the sun (it's spectrum is red-shifted).

The radial velocity can be measured for all stars which are bright enough for a spectrometric measurements and which have well defined spectral features. In particular, it is possible to measure very accurate radial velocities for very distant stars (e.g. at 50 kpc in the Magellanic Clouds) if they are bright and have suitable spectra.

The angular proper motion is measured in units of arcsec/yr in the direction of the right ascension μ_α (E–W) and declination μ_δ (N–S). Positive values are given for objects which move towards E and N, respectively.

At least two measurement taken at epochs separated by a few year, better many years, are required to determine the proper motion. In addition one needs also to correct

for the yearly parallax π because of the Earth's motion around the sun (see Slide 2-22). This parallax correction depends on the distance of the star. Therefore a proper motion measurement should include a parallax measurement π (trigonometric distance measurement) or at least a distance estimate of the object.

The angular proper motion is large and easy to measure for nearby stars. In fact a large proper motion is often used as criterion for the search or selection of nearby stars. The 5 stars with the largest proper motion, Barnard's star, Kapteyn's star, Groombridge 1830, Lacaille 9352, and Gliese 1, have a proper motion of about 5 – 10 arcsec/yr and are all nearby ($d < 10$ pc) stars.

Transverse velocity. From the measured angular motion $\mu = (\mu_\alpha^2 + \mu_\delta^2)^{1/2}$ and parallax π the tangential velocity can be derived

$$v_t = 4.74 \text{ km/s} \frac{\mu [\text{arcsec/yr}]}{\pi [\text{arcsec}]} \quad (2.23)$$

The determination of v_t is much less accurate for distant stars. For a given transverse velocity, say 10 km/s, the measurable angular proper motion μ and the parallax π decrease both proportional with the distance and therefore the measuring uncertainty rises rapidly.

2.3.2 Solar motion relative to the local standard of rest

The sun has like all other stars peculiar motion components. For this reason the solar motion is not an ideal reference.

For studying the dynamics of the galactic disk it is therefore useful to define a reference system which is more convenient. For this it is assumed that the galactic disk geometry and motion is rotationally symmetric. In this system one can define a motion vector composed of a radial component Π , an azimuthal component Θ ($= R \cdot \Omega$) and a vertical component Z (Fig. 2.7) which are defined by

$$\Pi = \frac{dR}{dt}, \quad \Theta = R \frac{d\theta}{dt}, \quad Z = \frac{dz}{dt}.$$

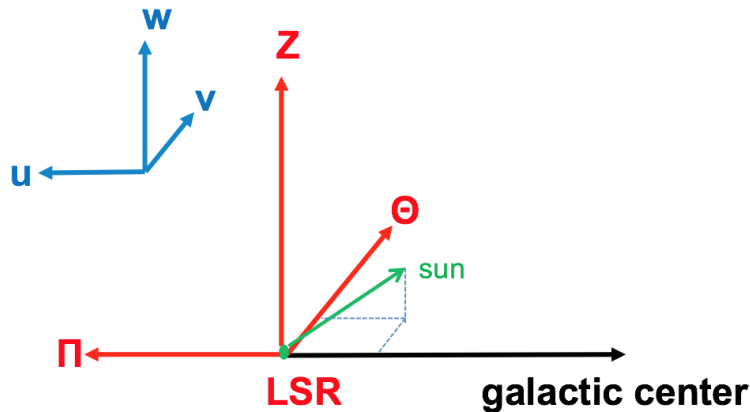


Figure 2.7: Illustration of the motion of the local standard of rest and the solar motion.

With this definition we can define for the solar neighborhood a velocity vector

$$(\Pi, \Theta, Z)_{\text{lsr}} = (0, \Theta(R_0), 0).$$

which represents the mean velocity of the stars near the sun on their circular orbit around the galactic center. This velocity vector is called local standard of rest (lsr). It represents a very useful reference for the investigation of the stellar dynamics in the galactic disk.

The vector $(u, v, w)_{\text{lsr}}$ is used for the peculiar velocity components of a star relative to the local standard of rest. For the peculiar velocity of the sun, one needs to measure the motion components (u, v, w) of a large group of stars relative to the sun for deriving the average velocity of the sun with respect to this sample:

$$\langle u_{\odot} \rangle = -\frac{1}{N} \sum_{i=1}^N u_i,$$

and similarly for $\langle v_{\odot} \rangle, \langle w_{\odot} \rangle$. However, the derived results vary by about ± 5 km/s depending on the selected reference star sample. The obtained values for the radial and vertical components $u_{\odot}$ and $w_{\odot}$ represent well the solar motion with respect to the local standard of rest. However, the azimuthal velocity Θ must be corrected for a bias effect because the orbits of the stars in the galactic disk have an eccentricity (see Fig. 2.8).

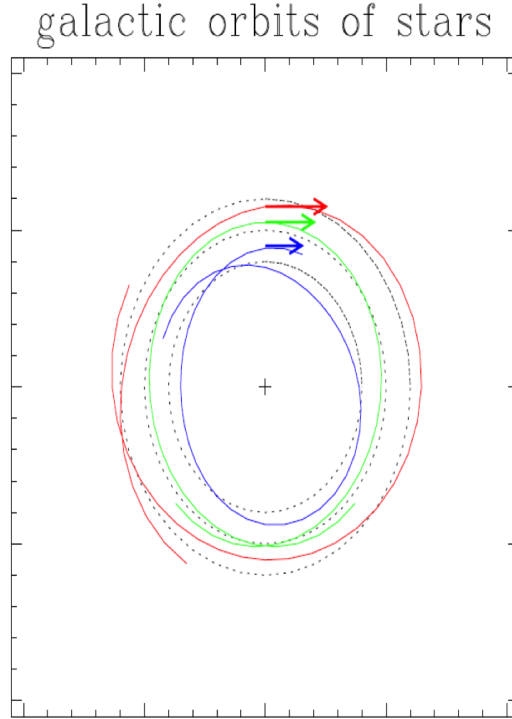


Figure 2.8: Orbits around the galactic center for stars near the sun with different eccentricities.

Stars with an orbits of type a in Fig. 2.8 have near the sun their maximum distance from the galactic center and will therefore have an azimuthal velocity component which is smaller than the local standard of rest $\Theta_a < \Theta_{\text{lsr}}$. Contrary to this the stars with orbits of

type c have their innermost point near the sun and will therefore move faster $\Theta_c > \Theta_{\text{lsr}}$. Stars with circular orbits (b) move with the same speed as the standard of rest $\Theta_b = \Theta_{\text{lsr}}$. Because there are more stars at small galacto-centric radius the average azimuthal speed of the stars will be slower than the local standard of rest $\langle \Theta \rangle < \Theta_{\text{lsr}}$. This bias effect, although difficult to quantify, needs to be considered for the definition of the Θ -component of the solar motion with respect to the local standard of rest.

The most recent result based on Gaia data, which is finally found, for the peculiar motion of the sun relative to the local standard of rest is:

$$(u_{\odot}, v_{\odot}, w_{\odot})_{\text{lsr}} = (-8.63 \pm 0.64, 4.76 \pm 0.49, 7.26 \pm 0.36) \quad [\text{km/s}].$$

This implies that the sun is moving with 16.5 km/s in the direction

$$\ell = -\arctan(v_{\odot}/u_{\odot}) = 29^{\circ} \quad \text{and} \quad b = \arcsin(w_{\odot}/(u_{\odot}^2 + v_{\odot}^2)^{1/2}) = 47^{\circ}$$

with respect to the local standard of rest. This is a slow motion toward smaller galactic radii and upwards towards the north galactic pole.

2.3.3 Velocity dispersion in the solar neighborhood

The velocity dispersion of galactic disk stars can be determined from the measured space motion and applying a correction for the solar motion with respect to the local standard of rest. If only stars in the solar neighborhood are considered then one can neglect the effects of the differential rotation in the Galaxy.

The data show that the peculiar velocities of the stars (with respect to lsr) show essentially a random or Gaussian distribution. It is useful to fit the measurements with a 3-dimensional ‘‘Gaussian ellipsoid’’ function

$$n(u, v, w) = \frac{\nu}{(8\pi^3 \langle u^2 \rangle \langle v^2 \rangle \langle w^2 \rangle)^{1/2}} \exp \left[- \left(\frac{u^2}{2\langle u^2 \rangle} + \frac{v^2}{2\langle v^2 \rangle} + \frac{w^2}{2\langle w^2 \rangle} \right) \right], \quad (2.24)$$

$n(u, v, w)$: is the number of stars per unit volume with velocities in an interval (du, dv, dw) ,

$\langle u^2 \rangle^{1/2}, \langle v^2 \rangle^{1/2}, \langle w^2 \rangle^{1/2}$: are the velocity dispersions along the three axes.

For one component of the velocity dispersion there is

$$\langle u^2 \rangle = \frac{1}{\nu} \int_{-\infty}^{+\infty} dw \int_{-\infty}^{+\infty} dv \int_{-\infty}^{+\infty} n(u, v, w) u^2 du,$$

and similarly for $\langle v^2 \rangle$ and $\langle w^2 \rangle$. This assumes that the principle axis of the velocity ellipsoid are along the coordinate axes. This is a simplification which is reasonable for basic results, but may be an over-simplification for more subtle studies.

Velocity dispersion for different stellar types. The components of the velocity dispersion have been determined for many different types of stars and a few results are given in Table 2.7. Interesting properties of the stellar velocity dispersion are:

- for all stellar types the three dispersion components behave like

$$\langle u^2 \rangle^{1/2} > \langle v^2 \rangle^{1/2} > \langle w^2 \rangle^{1/2},$$

with roughly $\langle w^2 \rangle^{1/2} \approx 0.5 \langle u^2 \rangle^{1/2}$. Thus the dispersion in the radial velocity component is twice as large as for the vertical velocity component.

Table 2.7: Velocity dispersion in km/s for different types of stars in the solar neighborhood.

stellar type	$\langle u^2 \rangle^{1/2}$	$\langle v^2 \rangle^{1/2}$	$\langle w^2 \rangle^{1/2}$
B0 main seq.	10	9	6
A0 main seq.	15	9	9
F0 main seq.	24	13	10
G0 main seq.	26	18	20
K0 main seq.	28	16	11
M0 main seq.	32	21	19
Class. Cepheids	13	9	5
G giants	26	18	15
M giants	31	23	16
planetary nebulae	45	35	20
white dwarfs	50	30	25
RR Lyr Var. (halo)	160	100	120

- B and A type main-sequence stars and classical Cepheids have the smallest velocity dispersion while evolved low mass stars, like M giants, planetary nebulae and white dwarfs show a much larger dispersion. Thus, there exists a clear correlation between the average age of stars and the velocity dispersion, mirroring the case of the disk scale height (Table 2.6).
- halo stars, in Table 2.7 represented by the RR Lyr stars, show a completely different velocity distribution than the disk stars.

2.3.4 Moving groups

Star clusters are gravitationally bound systems. For this reason all cluster members have essentially the same space velocity. The analysis of space velocities is therefore a very powerful tool to separate cluster members from non-cluster members.

The same method can also be applied to moving groups and associations of young stars. These systems are not gravitationally bound, but they were probably bound some time ago when they were formed in an interstellar cloud. The stars in a moving group are therefore moving still in the same space direction. Based on this property it is possible to identify young stars with the same age. Famous examples are the β Pic (age ≈ 20 Myr) and TW Hya (age ≈ 5 Myr) moving groups. Members of these two groups belong to the nearest young stars in the solar neighborhood.

Slide 2–16 shows the proper motion of the Hyades cluster. Because the stars have parallel space motions, they seem to converge in a common vertex point on the sky.

2.3.5 Dynamics in the galactic disk

The global dynamics of the stars near the sun is known since several decades but the Gaia satellite provides now much more accurate and complete measurements. Slide 2-24 gives a brief summary of results from Gaia which covers the dynamics of stars within

about 8 kpc around the sun. The predominant azimuthal velocity of the stars $\widetilde{V}_\phi$ is about 220 km/s. Above and below the midplane of the disk, especially at smaller galactic radii R , the median azimuthal velocity is significantly smaller. This points to a population of stars with a broader vertical distribution but less angular momentum. This distinguishes two types of disk stars, the thin disk stars on close to circular orbits and the thick disk stars with more elliptical orbits and larger orbital tilts with respect to the disk mid-plane.

Another way to show the stellar dynamics in the galactic disk provides the comparison of the radial and vertical velocity components (in Slide 2-24 called V_r and V_Z). Most stars move like the LSR or the sun. Stars with peculiar velocities have often a smaller azimuthal velocity component but a larger radial and/or vertical velocities in agreement with the thick disk stellar population.

2.3.6 High velocity stars

There are of the order $\lesssim 1\%$ of stars with a large peculiar velocity > 200 km/s. A small fraction of galactic stars have a very large space motion. They are obviously not moving with the general flow of stars around the galactic center. The orbits of these high velocity stars can be characterized from their motion in the solar neighborhood. According to their velocity vector (u, v, w) (radial, azimuthal, vertical velocities with respect to LSR) one can say qualitatively:

- a star with an azimuthal component $v < -250$ km/s is on a retrograde orbit,
- stars on a prograde orbit but with a large space velocity $v_s > 100$ km/s are on elliptical orbits with $\epsilon > 0.3$,
- the highest eccentricity have stars with $v \approx -250$ km/s because then they have only a very small azimuthal velocity component. They are on orbits which go close to the galactic center.

A classical analysis of high-velocity stars is shown in Slide 2–25 with a so-called Bottlinger diagram. The diagram distinguishes between old disk stars which are on prograde orbits with enhanced eccentricities $\epsilon \approx 0.4$, and stars which belong to the halo population which can have very high orbital eccentricities $\epsilon \approx 0.5 - 0.9$ or even retrograde orbits. Again, the space velocity is a good tool to distinguish and characterize different stellar populations.

Many high velocity stars are expected to originate from (dwarf) galaxies which were tidally disrupted by an interaction with the Milky Way while some halo stars may have been lost by the globular clusters.

2.3.7 Radial velocity dispersion in clusters

The measurable angular proper motion decreases rapidly with distance and therefore it becomes very difficult or impossible to determine the tangential velocity component v_t for distant stars and clusters. However, it is still possible to measure spectroscopically the radial velocity of stars and the velocity dispersion of a system. We consider this technique here for the measurement of the velocity dispersion in stellar clusters.

From the observational data we can distinguish three different cases:

- In nearby clusters it is possible to measure for many individual stars (essentially all bright objects) their radial velocity v_r and also their transverse velocity v_t from the proper motion. In this case one can carry out a detailed kinematic study of the

cluster and investigate radial dependencies and anisotropies in the velocity ellipsoid for the cluster stars. Nearby stellar clusters such as the Hyades or Pleiades can be studied in this way.

- In more distant clusters only radial velocities v_r of individual stars can be measured because the proper motion components are too small. For such cases one should also consider the distribution of the measured stars within the cluster. For example, in globular clusters, it is often difficult to get spectra of stars in the crowded central region and only the velocities of stars further out are measured. In this case one needs to consider the dependence of the stellar velocity with distance to the cluster center.
- For very distant clusters, e.g. in other galaxies, the individual stars cannot be resolved. In this case one can just measure the line width of the integrated cluster spectrum and determine a rough line broadening due to the velocity dispersion in the cluster. This analysis needs to consider which stars contribute most to the integrated spectrum, and preferentially they should have stable (non-pulsating atmospheres), and narrow spectral lines.

From the measured radial velocities one can derive the systemic radial velocity $\langle v_r \rangle$ and the velocity dispersion $\langle v_r^2 \rangle^{1/2}$ by fitting the data with a Gaussian distribution:

$$f(v) = \frac{1}{(2\pi\langle v_r^2 \rangle)^{1/2}} \exp\left[-\frac{(v - \langle v_r \rangle)^2}{2\langle v_r^2 \rangle}\right]. \quad (2.25)$$

Table 2.8 gives some values for the measured stellar velocity dispersion in clusters. It is visible that the dispersion is smaller for open clusters and larger for globular clusters. According to the virial theorem the velocity dispersion v_r is a measure of the ratio between cluster mass M_{cl} and the cluster radius, e.g. the core radius r_c :

$$\langle v_r^2 \rangle \propto \frac{M_{cl}}{r_c}.$$

Measurements of the radial velocity dispersion $\langle v_r^2 \rangle^{1/2}$ has the following diagnostic potential.

- estimates of the cluster mass including the invisible mass,
- investigation of a radial mass segregation in the cluster,
- search for signatures indicating transient processes.

Table 2.8: Key parameters for open and globular clusters: stellar velocity dispersion $\langle v_r^2 \rangle^{1/2}$ in km/s, core radius, and total mass.

cluster	type	$\langle v_r^2 \rangle^{1/2}$	r_c [pc]	M [$M_\odot$]
Pleiades	open	0.5	1.4	800
Praesepe	open	0.5	3.5	550
ω Cen	globular	9.8	3.8	$5 \cdot 10^6$
NGC 6388	globular	18.9	0.5	$1.3 \cdot 10^6$

2.3.8 Kinematics of the galactic rotation

Qualitative expectations for the solar neighborhood. For the solar neighborhood we consider the effect of the galactic rotation on the systematic motion of stars for the radial velocity $v_r(\ell)$ and transverse velocity $v_t(\ell)$ direction as function of galactic longitude ℓ . For this discussion we assume that the Milky Way is rotating differentially, in the sense that the orbital period is shorter for an object closer to the galactic center. Further, we assume that the stars move on circular orbits.

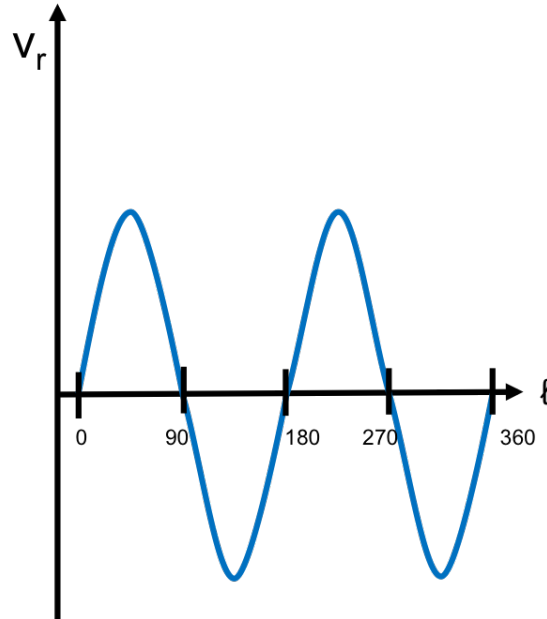


Figure 2.9: Sketch of the systematic radial velocity of stars relative to the local standard of rest because of the differential galactic rotation.

Radial velocity. Because of the differential rotation (shorter orbital period for smaller r) the stars with $R < R_0$ will overtake the sun, while stars with $R > R_0$ will be overtaken by the sun. According to Fig. 2.9 the systematic radial velocity is:

- v_r is positive for $0^\circ < \ell < 90^\circ$ and $180^\circ < \ell < 270^\circ$, and

- v_r is negative for $90^\circ < \ell < 180^\circ$ and $270^\circ < \ell < 360^\circ$,
- v_r is zero for $\ell = 0^\circ, 90^\circ, 180^\circ$, and 270° ,
- $v_r(\ell)$ is roughly a sine-curve $v_r(\ell) \approx c_1 \sin 2\ell$.

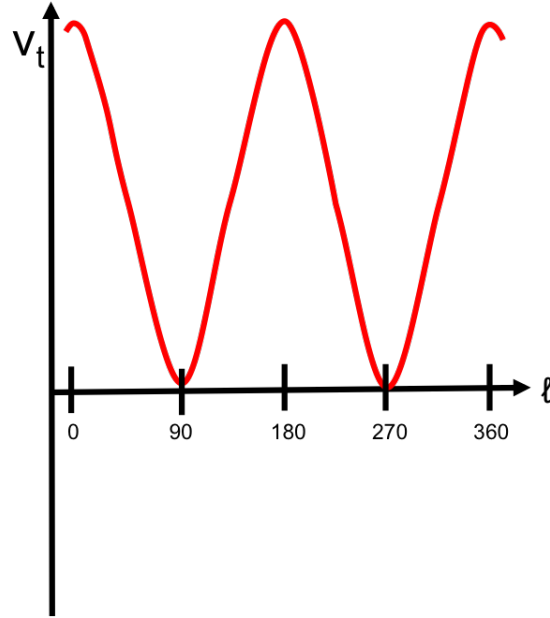


Figure 2.10: Sketch of the systematic transverse velocity of stars relative to the local standard of rest because of the differential galactic rotation.

Transverse velocity. Because all stars inside the solar orbit $R < R_0$ overtake the sun they move towards larger longitude. The stars outside the sun $R > R_0$ move backwards, but this is again in the direction of the galactic longitude angle and they have therefore again a positive angular motion. According to Fig. 2.10 the systematic transverse velocity is:

- v_t is zero or positive for all longitudes $0^\circ \leq \ell \leq 360^\circ$
- v_t has a maximum for the direction $\ell \approx 0^\circ$ and 180° ,
- v_t is zero for $\ell \approx 90^\circ$ and $\ell \approx 270^\circ$,
- $v_t(\ell)$ behaves like a shifted double cosine curve $v_t(\ell) \approx c_2 \cos 2\ell + c_3$

General rotation formula. We will now derive the general formula for the radial velocity v_r , the transverse velocity v_t and the angular proper motion in galactic longitude μ_ℓ for a differentially rotating disk. We consider again only circular orbits for the derivation. The definition of used parameters are given in the schematic geometric sketch shown in Fig. 2.11.

Radial velocity. The measured radial velocity for a star at point P is:

$$v_r = \Theta \cos \alpha - \Theta_0 \sin \ell. \quad (2.26)$$

According to the law of sines, and $\sin(\alpha + 90^\circ) = \cos \alpha$ there is:

$$\frac{\sin \ell}{R} = \frac{\sin(\alpha + 90^\circ)}{R_0} = \frac{\cos \alpha}{R_0}.$$

We can replace in Eq. 2.26 $\cos \alpha$ and use the angular orbital velocity $\Omega = \Theta/R$ or $\Omega R = \Theta$

$$v_r = \Theta \frac{R_0}{R} \sin \ell - \Theta_0 \sin \ell = \Omega R_0 \sin \ell - \Omega_0 R_0 \sin \ell = (\Omega - \Omega_0) R_0 \sin \ell. \quad (2.27)$$

This is a general result which assumes only that the galactic rotation is circular.

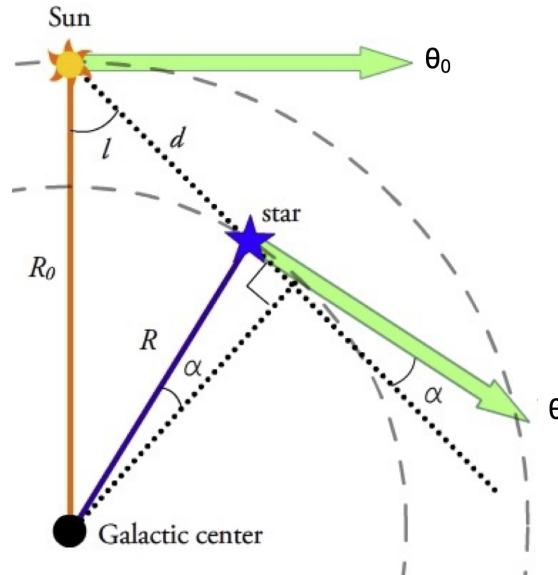


Figure 2.11: Geometry for the derivation of the relative stellar motions in a differentially rotating galactic disk.

From Eq. 2.27 we can derive the change in the angular rotation rate for different galactic radii $\Omega(R)$ for stars with known distances (if we know Ω_0 and R_0). We can also use this formula for the derivation of the galactic rotation curve from emission line observations of the interstellar gas (see next section).

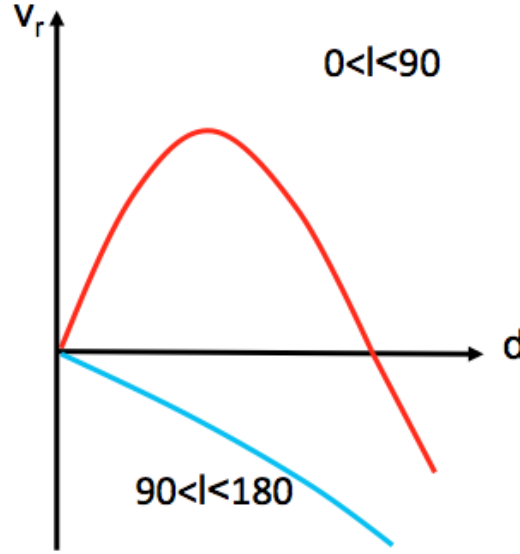


Figure 2.12: Illustration for the radial velocity of stars as function of distance due to their orbital rotation in the galactic disk.

Qualitatively, we can say for the dependence of the radial velocity $v_r(d)$ as function of distance (see Fig. 2.12):

- for the quadrant $0^\circ < \ell < 90^\circ$ the nearby stars are closer to the galactic center and they have therefore all a positive radial velocity.
- The maximum radial velocity $v_r(\text{max})$ is reached for the point along the sight-line where the distance to the galactic center is minimal,
- the radial velocity is zero $v_r = 0$ where the sight-line intersects the solar circle “on the other side”,
- further out the radial velocity v_r is negative and decreases further with distance.
- the quadrant $360^\circ > \ell > 270^\circ$ behaves similar, only the sign of the radial velocity is reversed.
- for the quadrant $90^\circ < \ell < 180^\circ$ we see only regions further out in the galaxy and therefore the difference $\Omega(R) - \Omega_0$ and the radial velocity v_r is always negative and decreases with distance.
- the quadrant $270^\circ > \ell > 180^\circ$ behaves similar, only the sign of the radial velocity is reversed.

Tangential velocity. For the tangential or transverse velocity of a star at point P there is:

$$v_t = \Theta \sin \alpha - \Theta_0 \cos \ell, \quad (2.28)$$

where v_t is positive for direction towards larger longitude ℓ . From Fig. 2.11 it is visible that

$$R \sin \alpha = R_0 \cos \ell - d.$$

One can now replace in Eq. 2.28 the term $\sin \alpha$, rearrange similar to the case of v_r , and it results

$$v_t = \frac{\Theta}{R}(R_0 \cos \ell - d) - \Theta_0 \cos \ell = (\Omega - \Omega_0)R_0 \cos \ell - \Omega d. \quad (2.29)$$

A similar discussion as for the radial velocity could be made for the transverse velocity of the stars in the galactic disk as function of distance. Essentially all stars on the other side of the galactic center would have a negative transverse velocity. However, the proper motions are very difficult to measure for distant stars and therefore this topic is not discussed here.

Oort's constants. In order to get accurate numerical values of the differential galactic rotation we can evaluate the general formulae 2.27 and 2.29 for the solar neighborhood and use the available, more accurate, velocity measurements of nearby stars.

Radial velocity. We consider first the radial velocity v_r given in Eq. 2.27 for a fixed longitude ℓ at the position of the sun R_0 . The only term which depends on distance is $\Omega - \Omega_0$. This difference can be approximated by a first-order Taylor expansion

$$(\Omega - \Omega_0) \approx \left(\frac{d\Omega}{dR}\right)_{R_0} (R - R_0). \quad (2.30)$$

The derivative of the angular rotation is:

$$\frac{d\Omega}{dR} = \frac{d}{dR} \left(\frac{\Theta}{R} \right) = \frac{1}{R} \frac{d\Theta}{dR} - \frac{\Theta}{R^2}$$

so that

$$\left(\frac{d\Omega}{dR}\right)_{R_0} = \frac{1}{R_0} \left(\frac{d\Theta}{dR}\right)_{R_0} - \frac{\Theta_0}{R_0^2}.$$

We can write Eq. 2.27 to first order

$$v_r \approx \left[\left(\frac{d\Theta}{dR}\right)_{R_0} - \frac{\Theta_0}{R_0} \right] (R - R_0) \sin \ell. \quad (2.31)$$

For $d \ll R_0$ the difference between R_0 and R can be approximated by

$$R - R_0 \approx -d \cos \ell$$

and it results using $\sin \ell \cos \ell = (1/2) \sin 2\ell$

$$v_r \approx \frac{1}{2} \left[\frac{\Theta_0}{R_0} - \left(\frac{d\Theta}{dR}\right)_{R_0} \right] d \sin 2\ell. \quad (2.32)$$

We then obtaine the double sine-wave variation of the radial velocity with galactic longitude as derived before from a qualitative discussion

$$v_r \approx A d \sin 2\ell \quad \text{with} \quad A = \frac{1}{2} \left[\frac{\Theta_0}{R_0} - \left(\frac{d\Theta}{dR}\right)_{R_0} \right], \quad (2.33)$$

where A is called the Oort's constant A .

Transverse velocity: Similarly we can evaluate the equation for the transverse velocity component v_t (Eq. 2.29) and get

$$v_t \approx \left[\left(\frac{d\Theta}{dR} \right)_{R_0} - \frac{\Theta_0}{R_0} \right] (R - R_0) \cos \ell - \Omega_0 d \approx \left[\frac{\Theta_0}{R_0} - \left(\frac{d\Theta}{dR} \right)_{R_0} \right] d \cos^2 \ell - \left(\frac{\Theta_0}{R_0} \right) d. \quad (2.34)$$

Using the trigonometric identity $\cos^2 \ell = (1 + \cos 2\ell)/2$ yields

$$v_t \approx \frac{1}{2} \left[\frac{\Theta_0}{R_0} - \left(\frac{d\Theta}{dR} \right)_{R_0} \right] d \cos 2\ell - \frac{1}{2} \left[\frac{\Theta_0}{R_0} + \left(\frac{d\Theta}{dR} \right)_{R_0} \right] d.$$

It results the shifted double wave cosine curve with the Oort's A defined above and Oort's B constant

$$v_t \approx d (A \cos 2\ell + B) \quad \text{with} \quad B = -\frac{1}{2} \left[\frac{\Theta_0}{R_0} + \left(\frac{d\Theta}{dR} \right)_{R_0} \right]. \quad (2.35)$$

This derivation was obtained in 1927 by Oort, who could prove that the Galaxy has a differential rotation.

Local rotation constants. Important is the result that the local value of the angular rotation rate and the radial derivative of the azimuthal velocity can be expressed with the Oort's constants:

$$\Omega_0 = \frac{\Theta_0}{R_0} = A - B \quad \text{and} \quad \left(\frac{d\Theta}{dR} \right)_{R_0} = -(A + B) \quad (2.36)$$

This describes well the local galactic rotation parameters.

Oort's rotation constant A can be derived from measurements of the radial velocity v_r of stars with known distance d for different longitudes ℓ using the formula

$$A = \frac{v_r}{d \sin 2\ell}.$$

Measured values for the Oort's constant A based on Gaia data (DR2) are

$$A \approx +15.1 \pm 0.1 \frac{\text{km}}{\text{s kpc}}.$$

The determination requires the measurement of radial velocities of a large, unbiased sample of stars. The measurement of radial velocities of many stars with an absolute precision of a few km/s is relatively easy.

Oort's constant B needs measurements of the transverse velocity (proper motion) of stars with known distance and for different galactic longitude ℓ using the formula

$$B = \frac{v_t}{d} - A \cos 2\ell$$

Because there is no transverse velocity component $v_t \approx 0$ for the direction of the galactic rotation and $\cos 2\ell \approx -1$, there is roughly $B \approx -A$. The transverse velocity of many stars has been measured with the Gaia satellite and the resulting B -value obtained is

$$B \approx -13.4 \pm 0.1 \frac{\text{km}}{\text{s kpc}}.$$

Angular rotation rate: Estimates of the angular rotation rate for the sun follow from

$$\Omega_0 = A - B \approx 28 \frac{\text{km}}{\text{s kpc}} = 28 \frac{10^{-9}}{\text{yr}}$$

The orbital period of the sun (or better the local standard of rest) around the galactic center is then roughly

$$P_0 = \frac{2\pi}{\Omega_0} \approx 224 \text{ Myr}.$$

Local velocity gradient. Also from the Oort's constants follows that the galactic rotation curve is essentially flat at the location of the sun. Explicitly:

$$\left(\frac{d\Theta}{dR}\right)_{R_0} \approx -(A + B) \approx -2(\pm 0.1) \frac{\text{km}}{\text{s kpc}}.$$

Galactic rotation velocity. The distance of the sun from the galactic center R_0 has been determined with various methods, like globular clusters, the motion of the star around the central black hole, and others which are not discussed here. The typical result of such studies yields $R_0 \approx 8 \text{ kpc}$. Because we know the angular velocity Ω_0 from the Oort's constant we get also the velocity of the galactic rotation Θ_0 for the local standard of rest:

$$\Theta_0 = R_0 \Omega = R_0 (A - B) \approx 8.2 \text{ kpc} \cdot 28 \frac{\text{km}}{\text{s kpc}} = 230 \text{ km/s}$$

2.4 Interstellar matter (ISM) in the Milky Way

In this section we consider the distribution of the interstellar matter in different regions of the Milky Way disc. For this we distinguish five gas-components:

- three diffuse components; atomic gas, photo-ionized gas, and collisionally ionized gas,
- two higher density components, molecular clouds and H II regions which are usually associated with star forming regions.

Table 2.9: Components of the ISM in the Milky Way disc.

	T [K]	$N(\text{H})[\text{cm}^{-3}]$	gas type	main particles
1.	10 – 100	$10^3 - 10^6$	molecular cloud	H_2 , dust, CO, ...
2.	100 – 1000	≈ 10	diffuse atomic gas	H^0 , dust, C^+ , e^- , N^0 , O^0 , ...
3.	≈ 10000	$10 - 10^4$	H II-regions	H^+ , e^- , dust, X^{+i} , ...
4.	≈ 10000	≈ 0.1	diffuse, photo-ionized gas	H^+ , e^- , dust, X^{+i} , ...
5.	$\gtrsim 10^6$	$\approx 10^{-3}$	diffuse, collisionally ionized gas	H^+ , e^- , X^{+i} , ..., (no dust)

In the Milky Way disc the components 1 and 2 contribute about 90 % to the baryonic mass of the ISM, while the components 2 and 5 fill essentially the space in the disk.

2.4.1 The ISM in the solar neighborhood

The distribution of the ISM in the solar neighborhood is determined observationally by interstellar absorption lines in the spectra of nearby stars and the reddening of stellar colors by dust. Thanks to observations along many sight-lines to stars which are at different distances it is also possible to estimate the distances to the absorbing gas structures.

Important absorption lines for the ISM in the solar neighborhood are:

- | | | |
|-------|---------------------------|--|
| 1. | molecular gas | H_2 and CO in the far UV, CH in the visual |
| 2. | atomic gas | H I Lyman lines, C I, C II, Si II, O I in the far UV;
Ca II, Na I in the visual |
| 3.+4. | photo-ionized gas | C IV, Si III, Si IV in the far UV |
| 5. | collisionally ionized gas | O VI in the far UV |

Emission lines and continuum emission from dust are not suited for the investigation of the ISM in the immediate solar neighborhood ($d < 200$ pc). This emission is very diffuse because there are no high density regions within this distance. Further it is almost impossible to determine the distance of the diffuse emission.

For larger distances ($d > 200$ pc) there are some well defined high density regions, which produce emission with higher surface brightness. Usually, this emission can be associated with molecular clouds or H II regions with young stars, which allow the determination of the distance.

Distribution of gas and dust within 250 pc. The strongest interstellar absorption lines are the far-UV absorption of H_2 and $\text{HI Ly}\alpha$. These lines trace the most abundant element and therefore they provide very reliable and very sensitive results. However, the H_2 and HI line absorptions require satellite observations and a hot $\gtrsim 20'000$ K background star. Observationally less demanding are dust extinction measurements using photometric data or absorption line data of Na I or Ca II in the optical spectra of A to K-type stars. An example of such absorption line data for the Na I line at 589.6 nm is shown in slide 2–26.

Local bubble. In the immediate neighborhood of the sun there is a low-density bubble which extends to about 60 pc. The observations for this local bubble show that

- the hydrogen column density up to about 60 pc is
 $N(\text{H}) = N(\text{HI}) + 2 \cdot N(\text{H}_2) \sim 1 - 5 \cdot 10^{18} \text{ cm}^{-2}$
 $\rightarrow$ mean density $\sim 0.01 - 0.1 \text{ cm}^{-3}$,
- the local HI -gas shows a systematic expansion of about 30 – 50 km/s,
- the O VI-absorption line increases with distance.

The average particle density in the Milky Way disk is more like

$$\bar{\rho}_{\text{ISM}} \approx 1 \text{ cm}^{-3}.$$

This indicates, that the sun is located in a hot bubble with a density far below (factor 10-100) the mean density in the disc.

An important conclusion for Galactic astronomy is:

- Hot bubbles, like the one around the sun, are quite frequent in the Milky Way disc. They are associated with supernova explosions. The supernova interpretation indicates that the local bubble has an age of $\approx 10^7 - 10^8$ years.

The dense clouds around 150 pc. The sun is surrounded by several dense interstellar clouds and star forming regions at distances of about ≈ 150 pc. The location of these clouds is traced with measurements of the dust extinction and the absorption by the $\text{Na I } \lambda\lambda 589.0, 589.6$ nm resonance lines along the sight lines towards nearby stars. Slides 2–23 and 2–24 show inversion maps for the dust extinction and the Na I line absorption perpendicular to the galactic disk and for the Gould belt in the galactic plane. In Slide 2–23 the maps for the dust extinction and Na I are compared. Essentially the same structures are seen in both maps. The Gould belt is a local disk ring structure of young stars and star forming regions, which is inclined by $\pm 18^\circ$ with respect to the disk plane. The young stars are ideal background targets for accurate measurements of the Na I line absorption and the dust extinction.

The maps in the Slides 2-27 and 2-28 show:

- the gas is predominantly distributed in the Milky Way disc,
- molecular clouds with high density are e.g. located in the direction of the galactic center (Sco-Oph), or towards the anticenter (Tau). In these regions star formation takes place.
- A “tunnel” of low density gas extends through the disk and gives us clear views towards the North and South galactic poles. This low density region is filled with hot, atomic and ionized gas.

This separation between well localized dense clouds, containing cold molecular clouds, and diffusely distributed atomic hydrogen in low density bubbles is quite typical for the entire Milky Way disc.

Dust is present in the diffuse, low density regions and in the dense clouds. The interstellar extinction by dust is therefore a good measure for the mass column density along the line of sight. Observations provide a good empirical relation between dust reddening E_{B-V} and hydrogen column density $N(\text{H})$:

$$N(\text{H}) [\text{cm}^{-2}] \approx 5.8 \cdot 10^{21} E_{B-V} [\text{mag}].$$

2.4.2 Global distribution of the ISM in the Galaxy

Well suited for the investigation of the global distribution of the ISM in the Milky Way are line and continuum emissions in spectral regions with little interstellar absorption. These are observations in the radio range, in the far IR, hard X-rays and gamma rays.

Important emission features for the different gas components are:

- | | | |
|-------|---------------------------|---|
| 1. | molecular clouds | CO-lines, IR-dust emission, γ -rays from π^0 -decay, created by collisions of cosmic rays |
| 2. | atomic Gas | H I 21 cm line, fine structure lines (e.g. C II), IR-dust emission, γ -rays (π^0 -decay) |
| 3.+4. | photo-ionized gas | H I-recombination lines (near-IR, radio range), bremsstrahlung (radio-range), collisionally excited lines |
| 5. | collisionally ionized gas | X-ray radiation (bremsstrahlung and X-ray line emission) |

The observations of emission lines provide one important advantage, when compared to continuum emission; from lines one can also measure the **radial velocity** v_r of the emission region.

Distribution perpendicular to the disc. The distribution of the interstellar matter can be derived from maps showing the emission of the different components, like the H I map shown in Slide 2–3 or the dust emission map in Slide 2–5.

The following table gives estimates of the disk scale height β (using $D(z) = D_0 e^{-|z|/\beta}$) for different components in the direction perpendicular to the disc:

- | | | |
|----|---------------------------|---|
| 1. | molecular clouds | $\beta \approx 30$ pc |
| 2. | atomic Gas (H I) | $\beta \approx 180$ pc |
| 3. | photo-ionized gas | difficult to determine
$\beta > \beta_{\text{Pulsar}} > 200$ pc based on dispersion measurements (not discussed yet) for pulsars |
| 4. | H II-regions | β like molecular clouds |
| 5. | collisionally ionized gas | $b > 250$ pc, hot gas extends far into the galactic halo |

2.4.3 Galactic rotation curve from line observations

Line emission regions along a line of sight in the Milky Way disc have different radial velocities due to the galactic rotation curve. In Sect. 2.3.8, we have derived the general

formula 2.27 for the radial velocity for a given galactic longitude ℓ :

$$v_r = (\Omega - \Omega_0)R_0 \sin \ell .$$

An emission line observations shows therefore many radial velocity components v_i depending on the angular velocity $\Omega(R_i)$ of the emitting clouds along the line of sight (see Fig. 2.13)

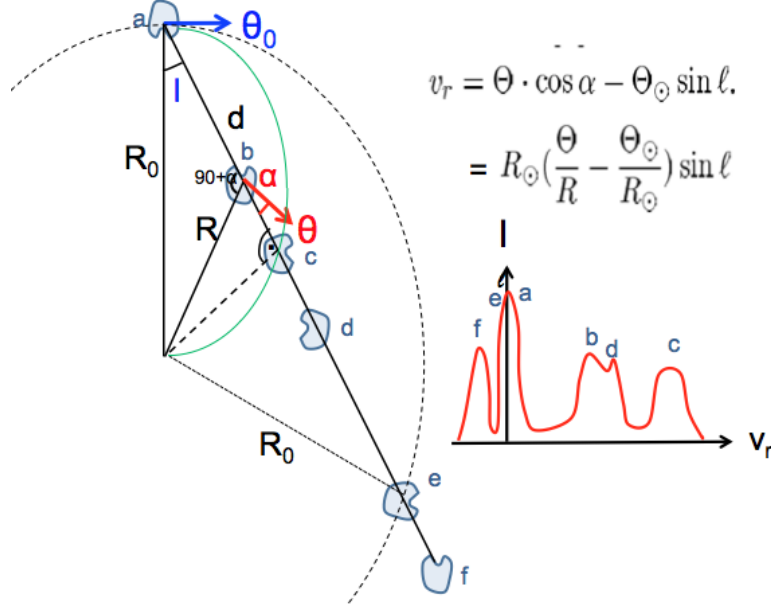


Figure 2.13: Illustration of the measured radial velocities of emission line clouds located along a sight line with fixed $\ell \approx 20^\circ - 70^\circ$.

Usually, it is not possible to derive an accurate distance to the cloud and derive its galactocentric distance R_i .

However, for longitudes $0^\circ < \ell < 90^\circ$ or $360^\circ > \ell > 270^\circ$ there exists a maximum (respectively minimum) radial velocity $v_{\max}$ for the point with the smallest galactocentric distance along the line of sight $R_{\min} = R_0 \sin \ell$. There, we see the galactic motion exactly along the radial velocity direction component and there is

$$v_{\max}(\ell) = (\Omega(R_{\min}) - \Omega_0)R_0 \sin \ell .$$

This maximum radial velocity will only occur for a differentially rotating galaxy for which Ω is steadily increasing with decreasing radius R (or $\Omega \cdot R \approx \text{const}$). This is the typical case for the disks in spiral galaxies but not for the central bulge region. Therefore it is possible to determine from the maximum velocity $v_{\max}(\ell)$ the Galactic rotation curve for $R_{\text{bulge}} < R < R_0$.

$\ell - v_r$ -maps: Galactic longitude – radial velocity. The radial velocity and the distribution of the gas in the Milky Way can be plotted in a diagram for the galactic longitude and the radial velocity. Slide 2–29 shows these maps for H I and CO. In the

exercises, we will derive the positions of different galacto-centric rings (inner ring $R < R_0$, outer ring $R > R_0$) in this diagram.

From the $\ell - v_r$ -maps of H I and CO one can deduce:

- the rotation curve of the Milky Way is essentially flat
 $v_r(R) = R\Omega(R) \approx \text{const.}$ in the range $R \approx 3 - 7$ kpc,
- the H I-gas extends from about 3 kpc out to about 15 kpc from the galactic center (outermost rim ~ 18 kpc),
- the CO molecular clouds are mainly located within a broad ring extending from about 3 kpc out to 8 kpc.

The v_{max} -method works best in the longitude range $\ell \approx 20^\circ - 70^\circ$ (respectively $\ell \approx 340^\circ - 290^\circ$). This yields the galactic rotation curve from about 3 kpc to 7 kpc. Inside 3 kpc is the galactic bulge with much less interstellar gas and the assumption of a differentially rotating system is also not valid. Between $\ell \gtrsim 90^\circ$ and $\ell \lesssim 270^\circ$ the method does not work, because there is no maximum radial velocity point.

Galactic rotation curve for $R > R_0$. The studies of the stellar dynamics in the solar neighborhood prove that the Galactic rotation curve is also flat near the solar cycle $R \approx R_0$. According to the Oort's constant there is:

$$\left(\frac{d\Theta}{dR}\right)_{R_0} \approx -(A + B) \approx -3(\pm 2) \frac{\text{km}}{\text{s kpc}}.$$

For $R > R_0$ the rotation curve $\Omega(R)$ can only be derived if the distance of a H I or a CO cloud with measured radial velocity can be determined. This can be achieved, if there are bright young stars, such as Cepheids, which can be associated with a CO cloud. The distance follows from the brightness of the Cepheid, the period-luminosity relation, and the correction for the interstellar extinction. From the distance d and the longitude ℓ follows R , so that $\Omega(R)$ can be derived from the measured radial velocity of the associated cloud.

2.4.4 H I and CO observations in other galaxies

The H I – 21 cm and the CO 2.3 cm lines are ideal for the investigation of the general distribution of the interstellar matter in disk galaxies. The line observations provide for resolved galaxies intensity and radial velocity maps. The maps provide also rotation curves using the $v_r(R)$ -profile along the major axis. For very distant, unresolved galaxies, one can measure the H I or CO velocity profile. This is sufficient for deriving the disk rotation velocities v_{rot} , if the inclination i of the disk can be determined from a resolved, e.g. optical image. The information which can be extracted from interstellar line observation is shown schematically in Slide 2–30, while Slide 2–31 and 2–32 give some examples for real data. Many nearby disk galaxies have been imaged in H I. CO data are still quite rare.

Sensitivity: A “deep” H I-map can reveal H I-gas with a column density down to $N(\text{H I}) \approx 10^{19} \text{ cm}^{-2}$. This corresponds to a mean surface density of $0.1 \text{ M}_\odot/\text{pc}^2$ per spatial resolution element.

The mean H I-surface density in disk galaxies is typically $1 - 4 \text{ M}_\odot/\text{pc}^2$. It seems that this is a kind of self-regulated value. If the surface density is larger than this value, then

the disk becomes optically thick for ionizing UV-radiation and atomic hydrogen H I is transformed into molecular H₂, because the radiative dissociation is strongly reduced.

In the centers of disk galaxies, the intensity maps show usually a minimum. Obviously, the density of atomic and molecular interstellar gas is strongly reduced in the bulges of disk galaxies.

In many galaxies the distribution of H I-gas can be traced to much larger distances than the visible star light. Edge-on galaxies show also the vertical distribution of H I which is usually broader than for the stars. On the other side, the CO is strongly concentrated to the disk mid-plane. Thus, the distribution of the interstellar matter in the Milky Way is typical for disk galaxies.

In external galaxies it is quite easy to recognize strong asymmetries in the distribution of the gas. Often the edge-on disks show warps, a tilt of the outermost disk ring regions with respect to the central disk. This phenomenon seems also to be present in the Milky Way.

H I-rotation curves. The main motion component of all spiral galaxies is the disk rotation. The motion of the H I or CO gas is always very smooth and deviates typically less than 20 km/s from the general rotation. The measured radial velocity at a given radius R and azimuthal angle ϕ (angle with respect to the line of nodes of the disk and the sky) is:

$$v_r(R, \phi, i) = v_{\text{sys}} + V(R) \cdot \cos \phi \cdot \sin i,$$

where v_{sys} is the systemic radial velocity of the galaxy and i the inclination of the disk. The intrinsic rotation curve $V(R)$ follows then from the measurements $v_r(R, \phi, i)$ with a cut through the major axis ($\phi = 0^\circ, 180^\circ$) of the galaxy and a radial velocity correction for $\sin i$. Usually it is assumed that the studied galaxy is rotationally symmetric.

The obtained $V(R)$ -rotation curves are flat ($V(R) \approx \text{const.}$) out to large distances for essentially all disk galaxies. This cannot be explained with the mass distribution of the stars and the interstellar medium. In all these galaxies, there must be an additional mass component which extends far into the halo.

Chapter 3

Galactic dynamics

Galactic dynamics can be divided into different regimes. First, there is the motion of the gas and the stars in the overall galactic potential. On these large scales the stars behave in a first approximation like test masses in a smooth potential for which dynamical gravitational interactions (near encounters with other stars) can be neglected. The stellar dynamics and the motion of the gas can be used to constrain the potential and the corresponding density distribution of the different galactic components: the stars in the bulge and the disk, the gas located in disk, and the extended dark matter halo.

Second, on small scales the motion of a star is determined by the gravitational potential of many stars in a smooth “background” potential. Depending on the case, it must be distinguished whether the dynamics of a star is strongly affected by individual dynamical interactions with other stars or not. In this context it is important to consider the difference between collisionless systems and systems with collisions.

In this chapter we describe first simple models for smooth gravitational potentials, the associated density distributions and the expected motion parameters and time scales. Then we consider relaxation (collision) time scales and discuss the impact of collisions on the dynamics.

3.1 Potential theory

In this section we describe the force field for a smooth distribution of mass. There exist simple but powerful analytic formula, which give a lot of insight on the motion of test particles in a smooth potential. In particular, we will discuss how the density structure of the Milky Way can be modelled. The description of this section follows the corresponding chapter in the book “Galactic Dynamics” from Binney and Tremaine.

3.1.1 Basic equations for the potential theory

The force $\vec{F}(\vec{x})$ at position $\vec{x}$ on a star with mass m_S is generated by the space distribution of mass $\rho(\vec{x}')$:

$$\vec{F}(\vec{x}) = m_S \vec{g}(\vec{x}) = m_S G \int \frac{\vec{x}' - \vec{x}}{|\vec{x}' - \vec{x}|^3} \rho(\vec{x}') d^3 \vec{x}'. \quad (3.1)$$

$\vec{g}(\vec{x})$ is the vector gravitational field, the force per unit mass or the gravitational acceleration.

The gravitational potential $\Phi(\vec{x})$ is defined by

$$\Phi(\vec{x}) = -G \int \frac{\rho(\vec{x}')}{|\vec{x}' - \vec{x}|} d^3\vec{x}', \quad (3.2)$$

which is the integral of the mass distribution weighted by the inverse distance to the point $\vec{x}$. The gradient for the inverse distance is

$$\vec{\text{grad}}_x \left(\frac{1}{|\vec{x}' - \vec{x}|} \right) = \frac{\vec{x}' - \vec{x}}{|\vec{x}' - \vec{x}|^3}$$

and therefore the gravitational field $\vec{g}(\vec{x})$ can be expressed by the gravitational potential according to

$$\vec{g}(\vec{x}) = -\vec{\text{grad}}_x \Phi(\vec{x}) = \vec{\text{grad}}_x \left(G \int \frac{\rho(\vec{x}')}{|\vec{x}' - \vec{x}|} d^3\vec{x}' \right). \quad (3.3)$$

The potential $\Phi(\vec{x})$ is very useful because it is a scalar field which can be described and analyzed based on equipotential surfaces. Φ contains the same information as the vector gravitational field or the acceleration $\vec{g}(\vec{x})$, which follows from the gradient of the potential.

3.1.2 Newton's theorems

Let's start with the simple case of spherical systems to get familiar with the mathematical procedures. Spherical systems are particularly simple because of Newton's theorems.

First theorem of Newton. A body inside a spherical shell experiences no net gravitational force from that shell.

Second theorem of Newton. A body outside a spherical shell experiences a gravitational force equal to the force of a mass point in the center of the shell with the mass of the shell.

Figure 3.1 illustrates the proof of the first theorem. A point P inside the shell is attracted equally strong by opposite shell section “seen” under the same solid angle $d\Omega$. This is obvious for radial sight lines through the center of the shell because the areas (with surface mass $m_{1,2}$) of the opposite regions are proportional to the distances squared $d_{1,2}^2$ from point P . Thus there is $F_1 = m_1/d_1^2 = m_2/d_2^2 = F_2$.

This is also valid for an arbitrary “sight” line (full line) because the tilt angles θ_1 and θ_2 between the tangential surfaces and the cone center lines are equal on both sides. Therefore the surface area defined by the solid angle cones are proportional to $d_1^2/\cos\theta_1$ and $d_2^2/\cos\theta_2$ and the attraction from the opposite sides is also equal.

Inside the shell the gravitational potential is constant because the gravitational force is zero

$$\vec{\text{grad}}_x \Phi = -g = 0.$$

The gravitational potential in the shell can be easily calculated for the central point from a radial form of Eq. 3.2 (see also Eq. 3.8)

$$\Phi(r < R) = -\frac{GM}{R}, \quad (3.4)$$

where $M = 4\pi\rho(r)dr$ is the total mass of a shell with thickness dr and R is the shell radius.

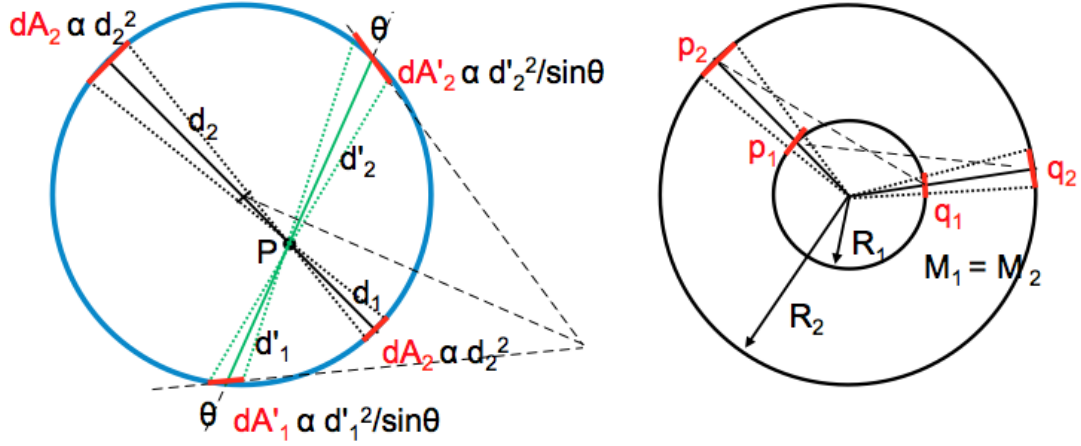


Figure 3.1: Figures for the proof of Newton's first theorem (left) and Newton's second theorem (right).

For the proof of Newton's second theorem a trick according to Fig. 3.1 with a special configuration of points p_1 , p_2 , q_1 and q_2 is needed. We consider two concentric shells with radius R_1 and R_2 and equal mass $M_1 = M_2$. Then one can write the potential for a point p_2 on the outer shell by a surface area region $\delta\Omega$ at point q_1 of the inner shell

$$\delta\Phi(\vec{p}_2) = -\frac{GM}{|\vec{p}_2 - \vec{q}_1|} \frac{\delta\Omega}{4\pi}.$$

This potential is equal to the potential for the point p_1 on the inner shell by a surface area region of the outer shell with the same angular dimensions $\delta\Omega$ at point q_2 .

$$\delta\Phi(\vec{p}_1) = -\frac{GM}{|\vec{p}_1 - \vec{q}_2|} \frac{\delta\Omega}{4\pi}.$$

Thus, there is $\delta\Phi(\vec{p}_2) = \delta\Phi(\vec{p}_1)$ because $|\vec{p}_2 - \vec{q}_1| = |\vec{p}_1 - \vec{q}_2|$ (symmetry) and the summation yields then that the potential due to the entire inner and outer shells are equal

$$\Phi_{\text{shell}_1}(\vec{p}_2) = \Phi_{\text{shell}_2}(\vec{p}_1).$$

We know $\Phi_{\text{shell}_2}(\vec{p}_1) = -GM/R_2$ (from Eq. 3.4) and therefore this is also the result for $\Phi_{\text{shell}_1}(\vec{p}_2)$ for the potential of a point at a radius $R = R_2$ outside a shell with $R_1 < R$ and mass M

$$\Phi_{\text{shell}_1}(R) = -\frac{GM}{R}. \quad (3.5)$$

This outside potential of a spherical shell is equal to the potential of a point with the same mass located at the center.

3.1.3 Equations for spherical systems

Simple equations can be derived for spherical systems using Newton's theorems.

The gravitational force of a spherical density distribution $\rho(r')$ on a star m_S at radius r is determined by the mass $M(r)$ interior to r

$$\vec{F}(r) = m_S \vec{g}(r) = -G \frac{m_S M(r)}{r^2} \vec{e}_r, \quad (3.6)$$

where

$$M(r) = 4\pi \int_0^r \rho(r') r'^2 dr'. \quad (3.7)$$

The gravitational potential of a spherical system is the sum of the potentials of spherical mass shells $dM(r) = 4\pi\rho(r)r^2dr$ with $r' < r$ (located inside r):

$$\Phi_{r' < r}(r) = -\frac{G}{r} \int_0^r dM(r')$$

and the mass shells at $r' > r$ (located outside r):

$$\Phi_{r' > r}(r) = -G \int_r^\infty \frac{dM(r')}{r'},$$

or written alternatively

$$\Phi(r) = -4\pi G \left[\frac{1}{r} \int_0^r \rho(r') r'^2 dr + \int_r^\infty \rho(r') r' dr \right]. \quad (3.8)$$

The circular speed $v_c(r)$, which is the speed of a test particle with negligible mass m_S in a circular orbit at radius r , is an important parameter for the characterization of the gravitational potential. The circular speed follows from the equilibrium $F_g(r) = -F_c(r)$ of gravitational force and centrifugal force $F_c = m_S v_c^2/r$:

$$v_c^2(r) = r g(r) = r \frac{d\Phi}{dr} = \frac{GM(r)}{r}. \quad (3.9)$$

This can also be expressed with angular velocity

$$\Omega(r) = \frac{v_c(r)}{r} = \sqrt{\frac{GM(r)}{r^3}}.$$

The potential energy of a system follows from an estimate of the change in potential energy if a small additional mass is added to the system with potential $\Phi(\vec{x})$.

$$\delta E_{\text{pot}} = \int \delta\rho(\vec{x}) \Phi(\vec{x}) d^3\vec{x}. \quad (3.10)$$

Integration of this from no mass to a full mass distribution of a system yields then the total potential energy.

For a homogenous sphere with radius R and mass M we may write for the work required to lift a small mass shell δM from the surface to infinity (work or energy required)

$$\delta E_{\text{pot}}(R) = \int_R^\infty \frac{GM\delta M}{r^2} dr = -G \frac{M\delta M}{R}.$$

For shells at different radii the mass is $M(r) = M_* (r/R_*)^3$ and $dM(r) = (M_*/R_*^3)3r^2 dr$ so that the total potential energy can be written as the following integral:

$$E_{*,\text{pot}} = -G \int_0^{M_*} \frac{M(r)}{r} dM(r) = -\frac{3GM_*^2}{R_*^6} \int_0^{R_*} r^4 dr = -\frac{3}{5} \frac{GM_*^2}{R_*} \quad (3.11)$$

3.1.4 Simple spherical cases and characteristic parameters

Potential of a point mass. This is a very simple case which is often referred as a Keplerian potential. For a point mass there is

$$\Phi(r) = -\frac{GM}{r}, \quad \text{and} \quad v_c(r) = \sqrt{\frac{GM}{r}}. \quad (3.12)$$

The potential energy of a point is $-\infty$ (or not defined).

Potential of a homogeneous sphere. Inside a homogeneous sphere with constant ρ there is:

$$M(r) = \frac{4}{3}\pi r^3 \rho. \quad (3.13)$$

The circular velocity increases linearly with radius

$$v_c(r) = \sqrt{\frac{GM(r)}{r}} = \sqrt{\frac{4\pi G\rho}{3}} r. \quad (3.14)$$

The orbital period is then defined by the density ρ

$$T = \frac{2\pi r}{v_c} = \sqrt{\frac{3\pi}{G\rho}} \quad (3.15)$$

The potential energy of a homogeneous sphere with radius a , density ρ and total mass $M = (4/3)\pi G\rho a^3$ follows from Eq. 3.11:

$$E_{\text{pot}} = -4\pi G\rho \int_0^a r M(r) dr = -\frac{16\pi^2 G\rho^2}{3} \int_0^a r^4 dr = -\frac{16}{15}\pi^2 G\rho^2 a^5 = -\frac{3}{5} \frac{GM^2}{a}. \quad (3.16)$$

The gravitational potential of a homogeneous sphere with radius a is

$$\Phi(r) = -2\pi G\rho(a^2 - \frac{1}{3}r^2) \quad \text{for} \quad r < a, \quad (3.17)$$

$$\Phi(r) = -\frac{GM}{r} \quad \text{for} \quad r > a. \quad (3.18)$$

Gravitational radius. The size of a system is sometimes characterized by the gravitational radius which is defined as ratio between mass squared divided by the total gravitational (potential) energy:

$$r_g = \frac{GM^2}{|E_{\text{pot}}|}. \quad (3.19)$$

For a homogeneous sphere, where $E_{\text{pot}} = -(3/5)GM^2/a$ the corresponding gravitational radius is $r_g = (5/3)a$. The gravitational radius can be a convenient quantity for the definition of the size of systems which have no sharp boundary (e.g. stellar cluster).

The dynamical time scale. The homogeneous sphere is a useful model for an estimate of the dynamical time scale of a system.

If a mass is released from rest in a gravitational field of a homogeneous sphere then its equation of motion is given by the gravitational acceleration

$$g(r) = \frac{d^2 r}{dt^2} = -\frac{d\Phi(r)}{dr} = -\frac{GM(r)}{r^2} = -\frac{4\pi G\rho}{3}r.$$

This is the equation of motion for a harmonic oscillator ($\ddot{x} = -\omega^2 x$) with oscillation period $T = (2\pi/\omega) = \sqrt{3\pi/G\rho}$. This is the same time as is required for a full circular orbit (Eq. 3.15).

Thus, there is for a homogeneous sphere not only an unique circular orbital period but also an unique free fall time t_{ff} , which is the time it takes for any particle released at rest to fall into the center. This time is

$$t_{\text{ff}} = \frac{T}{4} = \sqrt{\frac{3\pi}{16G\rho}} = 0.767 (G\rho)^{-1/2}$$

The dynamical time scale is defined as

$$t_{\text{dyn}} = (G\rho)^{-1/2}. \quad (3.20)$$

This quantity is of the same order as the free-fall time, the crossing time or the orbital time for a particle. According to our definition there is:

$$t_{\text{dyn}} = 1.3 t_{\text{ff}} = 0.65 t_{\text{cross}} = 0.33 t_{\text{orbit}}$$

The dynamical time scale is also a good parameter for the characterization of systems with not to extreme density gradients, as long as ρ is replaced by the mean density $\bar{\rho}$ inside the location of the particle.

$$t_{\text{dyn}} \approx (G\bar{\rho})^{-1/2}.$$

This relation is therefore used for the characterization of systems like open clusters, globular clusters, bulges of galaxies, or clusters of galaxies.

The Plummer model. Plummer proposed in 1911 a spherical density model with a “soft edge” which can be described by a simple gravitational potential

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + b^2}}. \quad (3.21)$$

The corresponding density can be described by

$$\rho(r) = \frac{3M}{4\pi} \frac{b^2}{(r^2 + b^2)^{5/2}}. \quad (3.22)$$

Thus, there is a density distribution like for a homogeneous sphere for $r < b$ without a sharp edge but with a steep density fall off like $\propto r^{-5}$.

3.1.5 Spherical power law density models

Many galaxies have luminosity profiles which can be fitted with power law profiles. Therefore it seems useful to investigate spherical potentials for density distributions which can be described by a power law of the form

$$\rho(r) = \rho_0 \left(\frac{r_0}{r} \right)^\alpha. \quad (3.23)$$

The mass inside r is then

$$M(r) = 4\pi \int_0^r \rho(r') r'^2 dr' = 4\pi \rho_0 r_0^\alpha \int_0^r r'^{2-\alpha} dr' = 4\pi \rho_0 r_0^\alpha \frac{r^{3-\alpha}}{3-\alpha}.$$

We consider only $\alpha < 3$, because only for such cases the mass interior to r is finite. On the other side the mass $M(r)$ diverges for $r \rightarrow \infty$ at large radii if $\alpha < 3$. The models are still useful because according to Newton's first theorem the spherical mass shells outside r do not affect the gravitational forces and dynamics inside r .

Thus, we can derive the circular velocity v_c for the power law models and obtain

$$v_c^2(r) = \frac{GM(r)}{r} = 4\pi G \rho_0 r_0^\alpha \frac{r^{2-\alpha}}{3-\alpha}. \quad (3.24)$$

This is a very interesting formula which can be used for the interpretation of the flat rotation curves observed in disk galaxies out to very large radii. The circular velocity $v_c(r)$ is constant if $\alpha \approx 2$ or for a dark matter density distribution which behaves at radii $\gtrsim 10$ kpc like

$$\rho_{\text{dm}}(r) \propto \left(\frac{1}{r} \right)^2.$$

Two-power density models. A spherical density model combining two power laws, one approximating the flatter central region and one approximating a steeper density fall-off at larger radius provides more modelling possibilities. Well studied is a analytical parameterization for which the density is described by

$$\rho(r) = \frac{\rho_0}{(r/a)^\alpha (1+r/a)^{\beta-\alpha}} \quad (3.25)$$

where a is a scaling radius. The α parameter is $\alpha < 3$ to avoid that the mass at small radius goes to infinity and $\beta \geq 3$ so that the total mass remains finite for large radius.

The following cases are simple and popular solutions:

- Hernquist model with $\alpha = 1$ and $\beta = 4$; this yields

$$\rho(r) \propto \frac{1}{(r/a)(1+r/a)^3}, \quad \Phi(r) \propto \frac{GM}{a+r}, \quad v_c(r) = \frac{\sqrt{GM}r}{b+r}.$$

- Jaffe model with $\alpha = 2$ and $\beta = 4$,

$$\rho(r) \propto \frac{1}{(r/a)^2(1+r/a)^2}, \quad \Phi(r) \propto \frac{GM}{a} \ln(1+a/r), \quad v_c(r) = \sqrt{\frac{GM}{b+r}}.$$

- Navarro, Frenk and White or NFW model with $\alpha = 1$ and $\beta = 3$.

$$\rho(r) \propto \frac{1}{(r/a)(1+r/a)^2}, \quad \Phi(r) \propto GM \frac{\ln(1+r/a)}{r/a}.$$

3.1.6 Potentials for flattened systems

Potential of a “Toomre” disk. A simple potential for a disk was introduced by Kuzmin in 1956 and independently by Toomre in 1963. The disk potential can be described by

$$\Phi(R, z) = -\frac{GM}{\sqrt{R^2 + (|a| + |z|)^2}}. \quad (3.26)$$

According to Fig. 3.2 the potential at point $(R, -z)$ is equal to a potential generated by a mass M located at the point $(0, a)$ or for points above the disk by a mass located at $(0, -a)$.

Such a potential can be generated by a razor-thin disk with the surface density distribution

$$\Sigma(R) = \frac{aM}{2\pi(R^2 + a^2)^{3/2}}. \quad (3.27)$$

The central surface density at $R = 0$ is $M/2\pi a^2$, while the surface density drops for large R like $\Sigma(R) \approx aM/R^3$. The constant a is just a scale parameter which indicates where the surface density changes from constant to a step gradient.

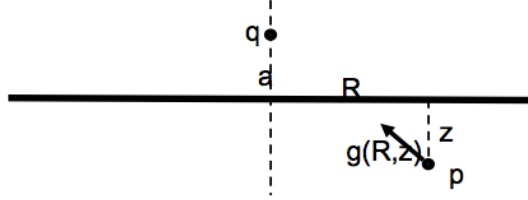


Figure 3.2: Illustration of the parameters for Toomre’s disk.

A hybrid model between Toomre’s disk and the Plummer sphere We can now generalize the disk model to include also a matter distribution in z -direction. This can be achieved with a parameterization of the potential according to

$$\Phi(R, z) = -\frac{GM}{\sqrt{R^2 + (|a| + \sqrt{z^2 + b^2})^2}}. \quad (3.28)$$

This potential has two extreme cases:

- for $b = 0$ the potential of a thin Toomre’s disk is obtained,
- for $a = 0$ and using $R^2 + z^2 = r^2$ yields the spherical Plummer potential.

Depending on the selection of the parameters a and b one can create a family of potentials covering density distributions from a thin disk to a sphere. The corresponding mass distributions for this types of potentials are described by Miyamoto and Nagai

$$\rho(R, z) = \left(\frac{b^2 M}{4\pi}\right) \frac{aR^2 + (a + 3\sqrt{z^2 + b^2})(a + \sqrt{z^2 + b^2})^2}{[R^2 + (a + \sqrt{z^2 + b^2})^2]^{5/2}(z^2 + b^2)^{3/2}} \quad (3.29)$$

Slide 3–1 shows contour plots of this density distribution for a few parameter cases. The case $b/a \approx 0.2$ is at least qualitatively a quite good representation for a disk galaxy, while $b/a \approx 1$ resembles a S0 galaxy (e.g. Sombrero galaxy).

Potential of spheroids. Many astronomical systems are spheroidals, flattened spheres, because of the presence of angular momentum. The evaluation of potentials for spheroids in general is very difficult, because we have to consider the 2D or 3D density distribution of the system.

An important simplification is possible if we consider, homoeoids, thin concentrically nested spheroidal shells. These shells are similar to the spherical shell used for spherical systems.

One homoeid shell is bound by an inner surface and an outer surface described by

$$\frac{R^2}{a^2} + \frac{z^2}{b^2} = 1 \quad \text{and} \quad \frac{R^2}{a^2} + \frac{z^2}{b^2} = (1 + \delta\beta)^2,$$

respectively. The perpendicular distance between the two surfaces varies with position. This happens in such a way that Newton's first theorem can be generalized to spheroidal (ellipsoidal) shells.

Newton's third theorem. A mass that is inside a homoeid experiences no net gravitational force from the homoeoids.

The potential theory of spheroids was further developed in order to model with high precision the potential of the Milky Way and other galaxies. Important for these models is Newton's third theorem and the theory of multipole expansions for the gravitational potential. This theory is not discussed in this lecture. Some of the important results are:

- many potentials for flattened (oblate) spheroid and triaxial ellipsoids have been derived and applied to galaxy bulges, bars, and elliptical galaxies,
- potentials of exponential galactic disks are successfully described by strongly flattened spheroid using Newton's third theorem,
- potentials for non-axisymmetric disks can be calculated using Bessel functions, and special potential functions are used for the description of logarithmic spiral structures.

3.1.7 The potential of the Milky Way

In this subsection the potential of the Milky Way is described. In particular the density distributions of the main mass components are given: the bulge, the disk with different distributions for the stars and the interstellar gas, and the dark halo. The described model is only partly derived from studies of the dynamical properties of the Milky Way. A lot of information on the mass distribution is also derived from photometric studies. In this description the Milky Way is an axisymmetric system given in cylindrical coordinates R and z . The model picked for this description has the parameters of Model I in the book of Binney & Tremaine. This is a Milky Way model with a relatively small disk but all parameters are compatible with the available observations. Slide 3–2 shows the equipotentials for this model as well as the different components and Slide 3–3 illustrates the corresponding circular velocities $v_c(r)$.

The central bulge. The bulge can be described by a oblate, spheroidal power law model which is truncated at an outer radius r_b :

$$\rho_b(R, z) = \rho_{b0} \left(\frac{m}{a_b} \right)^{\alpha_b} e^{-m^2/r_b^2}, \quad (3.30)$$

with

$$m = \sqrt{R^2 + z^2/q_b^2}.$$

The parameters describe:

- $\rho_{b0} = 0.43 \text{ M}_\odot/\text{pc}^3$ is the density normalization for the bulge
- $a_b = 1 \text{ kpc}$ is the size normalization of the bulge,
- $q_b = 0.6$ describes the bulge flattening,
- $\alpha_b = -1.8$ is the power law index for the density distribution,
- $r_b = 1.9 \text{ kpc}$ is the cut-off radius for the bulge.

The galactic disk. The Milky way disk consists of the stellar disk and a gas disks.

The stellar disk is described by an exponential fall-off in radial direction and two exponential laws for the vertical direction, one for the thin disk and one for the thick disks. The used formula is

$$\rho_s(R, z) = \Sigma_S e^{-R/R_S} \left(\frac{\alpha_0}{2z_0} e^{-|z|/z_0} + \frac{\alpha_1}{2z_1} e^{-|z|/z_1} \right). \quad (3.31)$$

The parameters describe:

- $\Sigma_S \approx 1500 \text{ M}_\odot/\text{pc}^2$ is the central surface density of the stellar disk which is not well known except for the solar radius R_0 . At R_0 the surface density of the stars is about $35 \text{ M}_\odot/\text{pc}^2$ to which the thick disk contributes about $3 \text{ M}_\odot/\text{pc}^2$.
- $R_S = 2.5 \text{ kpc}$ is the disk scale length,
- $\alpha_0 = 0.9$ and $\alpha_1 = 0.1$ are the relative normalizations of the thin and thick disk,
- $z_0 = 0.3 \text{ kpc}$ is the vertical scale height of the thin disk,
- $z_1 = 1 \text{ kpc}$ is the scale height of the thick disk.

The radial distribution of the interstellar gas disk is also described with an exponential law, but with a much larger scale length than for the stars. In addition, there is a hole with a radius of about 4 kpc in the center which is considered with an exponential cut-off. The vertical density distribution of the gas is much narrower than for the stars:

$$\rho_g(R, z) = \Sigma_g e^{-R/R_g} e^{-R_m/R} \frac{1}{2z_g} e^{-|z|/z_g}. \quad (3.32)$$

where the parameters are:

- $\Sigma_S \approx 500 \text{ M}_\odot/\text{pc}^2$, the surface density of the gas in the disk is not well known except for R_0 where the surface density is about $\Sigma_g(R_0) \approx 12 \text{ M}_\odot/\text{pc}^2$
- $R_g = 4 \text{ kpc}$ is the disk scale length for the gas (twice the value when compared to the stellar disk),
- $R_m = 4 \text{ kpc}$ is the radius of the inner hole,
- $z_g = 80 \text{ pc}$ is the scale height of the gas disk.

The dark halo. The dark halo can be described by an extension of the spherical two-power-law model to an oblate geometry.

$$\rho_h(R, z) = \frac{\rho_{h0}}{(m/a_h)^{\alpha_h} (1 + m/a_h)^{\beta_h - \alpha_h}} \quad (3.33)$$

where the flattening is described like for the bulge

$$m = \sqrt{R^2 + z^2/q_h^2}.$$

The parameters describe:

- $\rho_{h0} = 0.71\text{M}_\odot/\text{pc}^3$ is the density normalization for the halo,
- $a_h = 3.8$ kpc is the size normalization for the halo,
- $q_h = 0.8$ is a guess for the possible flattening of the dark halo,
- $\alpha_h = 2.0$ and $\beta_h = 3$ are the power law indices for the halo density distribution.

3.2 Motion in axisymmetric potentials

Stars moving in the equatorial plane of an axisymmetric potential behave like stars in a spherical potential, because one can always find a spherical gravitational potential which induces the same gravitational force on the stars in the equatorial plane as the axisymmetric potential. For this reason the orbits discussed in the previous chapter for spherical potentials apply also for the stars in the equatorial plane of an axisymmetric potential.

The motion of the stars located in or near the equatorial plane is an important problem for the investigation of disk galaxies.

3.2.1 Motion in the meridional plane

We assume that the potential is symmetric with respect to the plane $z = 0$. Then we can write the Lagrange equation with the following terms

$$\mathcal{L} = \frac{m_S}{2}(\dot{R}^2 + R^2\dot{\phi}^2 + \dot{z}^2) - m_S\Phi(R, z)$$

The 3-dimensional motion of a star in an axisymmetric potential can be reduced to a 2-dimensional motion of a star in the R - z -plane, the meridional plane.

The equation of motion in this co-rotating plane are:

$$\ddot{R} = -\frac{\partial\Phi_{\text{eff}}(R, z)}{\partial R}, \quad \ddot{z} = -\frac{\partial\Phi_{\text{eff}}(R, z)}{\partial z}, \quad (3.34)$$

where the effective potential is

$$\Phi_{\text{eff}}(R, z) = \Phi(R, z) + \frac{L_z^2}{2m_S^2 R^2} \quad (3.35)$$

Similar to the spherical case we can write the total energy equation, but now with an R and a z term for the kinetic energy

$$E_{\text{tot}} = \frac{m_S}{2}(\dot{R}^2 + \dot{z}^2) + m_S\Phi_{\text{eff}}(R, z). \quad (3.36)$$

The kinetic energy of motion in the R - z -plane is

$$\frac{m_S}{2}(\dot{R}^2 + \dot{z}^2) = E_{\text{tot}} - m_S\Phi_{\text{eff}}(R, z).$$

Orbits in the meridional plane are restricted to the area $E_{\text{tot}} > m_S\Phi_{\text{eff}}(R, z)$ and one can define contour lines or the zero velocity curves in the meridional plane where the kinetic energy term is instantaneously zero

$$m_S\Phi_{\text{eff}}(R, z) = E_{\text{tot}}.$$

The minimum of Φ_{eff} is in the equatorial plane $z = 0$ and the radial value follows from

$$0 = \frac{\partial\Phi_{\text{eff}}}{\partial R} = \frac{\partial\Phi}{\partial R} - \frac{L_z^2}{m_S^2 R^3}$$

This yields the radius for a circular orbit with angular speed $\dot{\phi}$ which is identical to the radius of the minimum of the effective potential. At this radial point, which is called the **guiding-center radius** R_g , there is

$$\left(\frac{\partial\Phi}{\partial R}\right)_{(R_g,0)} = \frac{L_z^2}{m_S^2 R^3} = R_g \dot{\phi}^2,$$

($L_z = m_S R^2 \dot{\phi}$). This is the condition for a circular orbit with angular speed $\dot{\phi}$ for a mass located at the radius R_g , which is at the minimum of the effective potential.

Example. Slide 3-4 shows as example the contour plot and orbits for the effective potential

$$\Phi_{\text{eff}}(R, z) = \frac{v_0^2}{2} \ln\left(\frac{R^2 + z^2}{q^2}\right) + \frac{L_z^2}{2m_S^2 R^2},$$

for $v_0 = 1$, $L_z = 0.2$ and axial ratio $q = 0.9$ and 0.5 . This represents the effective potential for an oblate, spheroidal mass distribution like a central bulge of a disk galaxy, an elliptical galaxy, or a dark matter halo with a constant circular velocity speed $v_c = \text{const.}$ The effective potential energy rises strongly near $R = 0$ because of the “centrifugal barrier” for the given angular momentum L_z .

The equations (3.34) for the relative motion in a co-rotating frame must be integrated numerically. Slide 3-4 shows the calculated motion. The given results are for stars in the same potential, same energy and same angular momentum but they still differ significantly. This problem is called the third integral problem and it is linked in this case to the precession of the angular momentum vector in a flattened potential.

3.2.2 Nearly circular orbits: epicycle approximation

In disk galaxies many stars are on nearly circular orbits. For this case we can simplify the equation of motion in the co-rotating system (Eq. 3.34)

$$\ddot{R} = -\frac{\partial\Phi_{\text{eff}}(R, z)}{\partial R}, \quad \ddot{z} = -\frac{\partial\Phi_{\text{eff}}(R, z)}{\partial z},$$

with a linearization of the corresponding effective potential at $R = R_g$ and $z = 0$. We introduce x as new variable in the radial direction

$$x = R - R_g$$

The effective potential in Eq. 3.35 can then be written as Taylor expansions for x and z . Because, we consider only the region, very close to the minimum of the effective potential, the first derivatives in the expansions are essentially zero, $\partial\Phi_{\text{eff}}/\partial R \approx 0$ and $\partial\Phi_{\text{eff}}/\partial z \approx 0$. Thus, the effective potential can be approximated by a constant value plus a quadratic dependence in x and z :

$$\Phi_{\text{eff}}(x, z) = \Phi_{\text{eff}}(R_g, 0) + \frac{1}{2} \left(\frac{\partial^2 \Phi_{\text{eff}}}{\partial R^2} \right)_{(R_g, 0)} x^2 + \frac{1}{2} \left(\frac{\partial^2 \Phi_{\text{eff}}}{\partial z^2} \right)_{(R_g, 0)} z^2 + O(xz^2) + \dots \quad (3.37)$$

This approximation, which is called the **epicycle approximation**, yields very simple, harmonic equations of motions for the radial x and vertical z directions:

$$\ddot{x} = -\kappa^2 x, \quad \ddot{z} = -\nu^2 z. \quad (3.38)$$

The oscillation parameters κ and ν describe the second derivatives of Φ_{eff} (curvature in the effective potential):

$$\kappa^2(R_g) = \left(\frac{\partial^2 \Phi_{\text{eff}}}{\partial R^2} \right)_{(R_g, 0)}, \quad \text{and} \quad \nu^2(R_g) = \left(\frac{\partial^2 \Phi_{\text{eff}}}{\partial z^2} \right)_{(R_g, 0)}.$$

The two time scales are called:

- the epicycle or radial frequency κ ,
- the vertical frequency ν .

These frequencies can be evaluated using Eq. 3.35 for the effective potential in a co-rotating system

$$\Phi_{\text{eff}}(R, z) = \Phi(R, z) + \frac{L_z^2}{2m_S^2 R^2}.$$

This yields for the vertical frequency the simple relation

$$\nu^2(R_g) = \left(\frac{\partial^2 \Phi}{\partial z^2} \right)_{(R_g, 0)} \quad (3.39)$$

Solution for the epicycle frequency. There are two terms for the epicycle frequency κ , a potential energy term and an angular momentum term (from $\partial^2/\partial R^2 (L_z^2/2m_S^2 R^2)$)

$$\kappa^2(R_g) = \left(\frac{\partial^2 \Phi}{\partial R^2} \right)_{(R_g, 0)} + \frac{3L_z^2}{m_S^2 R^4}. \quad (3.40)$$

We can now use the “global” angular velocity dependence for the circular motion at R_g which is (using also $v_c^2 = R(\partial\Phi/\partial R) = L_z^2/m_S^2 R^2$)

$$\Omega^2(R) = \frac{v_c^2(R)}{R^2} = \frac{1}{R} \left(\frac{\partial \Phi}{\partial R} \right)_{(R_g, 0)} = \frac{L_z^2}{m_S^2 R^4}$$

With this relation we can rewrite the equation for the epicycle frequency in terms of global, “galactic”, quantities:

$$\kappa^2(R_g) = \left(R \frac{d\Omega^2}{dR} + 4\Omega^2 \right)_{R_g} \quad (3.41)$$

using $d^2\Phi/dR^2 = d/dR(R\Omega^2) = \Omega^2 + R(d\Omega^2/dR)$. This relates the epicycle frequency to the radial dependence of the angular velocity $d\Omega^2(R)/dR$.

Comparison of epicycle period and orbital period. We can now compare the epicycle period T_r with the azimuthal orbital period T_{orb} which are simply:

$$T_r = \frac{2\pi}{\kappa} \quad \text{and} \quad T_\phi = \frac{2\pi}{\Omega}.$$

There are three useful approximate cases for a comparison between orbital frequency and epicycle frequency:

- Near the center of galaxies the circular speed v_c increases linearly and $\Omega(R)$ is essentially constant and therefore $d\Omega^2/dR \approx 0$. In this case there is

$$\kappa^2(R_g) \approx 4\Omega^2 \quad \text{or} \quad \kappa \approx 2\Omega,$$

This corresponds to the limiting case of a homogeneous sphere where the epicycle frequency is twice the orbital frequency, or the radial period is half the orbital period $T_r = T_\phi/2$.

- At large radii from the center the circular velocity falls off like (but usually less rapid) the Kepler law. For a Kepler law there is $\Omega \approx R^{-3/2}$ (using $R(d\Omega^2(R)/dR) = -3\Omega^2$). This limit implies

$$\kappa^2(R_g) \approx \Omega^2 \quad \text{or} \quad \kappa \approx \Omega.$$

Thus we have the case where the radial period and orbital periods are equal or $T_r = T_\phi$. This is as expected for a closed Keplerian orbit.

- At most points in a typical disk galaxy the circular velocity is constant or $\Omega \propto R^{-1}$. For this case the formula for the epicycle frequency is

$$\kappa^2 \approx 2\Omega^2 \quad \text{or} \quad \kappa \approx 1.4\Omega.$$

This indicates that in a disk the stars oscillate radially with a frequency of roughly 1.4 times the orbital frequency.

Application for the solar neighborhood. The third case, for intermediate separations, can be evaluated in detail for the solar neighborhood. As described in Chapter 2, we know quite well the Oort's constants A and B from the measurement of the radial and tangential velocities of stars in the solar neighborhood. We used the following formula for the Oort's constants:

$$A = \frac{1}{2} \left[\frac{\Theta_0}{R_0} - \left(\frac{d\Theta}{dR} \right)_{R_0} \right] \quad \text{and} \quad B = -\frac{1}{2} \left[\frac{\Theta_0}{R_0} + \left(\frac{d\Theta}{dR} \right)_{R_0} \right].$$

With $\Theta = R\Omega$ and $d\Theta/dR = d/dR(R\Omega) = \Omega + R(d\Omega/dR)$ we can write:

$$A = -\frac{1}{2} R \frac{d\Omega}{dR} \quad \text{and} \quad B = -\left(\Omega + \frac{1}{2} R \frac{d\Omega}{dR} \right)$$

The circular angular velocity is $\Omega = A - B$ while the epicycle frequency is

$$\kappa^2 = -4B(A - B) = -4B\Omega$$

or $\kappa^2/\Omega^2 = -4B/\Omega$, which yield the ratio between epicycle frequency and orbital period for the solar neighborhood

$$\frac{\kappa_0}{\Omega_0} = 2\sqrt{\frac{-B}{A - B}} \approx 1.3 \pm 0.1. \quad (3.42)$$

The result is obtained for the typical values for the Oort's constants $A \approx +15 \text{ km}/(\text{s kpc})$ and $B \approx -12 \text{ km}/(\text{s kpc})$. This means that the sun makes about 1.3 oscillations in radial directions within one orbit around the galactic center.

3.2.3 Resonances in disks

In the previous subsection we have introduced the following quantities for stars with almost circular orbits in disk galaxies:

- T_r : the epicycle or radial period for the in-and-out motion in radial direction,
- $\Delta\phi$: the azimuthal angle increase during the epicycle period,
- $\kappa = 2\pi/T_r$: the radial oscillation frequency,
- $\Omega = 2\pi/T_\phi$: the orbital frequency or orbital angular velocity where T_ϕ is the time for a full orbit around the galaxy center.

We now describe the motion of the stars in a frame which is rotating with some special angular velocity. The following quantities are defined in this system:

- Ω_P : angular velocity (or pattern speed) for the selected rotating frame,
- $\phi_P = \phi - \Omega_P t$: the azimuthal angle in the rotating reference system which changes with time,
- $\Delta\phi_P = \Delta\phi - \Omega_P T_r$: the azimuthal angle increase in the rotating system for one epicycle period.

One can always choose an angular velocity Ω_P for a rotating coordinate system in which the orbits are closed or $\Delta\phi_P/T_r = (n/m)\kappa$. This follows from the definition of $\Delta\phi_P$

$$\Omega_P = \frac{\Delta\phi}{T_r} - \frac{n}{m}\kappa. \quad (3.43)$$

For orbits close to circular orbits we can write $\Delta\phi/T_r = \Omega$ and

$$\Omega_P = \Omega - \frac{n}{m}\kappa. \quad (3.44)$$

Figure 3.3 illustrates the appearance of orbits with different m and n in a frame rotating with Ω_P .

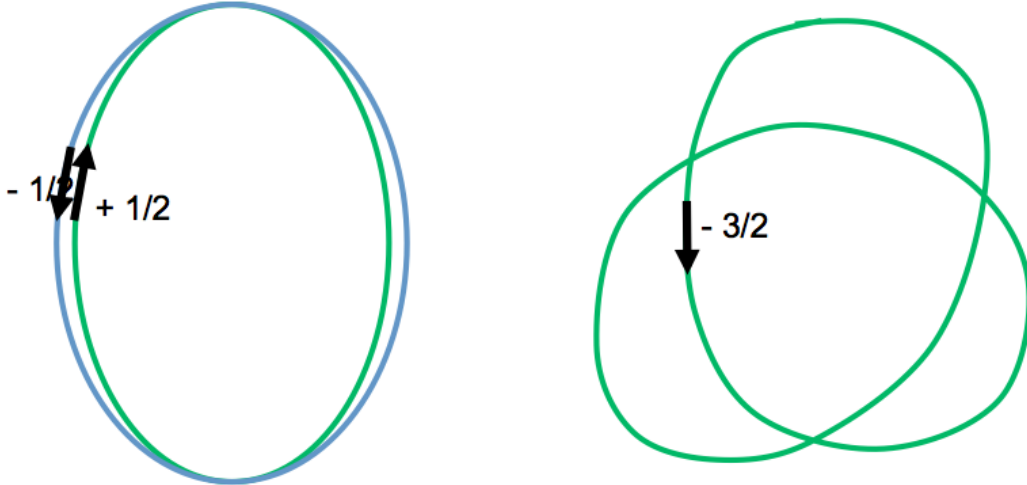


Figure 3.3: Closed orbits with different n and m in a rotating system.

In general, $\Omega - n\kappa/m$ is a function of radius, and no unique pattern speed Ω_P can be defined to close the orbits at all radii. Slide 3-6 shows curves for $\Omega - n\kappa/m$ for the Milky Way (model 1).

However, it was first noticed by Lindblad that the curve for $\Omega - \kappa/2$ is relatively constant for a wide range of galactic radii. A constant curve $\Omega - \kappa/2$ would mean that in a frame rotating at Ω_P all orbits would be ellipses, which are nested for a broad range of R . They would look like the ellipses shown in Slide 3-7. If stars move predominantly along these ellipses then they would create a **bar-like pattern**, which is stationary in a rotating frame. In a fixed frame this would then look like a **density wave** rotating with a pattern speed Ω_P .

In a real galaxy $\Omega - n\kappa/m$ depends on radius. Therefore, independent of the selected Ω_P , most orbits will not be exactly closed. The orientations at different radii will drift at slightly different speeds, and the pattern will twist, and might look like a spiral pattern (see Slide 3-7). This type of kinematic density waves, produce a non-axisymmetric disk pattern and an exact calculation of stellar orbits needs to take this into account.

Resonances. Resonances in a rotating disk occur if the circular frequency Ω and/or the epicycle frequency κ are in phase for different radii. The following resonances are distinguished:

- **Corotation resonance:** In this case the pattern frequency is equal to the circular frequency

$$\Omega_P = \Omega.$$

This resonance can be induced by an interaction with another galaxy or a strong, co-rotating density asymmetry in the disk.

- **Lindblad resonances:** They occur if the difference between the pattern and the orbital frequency is an integer of the epicycle frequency.

$$\Omega_P - \Omega = \pm \frac{1}{2}\kappa.$$

In this case a star encounters a weak perturbation in the potential in phase with its radial in-and-out motion. For this reason already a weak perturbation can build up a strong effect within several cycles. One distinguishes between inner $\Omega_P - \Omega = -\kappa/2$ and outer $\Omega_P - \Omega = +\kappa/2$ Lindblad resonances.

In a spiral galaxy the Lindblad resonances define the inner and outer boundaries of the spiral pattern.

- the inner Lindblad resonance is where the spiral structure starts. The stars are on elliptical orbits centered on the galactic center and they move faster than the disturbing spiral pattern. This happens in the Milky Way around 3 kpc.
- the outer Lindblad resonance is at the outer borders of the galaxy, where the spiral pattern ends. There, the orbits are again elliptical with the center in the middle. The disturbing spiral pattern is faster than the stars.

We give a simple numerical example for the inner and outer Lindblad resonances (LR) for $\kappa/\Omega = 1.6$ where the periodicities of the orbit and radial oscillations are 10 times longer for the outer LR than for the inner LR:

$\Omega_P =$	Ω	$\pm 1/2$	κ
inner Lindblad resonance for $P_\phi = 80$ Myr			
$\Omega_P =$	$2\pi/(80 \text{ Myr})$	$-1/2$	$2\pi/(50 \text{ Myr})$
one orbit for spiral pattern	400 Myr		
$2\pi/(400 \text{ Myr})$	$10\pi/(400 \text{ Myr})$	$-1/2$	$16\pi/(400 \text{ Myr})$
two radial oscillations	100 Myr		
$(\pi/2)/(100 \text{ Myr})$	$(5\pi/2)/(100 \text{ Myr})$	$-1/2$	$4\pi/(100 \text{ Myr})$
90° rotation	$360^\circ + 90^\circ$ orbit		2 radial oscillations
outer Lindblad resonance for $P_\phi = 800$ Myr			
$\Omega_P =$	$2\pi/(800 \text{ Myr})$	$+1/2$	$2\pi/(500 \text{ Myr})$
9 orbits for spiral pattern in	4000 Myr		
$18\pi/(4000 \text{ Myr})$	$10\pi/(4000 \text{ Myr})$	$+1/2$	$16\pi/(4000 \text{ Myr})$
two radial oscillations	1000 Myr		
$(9\pi/2)/(1000 \text{ Myr})$	$(5\pi/2)/(1000 \text{ Myr})$	$+1/2$	$4\pi/(1000 \text{ Myr})$
$720^\circ + 90^\circ$ rotation	$360^\circ + 90^\circ$ orbit		2 radial oscillations

This simple numerical example gives reasonable numbers for the resonance equation of a galaxy spiral, which would match for inner and outer Lindblad resonances with a global ratio epicycle to orbital frequency ratio $\kappa/\Omega = 1.6$ and a spiral pattern rotation period of about 420 Myr (400 Myr for inner disk 444 Myr for the outer disk).

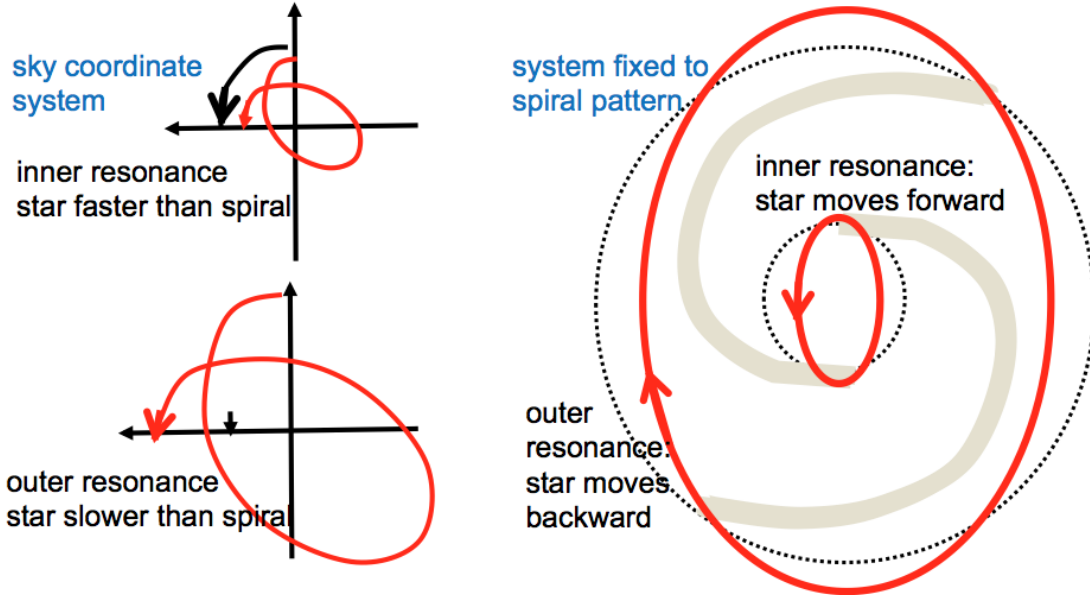


Figure 3.4: Illustration of the stellar orbits and the pattern motion in the sky (absolute) system and in the system co-rotating with the spiral pattern.

3.3 Two-body interactions and system relaxation

Up to now we have assumed that collisions, ie the interaction between individual stars, can be neglected. This is a reasonable assumption for galactic dynamics. We discuss now cases where such collisions between stars play an important role.

A star within a more or less homogeneous distribution of other stars “feels” the gravitational force of all these stars. The force $F = \Sigma F_i$ of all stars i in a given solid angle (see Fig. 3.5) behaves as follows:

- the force induced by an individual star is $F_i \sim 1/r_i^2$ and decreases with distance,
- the volume and therefore the number of stars in a fractional distance interval, e.g. $[r - r/2, r + r/2]$ increases like $\sim r^3$,
- the total force on the sample star is dominated by the more distant stars.

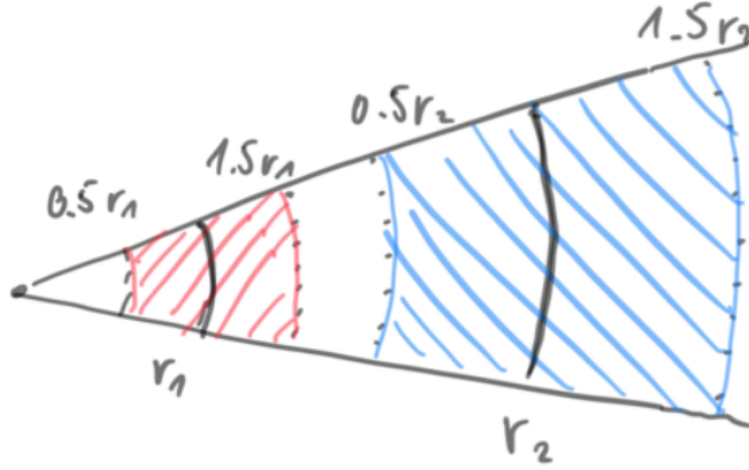


Figure 3.5: On the force induced by near and distant stars in a homogeneous distribution.

Therefore it is reasonable to assume that stars are smoothly accelerated by the force field that is generated by the Milky Way as a whole. In the following we investigate more quantitatively this simplification and consider cases where this approximation is no more valid.

3.3.1 Two-body interaction

We consider an individual star, called the **the subject star**, and investigate how much its velocity is disturbed by encounters with other stars, called two-body interactions, during the crossing through a system like a galaxy, or a star cluster. Thus we calculate the expected deflection of the trajectory of the subject star from the path it would have in the smooth overall potential. For our estimate we assume that all stars have the same mass m_s .

The velocity deflection $\delta\vec{v}$ induced by a two-body interaction can be simplified, if we consider only weak (distant) encounters which introduce small velocity deflections $|\delta\vec{v}|/v \ll 1$. Further it is assumed that the field star is stationary during the encounter. The velocity

deflection follows then from the perpendicular force $F_{\perp}$ of the field star induced onto the subject star, which is moving with velocity v along an essentially straight line with impact parameter b (Fig. 3.6).

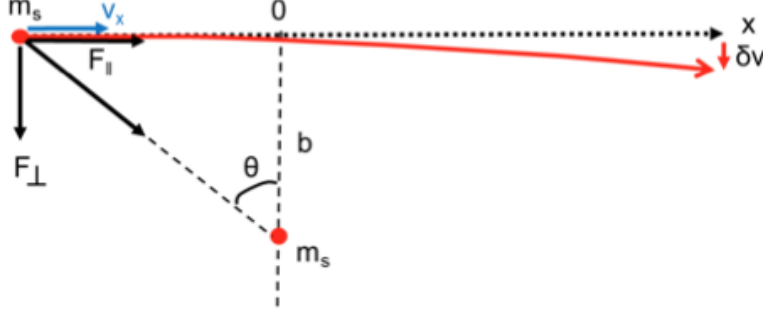


Figure 3.6: Geometry for an estimate of the deflection by a star-star interaction.

If both stars have the same mass, then the perpendicular force induced on the subject star is:

$$F_{\perp} \approx \frac{Gm_S^2}{b^2 + x^2} \cos\theta = \frac{Gm_S^2 b}{(b^2 + x^2)^{3/2}}, \quad (3.45)$$

using the trigonometric relation $\cos\theta = b/r = b/\sqrt{b^2 + x^2}$. The coordinate along the trajectory x can be expressed by the time and the velocity of the subject star $x = v \cdot t$ so that

$$F_{\perp} \approx \frac{Gm_S^2}{b^2} \left[1 + \left(\frac{vt}{b} \right)^2 \right]^{-3/2}.$$

According to Newton's law the acceleration or change in velocity $\vec{v} = \vec{F}/m_S$ is the time integral of the acting force, or

$$\delta v \approx \int_{-\infty}^{+\infty} \frac{F_{\perp}}{m_S} dt = \frac{Gm_S}{b^2} \int_{-\infty}^{+\infty} \frac{1}{[1 + (vt/b)^2]^{3/2}} dt \quad (3.46)$$

The integral is equal to $2b/v$ and the deflection is

$$\delta v \approx \frac{2Gm_S}{bv}. \quad (3.47)$$

This equation can be interpreted as follows:

- δv is proportional to the acceleration at closest approach Gm_S/b^2 times a characteristic duration of the acceleration $2b/v$,
- the derived approximative value is only valid for $\delta v \ll v$ or for an impact parameter larger than

$$b \gg Gm_S/v^2 = 900\text{AU} \frac{(m_S/M_{\odot})}{(v/(\text{km/s}))^2}.$$

As next step we estimate the number of encounters of the subject star in a stellar system for the impact parameter range $[b, b + db]$. We use an estimate for the surface density of field stars

$$\Sigma_{\text{stars}} \approx \frac{N}{\pi R^2},$$

where N is the total number of stars and R the radius of the considered system, e.g. the stellar cluster or galaxy. The subject star will have during one crossing of the system the following number of encounters

$$\delta n = \frac{N}{\pi R^2} 2\pi b db = \frac{2N}{R^2} b db. \quad (3.48)$$

with impact parameter between b and $b + db$. Each such encounter produces a deflection $\delta \vec{v}$ to the subject star, but these deflections are randomly oriented and their mean will be zero. But the mean square change δv^2 will not be zero after one crossing. The squared velocity deflection (change) for an impact parameter interval db will be:

$$\Sigma \delta v^2 \approx \delta v^2 \delta n = \left(\frac{2Gm_S}{bv} \right)^2 \frac{2N}{R^2} b db. \quad (3.49)$$

Now, we have to take into account all impact parameters by integrating from $b_{\min}$ to $b_{\max}$

$$\Delta v^2 = \int_{b_{\min}}^{b_{\max}} \Sigma \delta v^2 db = 8N \left(\frac{Gm_S}{Rv} \right)^2 \ln b \Big|_{b_{\min}}^{b_{\max}}. \quad (3.50)$$

The logarithm term can be written as

$$\ln \Lambda = \ln b_{\max} - \ln b_{\min}.$$

The maximum impact parameter is of the order $b_{\max} \approx R$, the smallest, where the small deflection approximation is still valid, is $b_{\min} \approx Gm_S/v^2$. These are only approximate values with an uncertainty of a factor of a few. For this reason we can write

$$\ln \Lambda = \ln \left(\frac{R}{b_{\min}} \right) + \ln(\epsilon_1/\epsilon_2).$$

In most systems $R \gg b_{\min}$ and the typical ratio is $R/b_{\min} \gg 10^4$ while the uncertainty term ϵ_1/ϵ_2 is much smaller, of the order of a few. This term can therefore be neglected with respect to the first term. Thus, the parameter Λ is approximately

$$\Lambda \approx \frac{Rv^2}{Gm_S} \approx N,$$

where we already used the next approximation for the typical stellar velocity v .

The encounters between the subject star and the field stars produce a diffusion of the star's velocity which is different from an acceleration induced by a smooth, large scale potential. This velocity diffusion is called **two-body relaxation**, because it is the result of a large number of mostly weak two-body interactions.

The typical speed v of the a field star can be approximated by the circular velocity of a star at radius R (at the edge) of the system

$$v^2 \approx \frac{GNm_S}{R}.$$

Equation 3.50 can be simplified with this velocity to

$$\frac{\Delta v^2}{v^2} \approx \frac{8 \ln N}{N}, \quad (3.51)$$

The subject stars makes typically many crossings until the velocity $\vec{v}$ changes by roughly Δv^2 . The number of crossings n_{relax} required for a change of the velocity by a value comparable to v is then

$$n_{\text{relax}} \approx \frac{N}{8 \ln N}.$$

3.3.2 Relaxation time

The relaxation time is defined as $t_{\text{relax}} = n_{\text{relax}} t_{\text{cross}}$, where the crossing time is $t_{\text{cross}} = R/v$. Using all the approximations from above we can express the relaxation time by the number of stars and the crossing time

$$t_{\text{relax}} \approx \frac{0.1N}{\ln N} t_{\text{cross}}. \quad (3.52)$$

Thus the relaxation time exceeds the crossing time in a self-gravitating system for $N \gtrsim 40$.

After the relaxation time the orbit of a (subject) star is changed significantly by all the small kicks induced by other stars, so that its velocity is now different from what one would expect in a smooth potential.

Table 3.1: Typical characteristic parameters for stellar systems

system	R	N	t_{cross}	t_{relax}	t_{lifetime}
clusters of galaxies	1 Mpc	1000	1 Gyr	14 Gyr	10 Gyr
galaxies	10 kpc	10^{11}	100 Myr	$\gg 100$ Gyr	10 Gyr
central pc of galaxies	1 pc	10^6	10^4 yr	100 Myr	10 Gyr
globular clusters	10 pc	10^5	10^5 yr	100 Myr	10 Gyr
open clusters	10 pc	100	1 Myr	10 Myr	100 Myr

Table 3.1 gives typical numbers for the different stellar systems using these approximations. The numbers show that the relaxation times are extremely long for galaxies. Therefore they can be treated as collision-less systems. Important to notice is, that the dynamics of stars in galaxies preserve at least partly information from past events.

On the other side there are the stellar clusters. Globular clusters relax in about 100 Myr and open clusters on a very short timescale of about 10 Myr. Their dynamics is dominated by relaxation and the star motions are rapidly randomized. For this reason it is often not possible to extract the past history of clusters from dynamical studies.

3.3.3 The dynamical evolution of stellar clusters

Galactic stellar clusters have a very short relaxation time. A disturbance of their dynamics is therefore rapidly randomized. In addition, galactic stellar clusters are also quite fragile and dissolve rapidly, typically within about 300 Myr. On the other side there are the globular clusters which have survived more than 10 Gyr. For these systems we have only

little information about their formation history and all signatures from the formation process in the stellar dynamics has been washed out.

Some important dynamical processes for the evolution of stellar cluster can be inferred from their current properties:

For globular clusters, we know that

- they have a long life time of 10 Gyr or more,
- they have typically $N \approx 10^5$ to 10^6 stars,
- many globular clusters show a dense core and a low density halo,
- there are often “hard binary systems” in the center.

For galactic open cluster, we know that

- they have typically 100 - 1000 star members,
- they dissolve in about 300 Myr,
- they show often a mass segregation with more massive stars in the center and lower mass stars further out.

In the following we discuss a few processes in stellar dynamics which influence the evolution of stellar clusters.

Cluster formation. We discuss the formation of a stellar cluster, considering a very young population of N stars which is still embedded in the gas cloud out of which the stars were formed. We define the total embedded stellar mass of the cluster M_{ecl} and use m_S as mean stellar mass

$$m_S = \frac{M_{\text{ecl}}}{N}.$$

Further we can define a fractional star-formation efficiency ϵ , the fraction of the total mass of the initial gas cloud M_{cloud} which ends up in newly formed stars

$$\epsilon = \frac{M_{\text{ecl}}}{M_{\text{ecl}} + M_{\text{gas}}}.$$

Here M_{gas} is the gas left over from the star-formation process ($M_{\text{cloud}} = M_{\text{ecl}} + M_{\text{gas}}$). Usually it is very difficult to determine observationally the mass of the remaining gas after the end of the star formation process. For this reason the existing “typical” fractional star formation efficiency parameter is very uncertain. A value in the range

$$0.2 \lesssim \epsilon \lesssim 0.4$$

is often quoted. This means that less than half of the mass of a collapsing cloud ends up in stars. This is a strong hint that the star formation in a collapsing cloud is terminated by the newly formed stars: this is called **feedback effect** in star formation. The following energetic processes can be responsible for the termination of star formation:

- the production of turbulence by the outflows from circumstellar disk around newly forming stars,
- photoionization and heating by the energetic radiation produced by the gas accretion processes of protostellar sources or the UV radiation from the hot photosphere of young, massive stars,
- shocks created by the stellar winds of young stars,
- shocks from supernova explosions of very massive, short lived stars.

Feedback: instantaneous gas removal. We can estimate what happens if there is an embedded cluster of protostars where the gas is removed in a short time by energetic stellar processes.

We assume that the embedded, proto-stellar cluster is in a dynamical equilibrium state what is a reasonable assumption for a 10 Myr young cluster (see Table 3.1). Then the total energy (or binding energy) is

$$E_{\text{ecl}} = -\frac{GM_{\text{init}}^2}{r_{\text{init}}} + \frac{1}{2}M_{\text{init}}\sigma_{\text{init}}^2 \quad (3.53)$$

where σ_{init} is the velocity dispersion which can be written for a virialized system $E_{\text{pot}} + 2E_{\text{kin}} = 0$ as

$$\sigma^2 = \frac{GM}{r}. \quad (3.54)$$

A virialized systems relates also the binding energy and the potential energy

$$E = +\frac{1}{2}E_{\text{pot}}.$$

The initial mass is $M_{\text{init}} = M_{\text{ecl}} + M_{\text{gas}}$ and this quantity can be the same as the total mass of the collapsing cloud $M_{\text{init}} = M_{\text{cloud}}$. The formalism is also valid for later stages where already some gas is lost, so that $M_{\text{init}} < M_{\text{cloud}}$ and $M_{\text{gas}}(t) < M_{\text{gas}}(t=0)$.

If energetic processes remove instantaneously the gas then the total mass of the cluster is changed from M_{init} to $M_{\text{after}} = M_{\text{ecl}}$ which includes only the total mass of the stars. The instantaneous gas removal does not change instantaneously the radial distribution r_{init} and the kinetic motion σ_{init} of the stars. The total binding energy of the cluster immediately after the gas removal is then

$$E_{\text{after}} = -\frac{GM_{\text{after}}^2}{r_{\text{init}}} + \frac{1}{2}M_{\text{after}}\sigma_{\text{init}}^2 = -\frac{1}{2}\frac{GM_{\text{after}}^2}{r_{\text{init}}}. \quad (3.55)$$

The cluster evolves now with a timescale of the order of the relaxation time scale to a new equilibrium state. If we assume that the mass $M_{\text{cl}} = M_{\text{after}}$ and energy $E_{\text{cl}} = E_{\text{after}}$ are conserved during this phase then the new equilibrium state can be described by a new radius r_{cl} and a new velocity dispersion σ_{cl}

$$E_{\text{cl}} = -\frac{GM_{\text{cl}}^2}{r_{\text{cl}}} + \frac{1}{2}M_{\text{cl}}\sigma_{\text{cl}}^2 = -\frac{1}{2}\frac{GM_{\text{cl}}^2}{r_{\text{cl}}}. \quad (3.56)$$

The resulting cluster radius follows from $E_{\text{cl}} = E_{\text{after}}$. $M_{\text{cl}} = M_{\text{after}}$, and using $\sigma_{\text{init}}^2 = GM_{\text{init}}/r_{\text{init}}$ yields another form of E_{after} :

$$E_{\text{after}} = -\frac{GM_{\text{after}}}{r_{\text{init}}}\left(M_{\text{after}} - \frac{1}{2}M_{\text{init}}\right)$$

With the relations from above $M_{\text{init}} = M_{\text{cl}} + M_{\text{gas}}$ we obtain

$$\frac{r_{\text{cl}}}{r_{\text{init}}} = \frac{1}{2}\frac{M_{\text{cl}}}{M_{\text{cl}} - M_{\text{init}}/2} = \frac{M_{\text{cl}}}{M_{\text{cl}} - M_{\text{gas}}}. \quad (3.57)$$

This equation implies that the cluster radius goes to ∞ , or becomes unbound

- for $M_{\text{gas}} \rightarrow M_{\text{cl}}$, or if the removed gas contains equal or more mass than the stellar mass of the cluster,
- this is potentially the case for inefficient star formation when the fractional star formation efficiency $\epsilon \leq 0.5$ is low and a lot of gas is still present in the newly formed clusters.

Since, the inferred fractional star formation rate is low $\epsilon < 0.5$, and there are many gas-less clusters observed there must be alternatives to the instantaneous gas removal model. Instantaneous gas removal would lead to the destruction of many galactic clusters, but this did not happen for all the known stellar clusters in our Milky Way.

Feedback: continuous removal of gas. One can talk of a continuous mass loss if the time scale for gas removal is much longer than the relaxation time or the cluster crossing time:

$$\tau_{\text{gas}} \gg \tau_{\text{relax}}.$$

In this case the cluster adjusts its dynamics continuously according to the virial equilibrium. The increase of the cluster radius can then be described as a result of small (infinitesimal) mass removals:

$$\frac{r_{\text{init}} + \delta r}{r_{\text{init}}} = \frac{M_{\text{init}} - \delta M_{\text{gas}}}{M_{\text{init}} - \delta M_{\text{gas}} - \delta M_{\text{gas}}}. \quad (3.58)$$

which can be written as

$$\frac{r + dr}{r} = \frac{M - dM}{M - 2dM},$$

We search now for the formula for the relative radius increase of the cluster because of a small mass loss. Useful formulae are obtained by rearranging

$$\left(\frac{dr}{r} + 1\right)(M - 2dM) = M - dM \quad \text{or} \quad \frac{dr}{r}(M - 2dM) = -(M - 2dM) + M - dM = dM.$$

Above we used δM and dM for mass which was removed. In the following equation the sign is switched for the case where the mass loss dM is a negative quantity:

$$\frac{dr}{r} \frac{M + 2dM}{M} = -\frac{dM}{M} \quad \text{or} \quad \frac{dr}{r} = -\frac{dM}{M} \left(\frac{M}{M + 2dM}\right)$$

For slow mass loss, there is $|dM| \ll M$. and we can approximate

$$\frac{dr}{r} \approx -\frac{dM}{M}. \quad (3.59)$$

Integration yields $\ln(r_{\text{cl}}/r_{\text{init}}) = -\ln(M_{\text{cl}}/M_{\text{init}})$ or

$$\frac{r_{\text{cl}}}{r_{\text{init}}} = \frac{M_{\text{init}}}{M_{\text{cl}}} = \frac{M_{\text{ecl}} + M_{\text{gas}}}{M_{\text{cl}}} = \frac{1}{\epsilon}. \quad (3.60)$$

If the mass-loss is slow, then one can have a low fractional star formation efficiency (say 0.2) and loose a lot of gas (80 %) from the initial cloud mass and still end up with a bound cluster. The radius of the cluster expands like $1/\epsilon$. For example, if 80 % of the mass is lost by a continuous gas removal then the initial radius of the cluster expands by a factor of 5.

The conclusion is that with a slow mass loss, which allows a continuous re-virialization of the cluster dynamics, the mass loss causes less expansion and a more likely survival of a cluster compared to an instantaneous mass loss.

Mass segregation and core-formation. A cluster contains stars with a range of masses. The interactions of stars in a cluster induces, like in the kinetic gas theory, an evolution towards **equipartition**:

- in two-body interactions, the more massive stars transfer a significant amount of their large kinetic energy to less massive stars i , until $m_1 v_1^2 \approx m_2 v_2^2$,
- this leads in a self-gravitating star clusters to a mass segregation, the more massive stars have less specific (per unit mass) kinetic energy and sink towards the cluster center, while less massive stars gain kinetic energy and diffuse outwards to larger radii.

The concentration of massive stars towards the center would just continue and lead to a core collapse. A relatively small number of massive stars concentrate in a very compact cluster core while the halo expands. This evolution would lead to a singularity if hard binaries would not counteract to this process.

Compact binary stars. Binary stars can transfer a lot of energy to a dense stellar system by dynamic interactions. We consider here only a simple energy argument.

A virialized system has a binding energy of

$$E_{\text{cl}} \approx -\frac{GM^2}{R_{\text{cl}}} \approx -\frac{G N^2 m_S^2}{R_{\text{cl}}}.$$

We can compare this to the binding energy of a binary star which is

$$E_{\text{bin}} \approx -\frac{G m_S^2}{a}$$

where a is the orbital separation (semi-major axis).

If the binary is sufficiently compact then its binding energy (negative total energy) is equal to the total binding energy of the entire cluster. The corresponding binary separation is

$$a_{\text{eq}} \approx \frac{R_{\text{cl}}}{N^2}. \quad (3.61)$$

This separation corresponds to

- $a_{\text{eq}} \approx 2$ AU for a open cluster with 1000 stars and a radius of 10 pc,
- $a_{\text{eq}} \approx 10^{-4}$ AU (or $0.1 R_{\odot}$) for a globular cluster with 10^5 stars and 10 pc radius.

This comparison shows that compact binaries, also called hard binaries, can stabilize a stellar cluster against collapse. Interaction of a hard binary star with a single star can transfer orbital energy of the binary to the third star, which gains then kinetic energy and moves outward in the cluster. This interaction reduces of course the separation and the total energy of the binary. However, a compact binary can have more binding energy than an entire open cluster. Such binary star interactions act against the cluster core collapse due to two-body interactions and equipartition. Of course, the binaries become more and more compact with time and they may even merge. This scenario can also explain the presence of the blue stragglers in the HR-diagram of clusters.

In globular cluster several hard binaries are required to stabilize the system. With X-ray observations such hard binaries were indeed found in several globular clusters. There

are cases with about 10 or even more such binaries in one globular cluster. These X-ray binaries have characteristics of low mass X-ray binaries, which are composed of a neutron star and a companion, often a white dwarf, in a very compact orbit with an orbital period of about an hour. Thus the orbital separation is indeed very small, of the order 10^{-3} AU, or even less. Several such binaries are capable to stabilize a globular cluster against collapse of the compact core.

Evaporation. The stars in the cluster halo can escape from a cluster if the encounters with other stars transfer enough energy so that they can escape from the system. For this a star must reach a velocity above the **escape speed** $v_e(r)$ or its total energy must become positive:

$$E_{\text{kin}} + E_{\text{pot}} = \frac{1}{2}m_S v^2(r) + m_S \Phi(r) > 0.$$

or

$$v(r) > v_e(r) = \sqrt{2\Phi(r)}.$$

This can be generalized to an expression $v_e^2(\vec{x}) = -2\Phi(\vec{x})$ so that we can write a general mean-squared escape velocity for a system with a density $\rho(\vec{x})$ according to

$$\langle v_e^2 \rangle = \frac{\int \rho(\vec{x}) v_e^2(\vec{x}) d\vec{x}}{\int \rho(\vec{x}) d\vec{x}} = -2 \frac{\int \rho(\vec{x}) \Phi(\vec{x}) d\vec{x}}{M} = -4 \frac{E_{\text{pot}}}{M}$$

According to the virial theorem $2E_{\text{kin}} + E_{\text{pot}} = 0$, where E_{kin} is the total kinetic energy $M\langle v^2 \rangle/2$, the root mean squared (rms) escape speed is just twice the rms speed:

$$\langle v_e^2 \rangle = 2\langle v^2 \rangle.$$

We may assume that the velocity distribution behaves in a collisionally dominated system ($t > t_{\text{relax}}$ like a Maxwellian distribution, where a fraction of about $\gamma = 0.7\%$ of particles have a velocity which is $v > 2\langle v \rangle$. Thus we can assume that the two-body interaction removes about a fraction γ of stars by evaporation every relaxation time:

$$\frac{dN}{dt} = -\frac{\gamma N}{t_{\text{relax}}} = -\frac{N}{t_{\text{evap}}}.$$

Thus the **evaporation time** is of the order

$$t_{\text{evap}} = \frac{t_{\text{relax}}}{\gamma} \approx 140 t_{\text{relax}}.$$

Thus any system with an age comparable to $\tau \approx t_{\text{relax}}$ will have lost a substantial fraction of its stars. If we use the characteristic relaxation time scale for open cluster $t_{\text{relax}} \approx 10$ Myr then we obtain an evaporation time scale of the order 1.5 Gyr. This is of the same order of magnitude, although a bit higher, than the estimated typical age of stellar cluster of about $t \approx 0.3$ Gyr.

Most likely, there exist additional processes, which accelerates the evaporation of open clusters in the galactic disk. A possible process is the gravitational interaction of clusters with molecular clouds which enhances the stellar velocity dispersion in the cluster and shortens the evaporation time scale.

Chapter 4

Milky Way formation and evolution

A description of the Milky Way formation and evolution requires the understanding of many different processes:

- the hydrodynamics of proto-galaxies,
- the transport of angular momentum,
- the incidence of gravitational instabilities,
- the star formation, nucleosynthesis and the chemical enrichment of the gas,
- the accretion history of the galaxy,
- the effects of a time-dependent gravitational potential,
- the origin, growth and effects of magnetic fields.

Many of these processes can only be described with significant uncertainties because we do not know well essential parameters. Nonetheless one can try to estimate the properties of certain processes with idealized models and comparisons with observations of the Milky Way and other galaxies.

4.1 Virial theorem and galaxy formation

Some basic constraints on the galaxy formation process can be gained from simple considerations of the virial theorem

$$E_{\text{kin}} = -\frac{1}{2}E_{\text{pot}}.$$

The standard picture is that galaxies form in growing dark matter concentrations. If gas cooling can be neglected then the virial temperature for the gas falling into the proto-galactic potential will have a temperature of

$$E_{\text{therm}} \approx \frac{M kT}{m_{\text{H}}} \approx \frac{GM^2}{R}.$$

This yields for typical quantities of a galaxy ($R = 10$ kpc, $M = 10^{11}M_{\odot}$), a virial temperature of

$$T_{\text{virial}} = \frac{GMm_{\text{H}}}{kR} \approx 5 \cdot 10^6 \text{ K} \frac{M/10^{11}M_{\odot}}{R/10 \text{ kpc}}.$$

The fact that the Milky Way disk is a significant component of the galaxy consisting of co-rotating stars and gas moving in an orbital equilibrium and a small velocity dispersion ($\sigma \approx 30$ km/s) around the galactic center implies the following:

- the gas must dissipate a lot of energy by collisional excitation of atoms and molecules and subsequent radiative emission in order to cool to low temperature,
- the gas has settled into a rotating disk by gas friction and angular momentum conservation,
- most disk stars share the same overall circular motion around the galactic center indicates that they were all born over many Gyr out of a cold rotating disk.

A very similar estimate based on the virial theorem can be made for the kinetic velocity dispersion σ of N collisionless particles (= stars) with mass m_S ($M = Nm_S$) falling into such a proto-galactic potential

$$E_{\text{kin}} = \frac{1}{2}Nm_S\sigma^2 \approx -\frac{1}{2}\frac{GM^2}{R}.$$

For the velocity dispersion of the stars it is not so important whether the mass falling into the potential consists only of stars or a mixture of stars and gas. The estimate for the resulting velocity dispersion is then for the galaxy parameters given above

$$\sigma \approx \sqrt{\frac{GM}{R}} \approx 200 \text{ km/s} \sqrt{\frac{M/10^{11}M_\odot}{R/10 \text{ kpc}}}.$$

This is about the velocity dispersion observed for the system of globular clusters and halo stars. Thus they have been formed far from the center of the potential and seem to have “fallen” into the system without much energy dissipation.

On the other side one can say that the stars in the central bulge of the Milky Way have much smaller kinetic velocities than expected for a population of infalling “collisionless” stars. This indicates that these stars were formed after a lot of energy dissipation from cold gas which had a more spherical distribution and less angular momentum than the disk.

Time-scales. One should also consider the typical time-scales for the mass infall and the energy dissipation for a proto-galaxy. The time-scale for the collapse of a gas cloud is derived in the section on star formation

$$t_{\text{coll}} \approx 5 \cdot 10^7 \text{ yr} \frac{1}{\sqrt{n_{\text{H}}}}.$$

If we consider that the dark matter mass is about a factor 5 higher than the baryonic matter mass, the collapse time scale is shorter by a factor of a few.

It can be assumed that the infalling gas was either neutral or photo-ionized as can be inferred from the intergalactic Ly α absorption lines in quasar spectra. Thus the infalling gas has a temperature of 10^4 K or less and the corresponding sound speed is about $c_s \approx 10$ km/s or less. Thus the infalling gas will produce shocks which results in significant gas heating. This needs to be radiated away by radiative cooling which happens on a time scale of

$$t_{\text{cool}} \approx \frac{nkT}{n^2\Lambda(T)}.$$

It can be shown that the cooling is very roughly proportional to the gas temperature

$$\Lambda(T)/T \sim 10^{-28 \pm 1} \text{erg cm}^3 \text{s}^{-1} \text{K}^{-1}$$

with deviations of about one order of magnitude. Thus the cooling time scale is roughly

$$t_{\text{cool}} \approx 4 \cdot 10^4 \text{yr} \frac{1}{n_{\text{H}}}.$$

This is typically much shorter than the collapse time scale. This means that gas falling into the dark matter potential of a proto-galaxy is cooling fast, and can dissipate efficiently its energy. Thus, gas will settle rapidly in the center of the galaxy where it can cool and form stars.

4.2 Milky Way evolution and high redshift observation

From Galactic studies it is difficult to derive the temporal history of its formation and evolution. Stars can be used as age indicators but it is hard to associate individual stars with past evolutionary events.

With modern large telescopes it is possible to follow the evolution of galaxies with high redshift observations. The look-back times for galaxies at redshift $z = 0.5, 1, 2$, and 5 are about $\tau_{\text{b}} \approx 5, 8, 11$ and 13 Gyr. These high redshift galaxy observations trace mainly the phases with strong star formation because then the objects are bright. Galaxy surveys at different red-shifts show:

- galaxies like the Milky Way show for $\tau_{\text{b}} > 5$ Gyr a significantly higher star formation rate, which is about an order of magnitude enhanced for red-shifts between $z = 1$ and 3 when compared to galaxies in the local Universe,
- galaxies at $z > 1$ show less well defined spiral structures, but often very bright knots located near the galaxy center, which shine so bright because of very strong starburst events,
- the estimated stellar mass of objects which are expected to evolve into a disk galaxy like our Milky Way is typically about 1000 times smaller for $\tau_{\text{b}} \approx 13$ Gyr ($z = 5$), about 100 times smaller for $\tau_{\text{b}} \approx 10$ Gyr, and about 10 times smaller for $\tau_{\text{b}} \approx 7$ Gyr than the current mass of our Milky Way.

The general picture on the evolution of the Milky Way galaxy which emerges from the study of distant galaxies can be described as follows:

- 10 Gyr ago, the Milky Way was continuously fed by streams of fresh gas and merging galaxies as it is observed for high redshift galaxies $z > 2$,
- the constant inflow of gas and minor mergers produce disk instabilities and disk contractions due to counter-rotating streams, inducing phases of violent star formation in the central region (< 2 kpc) which become “in the end” the central bulges,
- since about 5 Gyr up to about 2 Gyr the gas inflow is significantly reduced and the star formation in the central bulge region is suppressed by e.g. rapid gas consumption, the activity of the central supermassive black hole, and/or blow out of gas by stellar winds and supernovae, while the rotationally supported extended disk shows continuous star formation and growth (inside-out growth) but at a slow rate,

- since about 2 Gyr the disk structure is well established, gas inflow and minor mergers with other galaxies still takes place but at a much reduced rate (100 times less) than in the early Universe and the star formation is much reduced but steadily continuing in the disk outside the bulge.

The high redshift galaxy surveys provide an excellent overall picture of the evolution of galaxies of the type of our Milky Way. On the other side it is impossible to deduce from high redshift observations detailed information about the properties of the disk, bulge and halo of the Milky Way.

4.3 Gas infall and minor mergers today

Galaxy surveys demonstrate that the life of galaxies was much “wilder” in the past, with many gas accretion events and minor mergers with small galaxies. Similar events are still happening in the Milky Way today but at a lower rate.

4.3.1 Gas inflow

With H I radio observations many hydrogen clouds were detected in the galactic halo. These clouds are special because they show high radial velocities $|v_r| > 70$ km/s with respect to the local standard of rest. This property allows us to distinguish them from local H I high latitude clouds. Slide 4–1 shows a galactic map of these high velocity clouds.

The distance to the high velocity clouds is typically between 2 and 15 kpc and they are located up to 10 kpc above or below the galactic plane. The distance can be determined by measurements of the Ca II or Na I absorptions in the spectra of halo stars with known distances. From the column density of about $n_H = 10^{18} \text{cm}^{-2}$ and the typical cloud extensions the total mass in the high velocity clouds is estimated to be of the order $\approx 10^8 M_\odot$.

The overall gas inflow rate is estimate to be at a level of about $\approx 1 M_\odot/\text{yr}$ or a bit less. This inflow of fresh gas keeps the star formation rate at a constant level (also about $1 M_\odot/\text{yr}$) in the Milky Way disk.

Besides the high velocity clouds, there are two other types of high galactic latitude H I clouds:

- The Magellanic stream which is made of H I gas stripped off from the two Magellanic clouds due to tidal interaction between SMC and LMC. The Magellanic stream is at a distance of about 55 kpc (like the LMC and SMC) and it contains a mass of about $\approx 10^9 M_\odot$.
- Intermediate velocity clouds ($|v_r| \approx 40$ km/s) which could be the results of the ejection of gas from the galactic disk by supernova explosions, which cooled and falls now back onto the disk.

4.3.2 Mergers with dwarf galaxies

There are about 10 galaxies with are located very close, within 100 kpc, of the Milky Way. This list includes the Magellanic Clouds, LMC at 50 kpc and SMC at 65 kpc, but also several dwarf spheroidals and dwarf elliptical galaxies (UMa dwarf galaxy I and II, Sgr dwarf, UMi dwarf, Sex dwarf, Scu dwarf, Dra dwarf and a few others). There is a quite

large probability that some of these galaxies will collide in the coming few Gyr with the Milky Way.

Actually the Sagittarius dwarf spheroidal (Sgr dSph) galaxy is currently colliding with the Milky Way. The collision takes place at a distance of about 20 kpc from the sun on the “other side” of the galactic center. Sgr dSph is a small galaxy with only about 10^7 population II stars ($[\text{Fe}/\text{H}] < -1.6$) and apparently no gas. The system moves through our Milky Way in a roughly polar orbit. It is elongated, most likely due to the strong tidal forces, and many stars were already lost by the galaxy and they spread along a stellar stream in the halo (Slide 4–2 and 4–3). The orbit of Sgr dSph extends to about 40 kpc with a period of the order 600 Myr. Several globular clusters are associated with the Sgr dSph galaxy and one of them, M54, could be the core of the galaxy.

This example shows that the interpretation of stellar properties of stars in the halo and the thick disk can be very complex. Such merger events have happened often in the past and in the end it is very difficult to find out for a given star or stellar group whether they were “born” in the Milky Way, or whether they were accreted by a merger event. Slide 4–3 illustrates with an artist impression the collision of the Sgr dSph with the Milky Way. A most impressive example (and therefore not representative) are the observations of the edge-on galaxy NGC 5907 where the stellar stream created by the collision can be nicely observed with very deep observations (Slide 4–4).

4.4 The chemical evolution of the Milky Way

The metallicity of the stars and gas in the Milky Way is far from homogeneous and the distribution provides interesting information about the Milky Way evolution. Important features are:

- the old stars $\tau_{\text{age}} \gtrsim 10$ Gyr in the globular clusters and the halo are metal poor with a metallicity $[\text{Fe}/\text{H}] \lesssim -1$,
- the galactic bulge consists mainly of intermediate age and old stars $\tau_{\text{age}} \gtrsim 2$ Gyr, but hardly any young stars $\tau_{\text{age}} \lesssim 300$ Myr. The galactic bulge stars show typically a metallicity which is higher than the solar metallicity $[\text{Fe}/\text{H}] > 0$,
- the Galactic disk shows two metallicity gradients:
 - one as function of radius where the metallicity decreases from $[\text{Fe}/\text{H}] \gtrsim 0$ inside the sun ($R < R_0$) to $[\text{Fe}/\text{H}] \lesssim 0$ for the outer disk $R > R_0$,
 - one in vertical direction with older and lower metallicity stars located in a thick disk while the young, higher metallicity stars are located close to the disk mid-plane.

This metallicity distribution is in qualitative agreement with the evolution history derived from the galaxy surveys. However, it is possible to extract much more details from accurate abundance measurements and detailed modelling of the gas enrichment with heavy elements.

4.4.1 Nucleosynthesis and stellar yields

The stars produce heavy elements by nuclear fusion and some fraction of these products are expelled at the the end of the stellar evolution in stellar winds and explosions. This

“lost” stellar material enriches the interstellar gas with heavy elements. This production of heavy elements depends on the mass of the star. This subsection on stellar yields follows the paper from A. Maeder (1992, *Astron. & Astrophys.* 264, 105). Calculations for the stellar yields are reproduced on Slide 4–5, 4–6 and 4–7. Slide 4–5 shows the end products resulting from the stellar nuclear burning for gas which has initially a metallicity of 0.02 (or 2 %).

There are three main components which define the resulting heavy element enrichment by a star of a certain star:

- the fraction of the initial mass which is in the end locked in the stellar remnants,
- the fraction of the gas which is lost by the star but which is not converted to heavy elements by nuclear processes,
- the fraction of the gas which is processed into heavy elements by nuclear processes and subsequently ejected to the interstellar medium by stellar winds or/and explosions.

We discuss the enrichment of the gas with heavy elements for three different masses according to Slide 4–5.

- **1 $M_{\odot}$:** low mass stars produce a white dwarf composed of C and O with a mass of about $0.5 M_{\odot}$. Most of the nuclear burning products from the H- and He-burning phase are concentrated in this stellar remnants. The stars loses at the end of its evolution the outer envelope, but these outer layer were not enriched with the nuclear burning products and therefore low mass stars contribute very little to the chemical enrichment of the Milky Way.
- **6 $M_{\odot}$:** intermediate mass stars produce also a white dwarf but with a mass of about $1 M_{\odot}$. During the red-giant phase the outer convective layer penetrates deeply into the star and “dredges-up” the products of the hydrogen burning layer. For this reason the mass lost during the red giant phase is enriched in helium which enhances the He abundance of the interstellar medium.
- **40 $M_{\odot}$:** high mass stars are important for the enrichment of the interstellar matter with heavy elements. Objects with such high initial mass lose a lot of mass via stellar winds. First, during the blue supergiant phase, the lost mass is hardly enriched by heavy elements. But, during later evolutionary phases, as luminous blue variable or Wolf-Rayet star, the stellar wind is so strong ($\approx 10^{-5} M_{\odot}/\text{yr}$) that it peels off subsequent layers of the star and expels about $10 M_{\odot}$ of He and N rich layers (WN-phase) where previously the H-burning took place, and then about $5 M_{\odot}$ of C and O rich material (WC and WO-phase) from the He-burning zone. The star has only about $5 M_{\odot}$ left, when it explodes as SN Ib.

For stars with an initial mass in the range $10\text{--}25 M_{\odot}$ the enrichment is much less important for the enrichment of the α -elements. is mainly due to SN II explosions, which produces predominately α -elements such as Ne, Mg, Si, S, and Ca. Stellar wind mass loss is much less important for the enrichment of the α -elements.

Metallicity dependence. The stellar yields are different for low metallicity stars (see Slide 4–6). The main reason is that the gas opacities (metal line absorptions and e^{-} -scattering) are much reduced for low metallicity gas so that the radiation driven stellar

winds are much weaker. This means that a star with a given initial stellar mass loses much less mass during its evolution and the star will be much more massive and has still a hydrogen rich envelope when the “final” supernova (type II) explosion occurs. This leads to some variation in the production rates of heavy elements.

Weighting with the IMF. The stellar yield shown in Slide 4–5 and 4–6 must be weighted with the initial mass function. There result the IMF-weighted stellar yields diagrams shown in Slide 4–7.

This diagram shows that the low and intermediate mass stars with $M < 8 M_{\odot}$ dominate strongly the enrichment of the Milky Way in He, while these objects have no impact for heavier elements (but see below the section on SN Ia). The more massive stars $M > 8 M_{\odot}$ are the dominant producers of all heavy elements and they contribute also to the He enrichment.

4.4.2 The role of SN Ia.

The previous section discussed the yields of heavy element from single star stellar evolution. However, there is a small group of strongly interacting binary stars which explode at the end of their evolution as supernova Ia and they have a significant impact on the abundances of the “iron peak” (Cr, Fe, Co, Ni, ...) elements in the interstellar medium. Most binary stars are not strongly interacting because they have wide orbits $a > 100$ AU or they lose predominantly unprocessed material. Such systems can be counted like single stars.

SN Ia explosions. A SN Ia is an explosion of a CO white dwarf. These explosions convert the carbon and oxygen in explosive thermonuclear processes into mainly ^{56}Ni and other “iron peak” elements and the whole white dwarf explodes without remaining compact remnant. This means that the fusion products (Ni, Fe, ...) are all distributed into the interstellar medium. This is unlike to the core collapse supernovae (all other supernova types) where the “final” Ni–Fe-core collapses into a neutron star or black hole. For this reason the SN Ia are a main contributor to the Fe abundance, while high mass stars produce mainly the “ α -elements” such as O, Ne, Mg, Si, S, Ca.

The models for a SN Ia predict that the explosion takes place when a CO white dwarf accretes matter and approaches the Chandrasekhar mass limit of $1.4 M_{\odot}$. This can happen in close binary stars where mass is transferred from a companion to the white dwarf or if two very close white dwarfs merge due to the loss of orbital energy by the emission of gravitational radiation. Several different types of progenitor system have been identified, but it is not clear yet, which are the best candidates for becoming in the end a SN Ia event. Nonetheless SN Ia are good standard candles for cosmological studies, because the explosions are at least in most cases due to a nuclear explosion of a rather well defined CO white dwarf “bomb” of $1.4 M_{\odot}$.

SN Ia abundance effect. SN Ia are produced by intermediate mass binary systems which live about 1 Gyr before they explode. This is much longer than the core-collapse supernovae of massive stars which explode after less than 100 Myr. For this reason one observes for metal poor stars in the Milky Way a strong α/Fe -element overabundance because during the first Gyr no SN Ia contributed to the iron-abundances. This effect is illustrated in Slide 4–8.

The $[\text{O}/\text{Fe}] - [\text{Fe}/\text{H}]$ plot (Slide 4–8) gives the abundance ratios of the elements relative to solar abundance ratios. The important features in this plot are:

- low metallicity stars $[\text{Fe}/\text{H}] < -1$ have typically an overabundance of 0.5 dex in $[\text{O}/\text{Fe}]$ (= factor 3) with respect to the sun,
- $[\text{Fe}/\text{H}] \approx -1$ corresponds roughly to the epoch where the Milky Way was about 1 Gyr old and where the SN Ia started to add significant amounts of Fe,
- since then, the $[\text{O}/\text{Fe}]$ ratio evolved steadily towards the solar value,
- the slope in the distribution of stars in the $[\text{O}/\text{Fe}] - [\text{Fe}/\text{H}]$ diagram for $[\text{Fe}/\text{H}] \gtrsim -1$ can be explained with a supernova ratio of SN Ia/SN II+Ib $\approx 1.5/1$ as indicated by the line in the diagram.

Similar plots were compiled for many different element ratios for investigations of the origin of individual elements (SN Ia or SN II ?).

4.4.3 Modelling the chemical evolution of the Milky Way

The chemical evolution of the Milky Way is often modelled with simple stellar population models which consider for a very basic first approximation at least the following processes (see also corresponding exercise):

- the star formation rate, which is for the Milky Way of the order $\approx 1 - 10 \text{ M}_\odot/\text{yr}$, what yields about 10^{11} stars in 10^{10} yr,
- the initial mass function which describes the mass distribution of the newly formed stars ($f(M) \propto M^{-2.35}$)
- stellar lifetimes which are roughly $t_{\text{age}} \propto M^{-3}$
- the stellar yields as function of mass as shown in Slides 4–5 to 4–7.
- a simple history for the description of the low metallicity mass accretion by using e.g. a closed box model $M_0 = 10^{11} \text{ M}_\odot$ or a continuous mass inflow rate in the range $\approx 1 - 10 \text{ M}_\odot/\text{yr}$.

This modelling can be elaborated in much more detail with the consideration of more realistic descriptions and including additional processes. The models can then be compared with the available observational data. However, often the modelling is ambiguous or the uncertainties in the data are too large for firm conclusions.

Despite this, several important results are based on the study of elemental abundances of the Milky Way. The models can constrain in particular:

- the chemical evolution of the stars in the halo, the bulge, and the thick and thin disk; halo stars are old, metal poor and a large fraction of them were accreted by infall or merger events, the bulge stars originate mainly from the early galactic evolution, while the disk stars were formed later.
- continuous gas inflow is important because there are much less low metallicity G-stars than expected from a closed box model. The existence of less gas in the beginning allowed for a faster rise of the metallicity early in the galactic history, explaining the low frequency of metal poor, 10 Gyr old, G-dwarfs,
- the outward metallicity drop in the Milky Way disk is simply a result of the inside-out growth of the Milky Way as observed directly from the changing average properties of galaxies with red shift.